



**SORBONNE
UNIVERSITÉ**
CRÉATEURS DE FUTURS
DEPUIS 1257



Mécanismes Moléculaires de la Signalisation Cellulaire (MMSC) Master Biologie Intégrative et Physiologie, M1S1

(2026/27)

Cours - Calcium Signaling

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Course outline

1 - Generalities

2 - Extracellular calcium: Sources and Regulation

3 - Intracellular calcium: Function

Excitation-Contraction Mechanism

Exocytosis

Plasticity

...

4 - Intracellular calcium: Regulatory Mechanisms

ON/OFF mechanisms

Membrane channels: IP₃, RyR, PMCA, NCX

Reticulum channels: SERCA

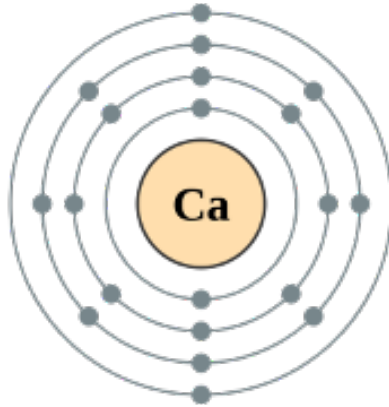
5 - Spatio-temporal profile of Ca²⁺ signals (transients, puff, spark, wave)

Non-excitabile vs Excitable cells

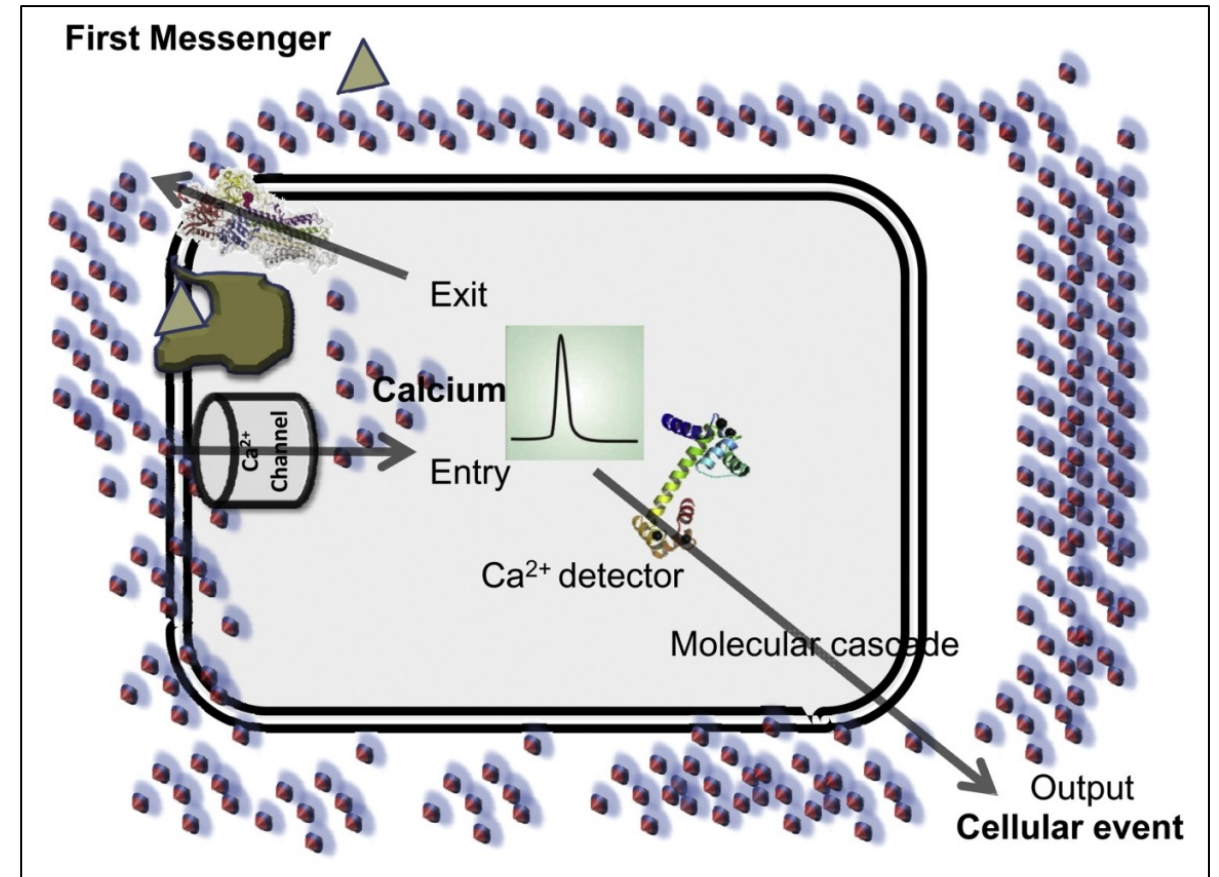
1- Generalities



^{20}Ca
40g/mol



- Alkaline earth metal, never in its pure state.
- 5th most abundant element (after Na, Cl, Mg, sulphate).
- Isolated in 1808 from lime (calcium oxide).
- Divalent cation Ca^{2+} , called free calcium, when it is not bound to inorganic anions (CaCl_2 ...) or organic molecules (proteins).

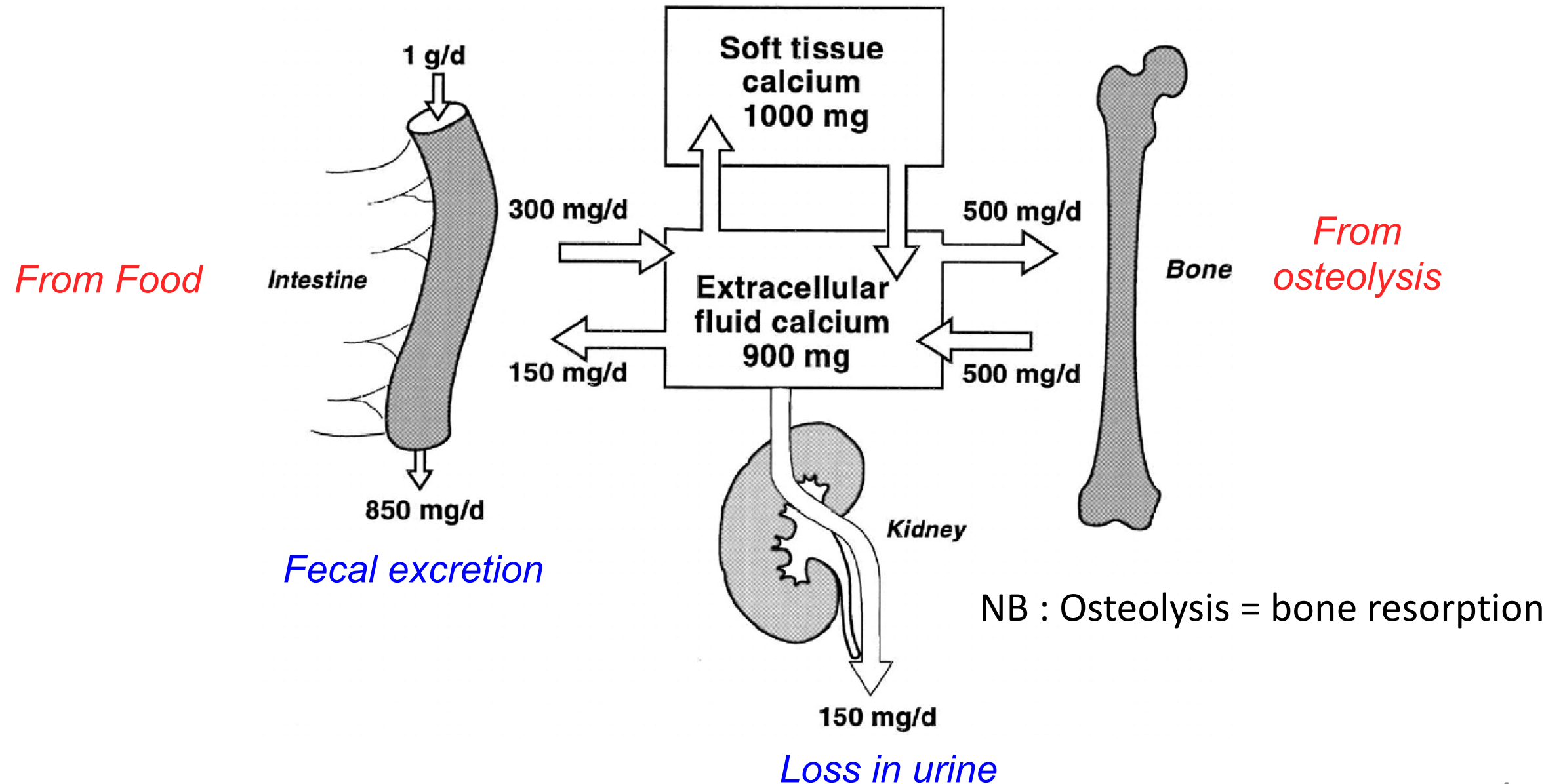


In physiology,

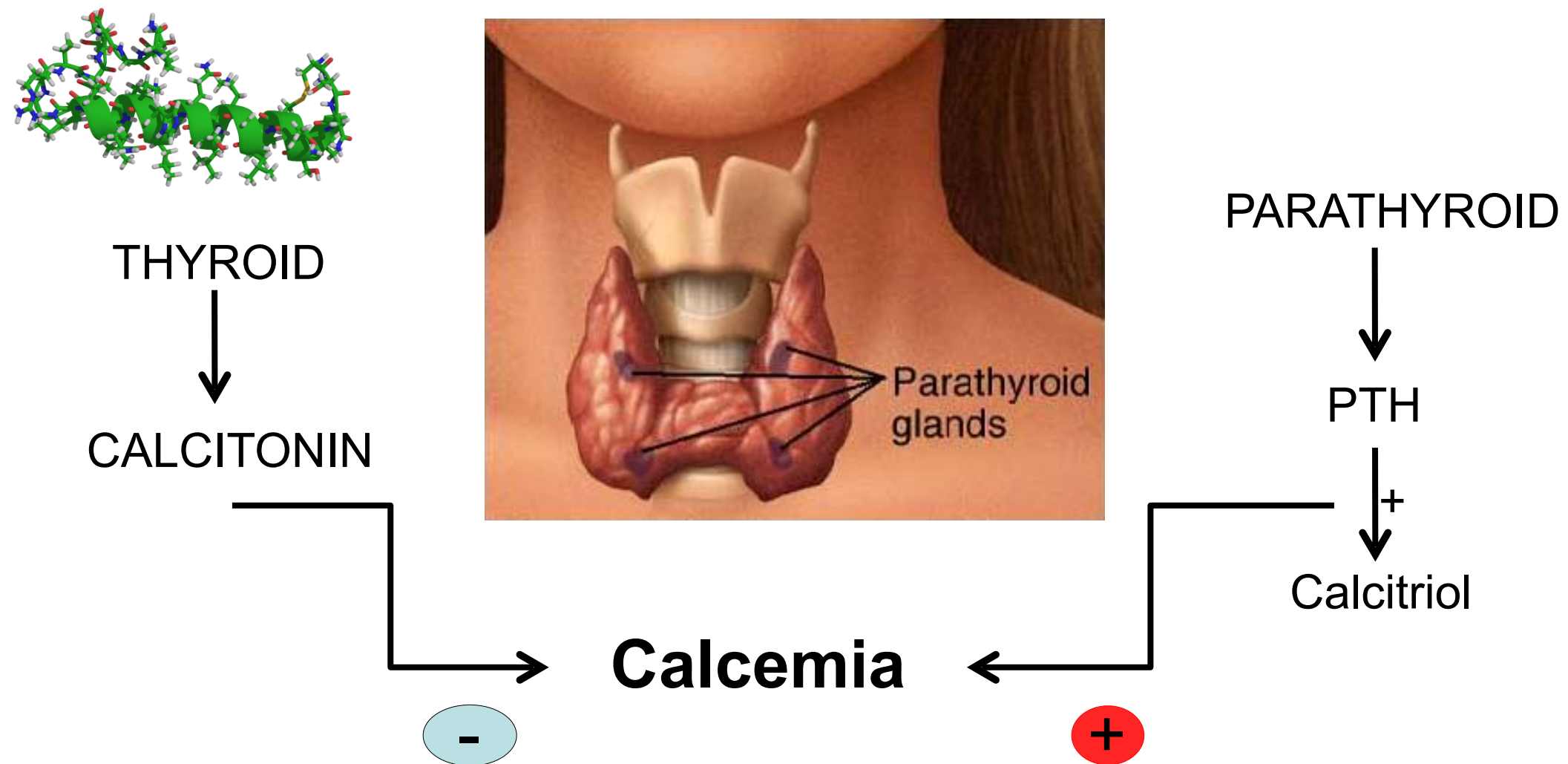
- Most abundant metal in human body (1 Kg, 99% deposit in bone)
- High extracellular (1.2 mM) / Low intracellular concentration (100 nM).
- Behaves as a 2nd messenger (signaling molecule)
- Calcium signal = "**calcium transients**" = rapid and transitory variations in intracellular calcium concentrations = $[\text{Ca}^{2+}]_i$

2- Extracellular Calcium: sources and regulation

Homeostasis = Physiological process that maintains the concentrations of certain organic molecules (minerals, fats, carbohydrates, hormones, etc.) within a range compatible with the optimal functioning of the organism/cell. **Very dynamic!**

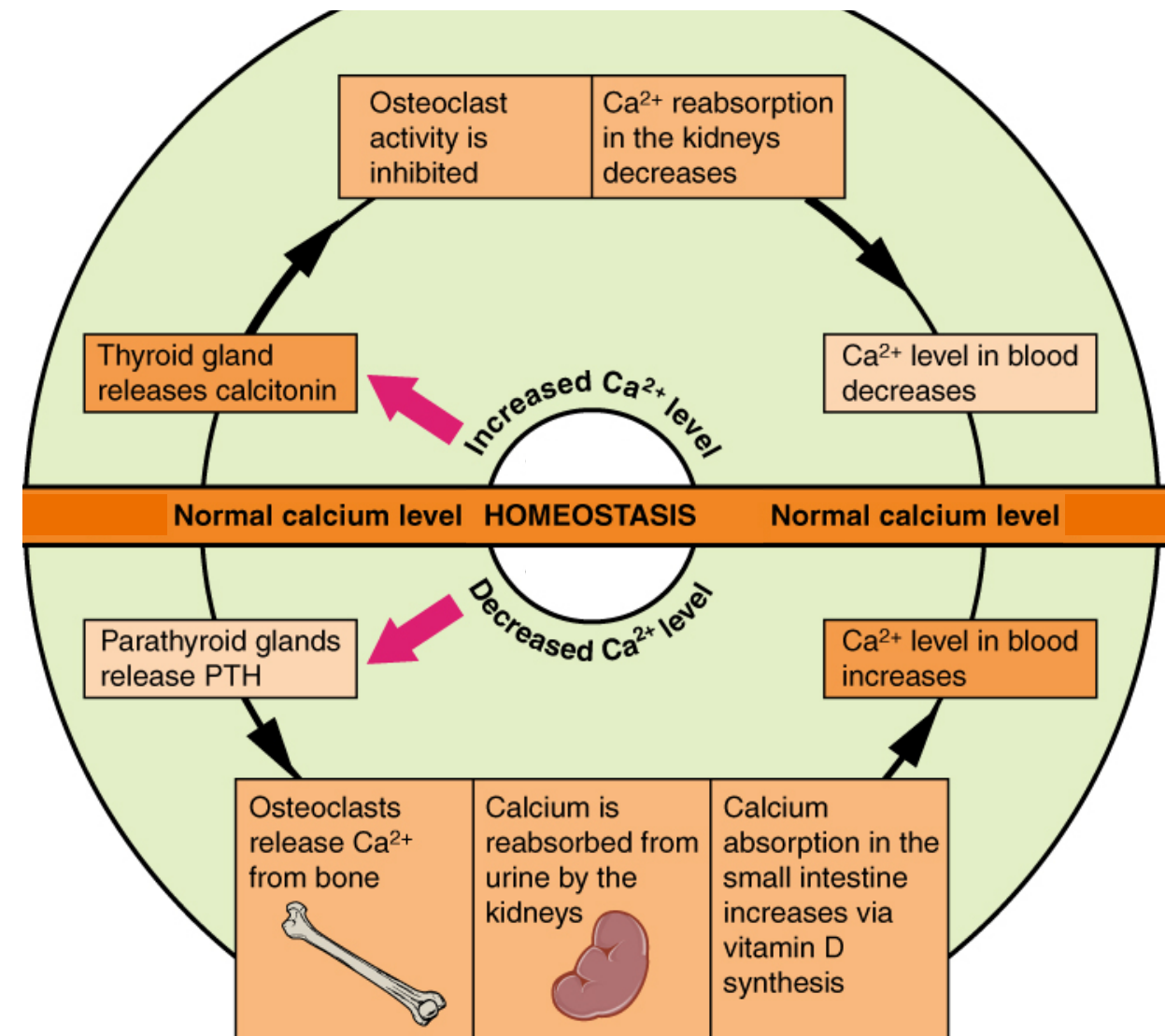
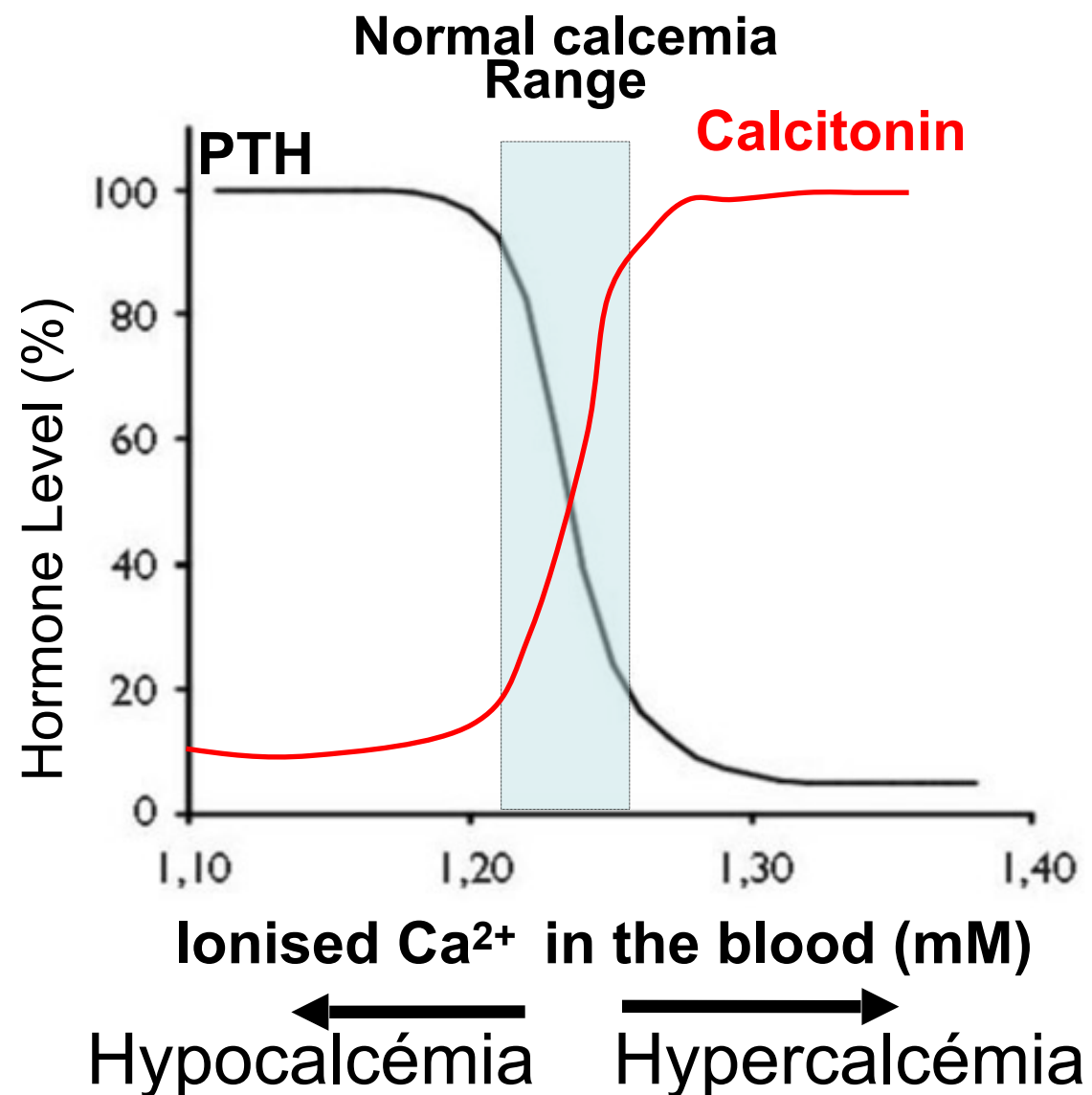
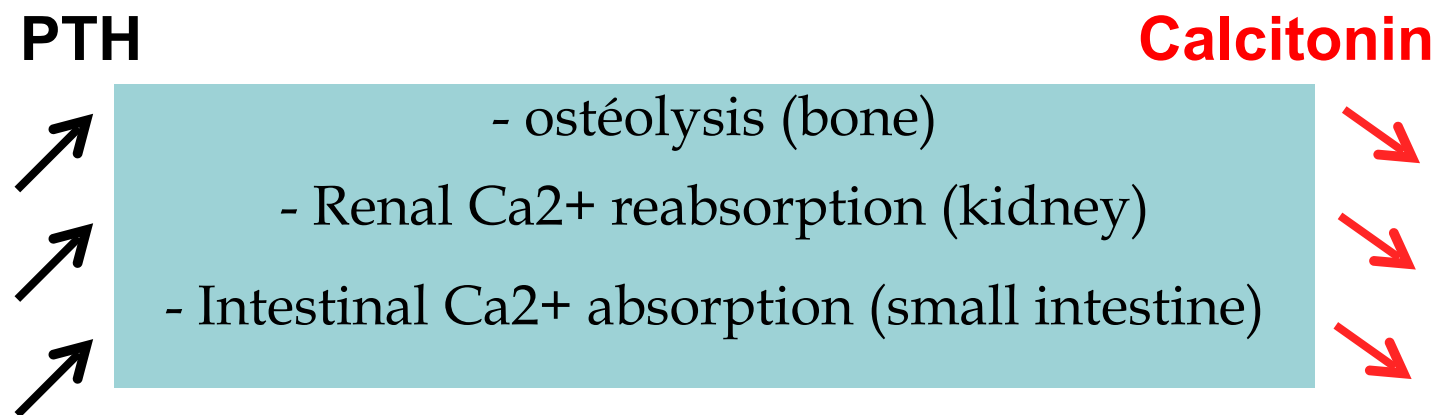


2- Extracellular Calcium: sources and regulation



- **calcitonin** = hypocalcemic peptide hormone (helix α , 32 aa) / thyroid gland
- **parathyroid hormone (PTH)** = hypercalcemic peptide hormone (84 aa) / parathyroid glands
- **calcitriol or 1.25 dihydroxyvitamin D3** = hypercalcemic steroid hormone / production in the kidney by hydroxylation of vitamin D (promoted by PTH)

2- Extracellular Calcium: sources and regulation

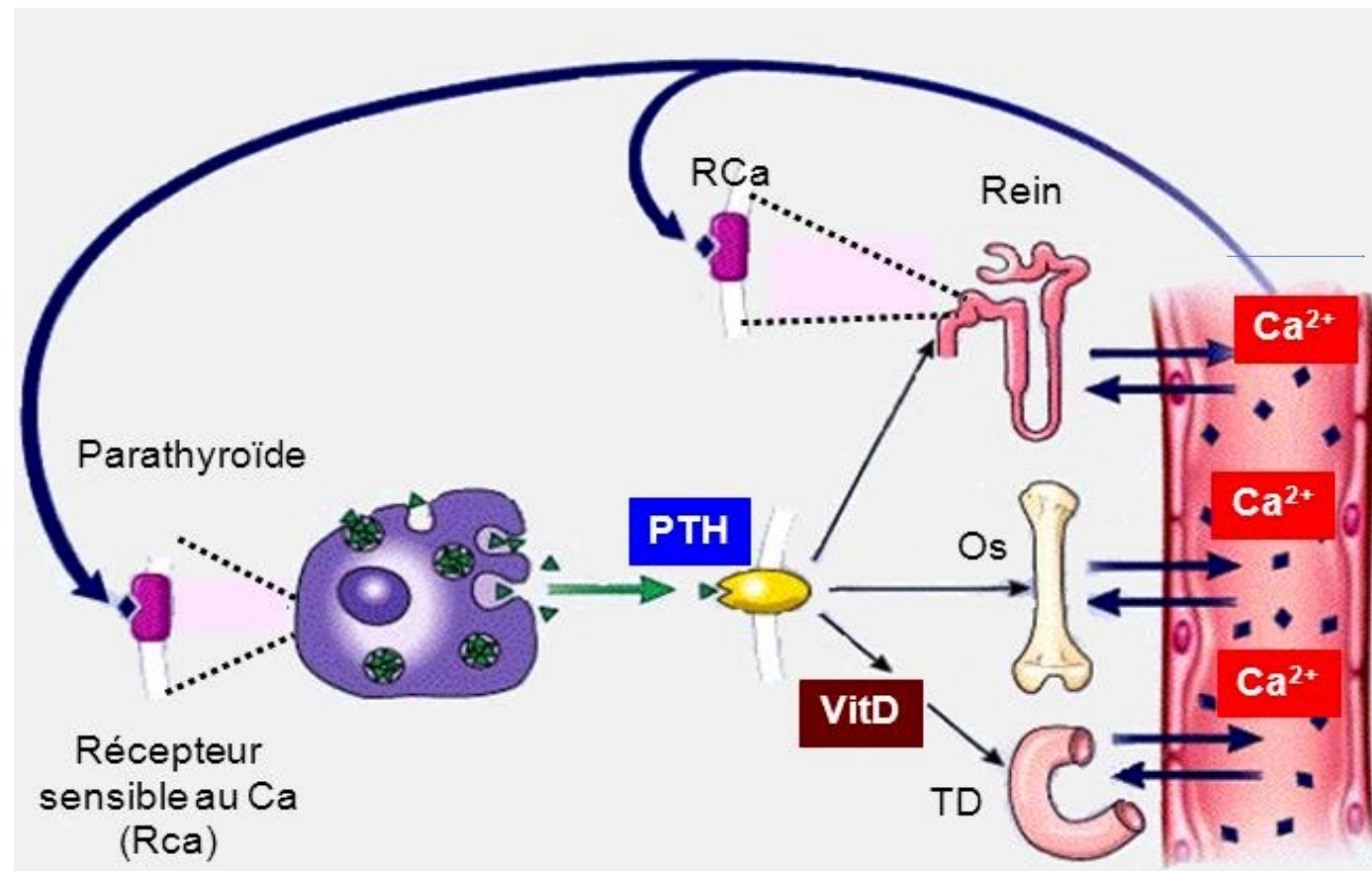


How Ca^{2+} regulates PTH and Calcitonin Levels?

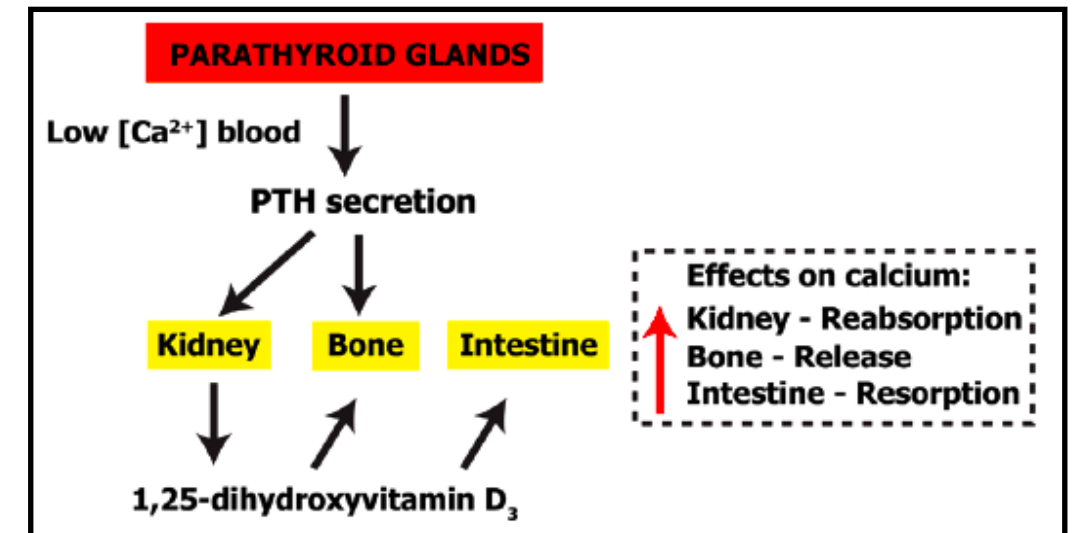
2- Extracellular Calcium: sources and regulation

Calcium Sensing Receptor (CaSR) on PTH cells Mediates the regulation of extracellular calcium homeostasis

CaSR is expressed in bone, kidney, intestine cells, C cells (thyroid)



Note: When calcium bind to CaSR receptors (high Concentration)=> PTH release by parathyroid cells is **blocked**. Activation of CaSR in thyroid cells by high Ca²⁺ increases Calcitonin release.



1. If Calcemia is low => Parathyroid cells release PTH :
 - increases Ca reabsorption in kidney
 - Ca release from bone
 - promotes the production of VitD3 in kidney that increases:
 - Ca absorption in small intestine
 - Ca reabsorption in kidney
 - Ca release from bones (osteolysis)

↓
Calcemia increase

2. Feedback inhibition : Calcium increase blocks PTH secretion

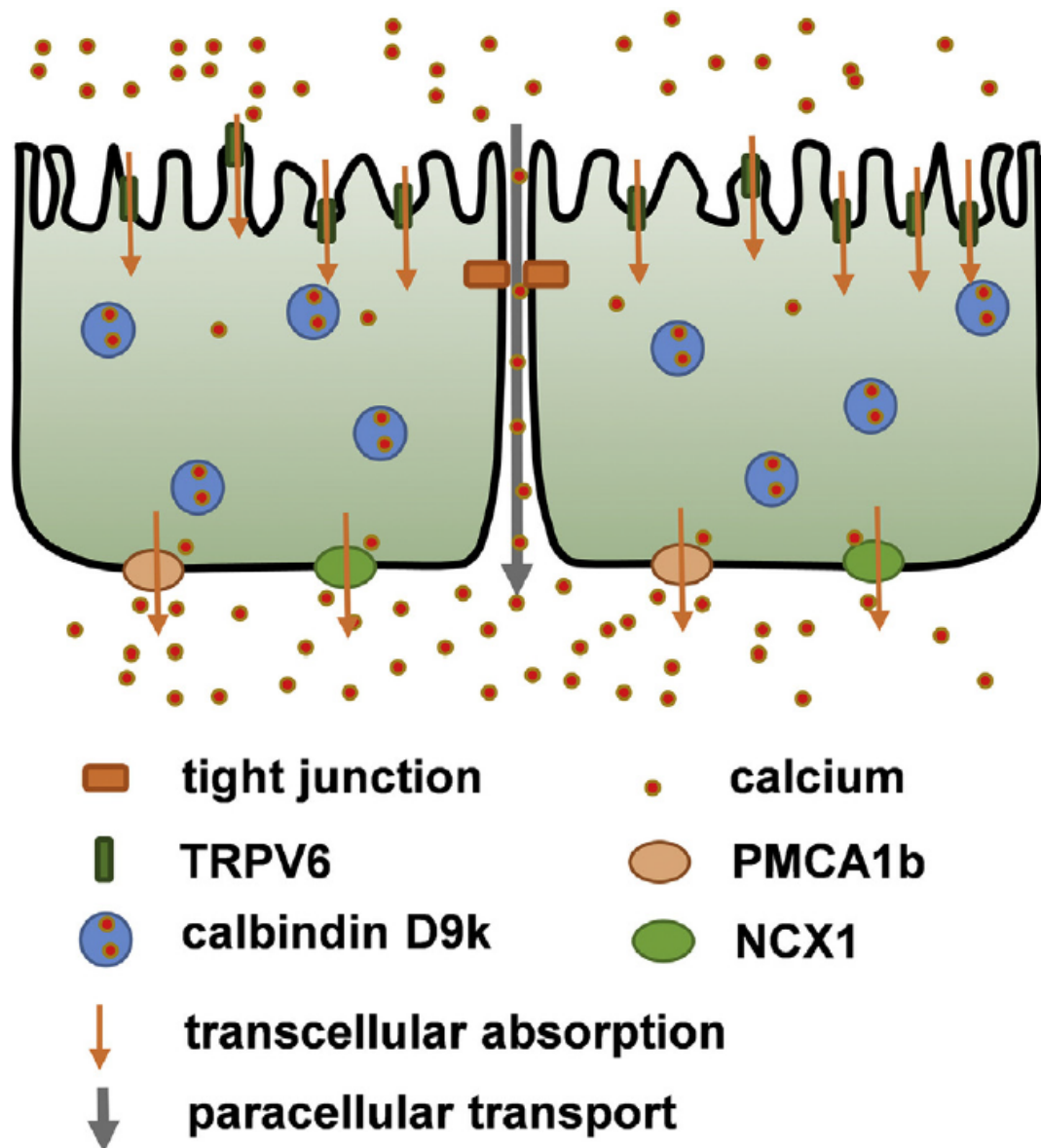
2- Extracellular Calcium: sources and regulation

2. Intestinal calcium absorption by transcellular absorption and paracellular transport

Intestine is the major organ responsible for calcium uptake.

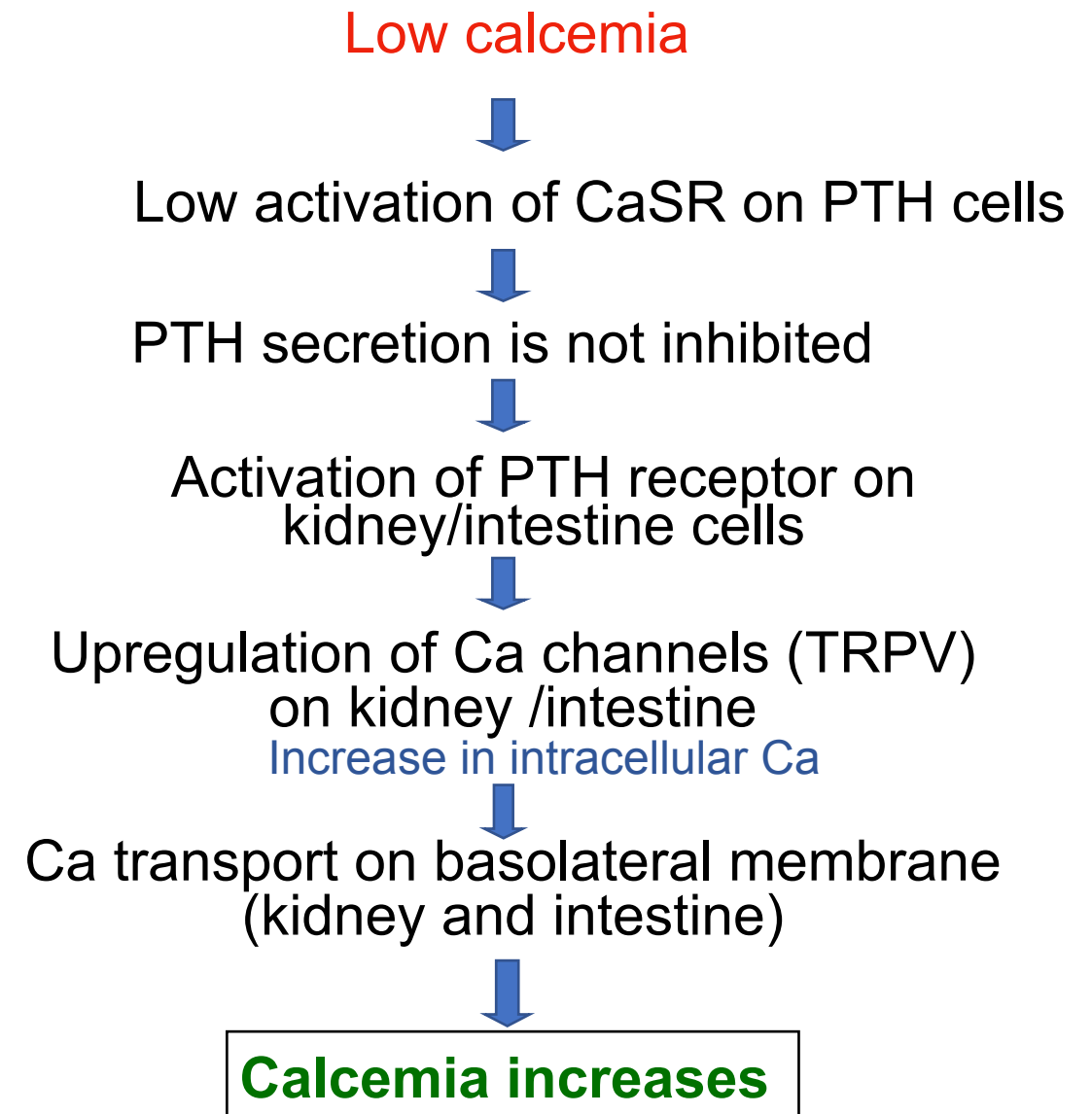
1- Entry by TRPV6 Ca channel

2 - Exit by PMCA1b in Basolateral membrane



Pu et al, 2016

Steps in calcemia regulation



NB : Transgenic mice with TRPV6 overexpression can result in hypercalcemia and soft tissue calcification

2- Extracellular Calcium: sources and regulation

Link between extracellular and intracellular calcium

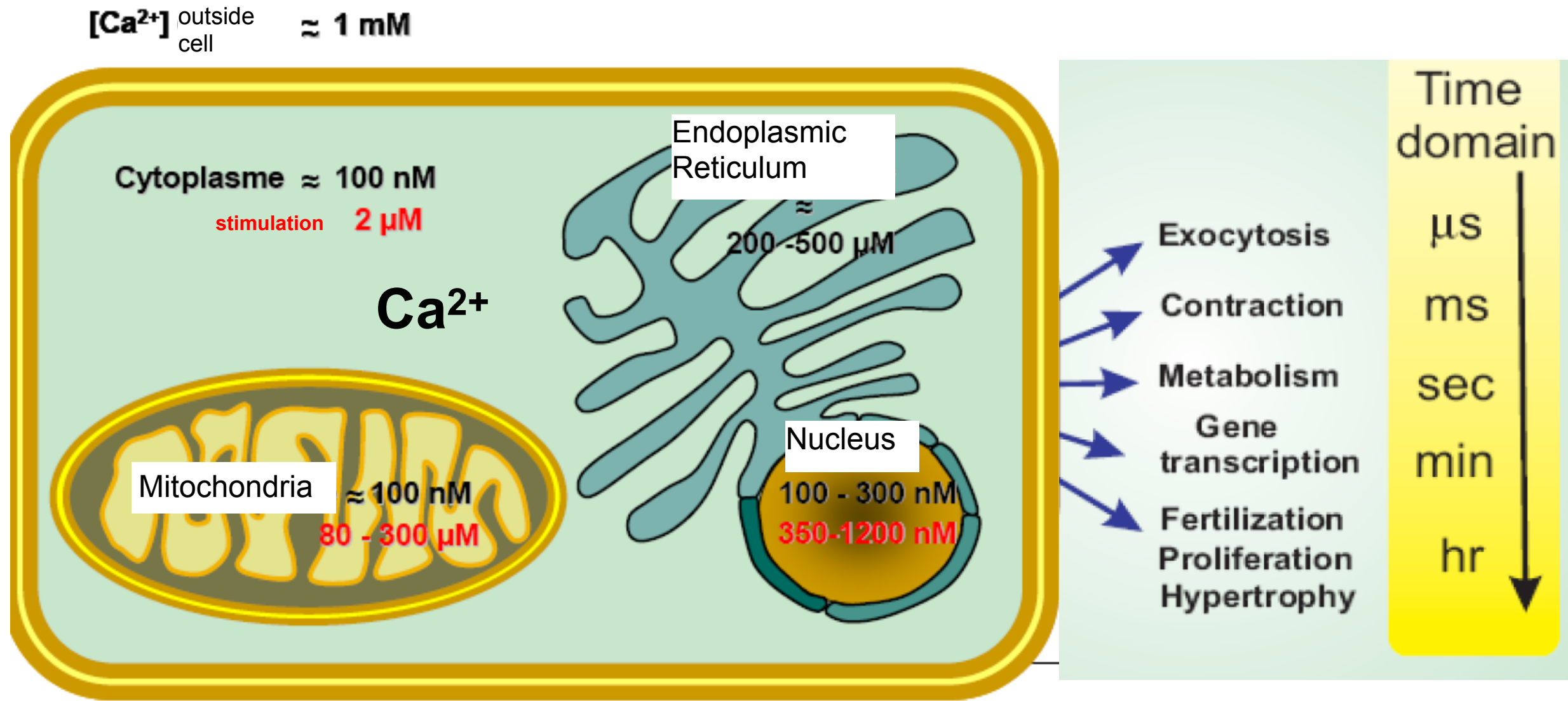
Disregulation in blood calcium levels :

- **Excess** (hypercalcemia) => kidney stones, increased blood pressure (increased cardiovascular risk?)
- **Deficiency** (hypocalcemia) => osteoporosis, growth problems (rickets), fractures, diabetes consequence of a thyroidectomy or low dietary calcium

And intracellular Ca^{2+} : What function?

3 -Intracellular Calcium: Function

MULTIPLE CALCIUM FUNCTIONS

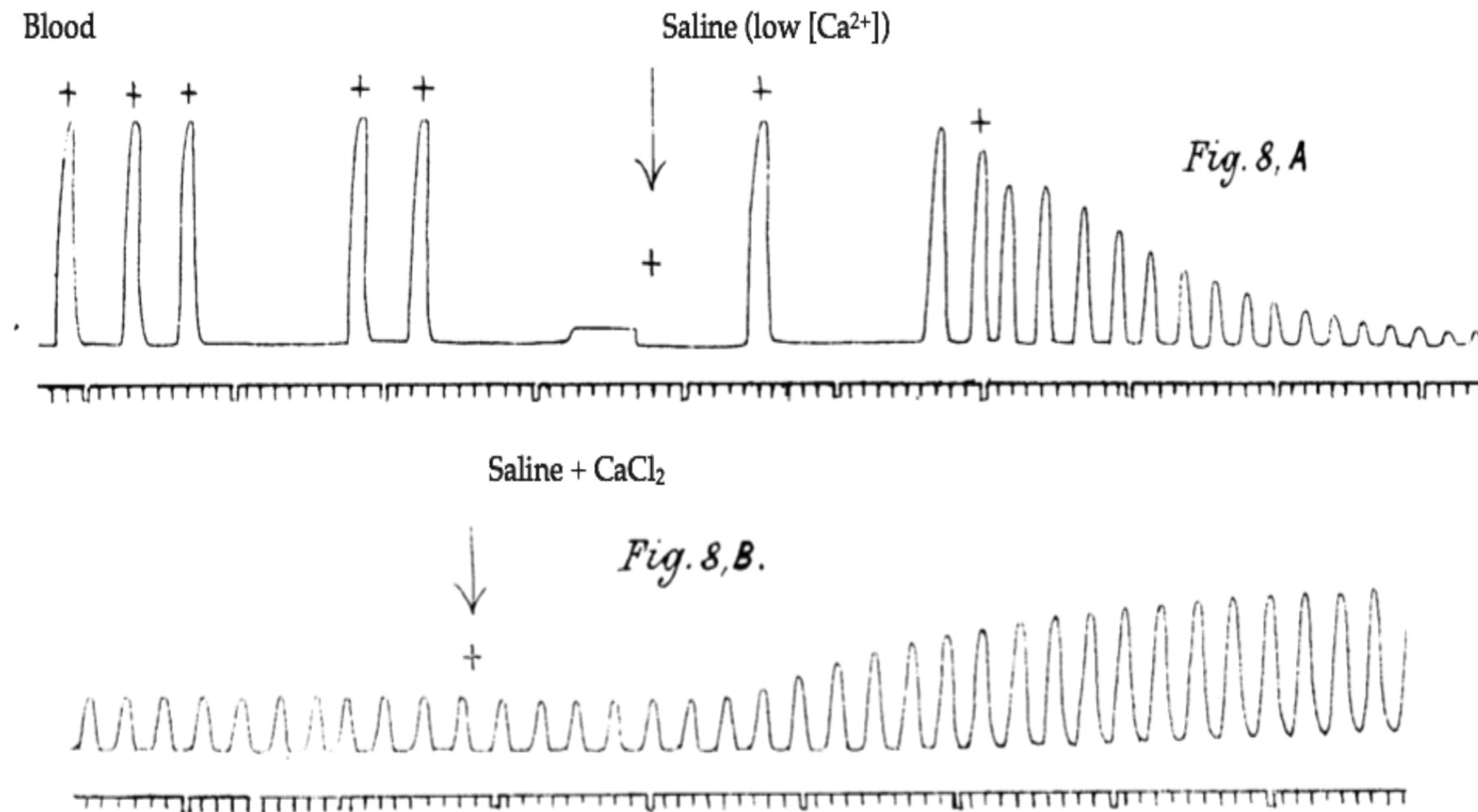


Values at rest

10

3 -Intracellular Calcium: Function

Early demonstration of the importance of Ca^{2+} for muscle contractility – Sydney RINGER 1883



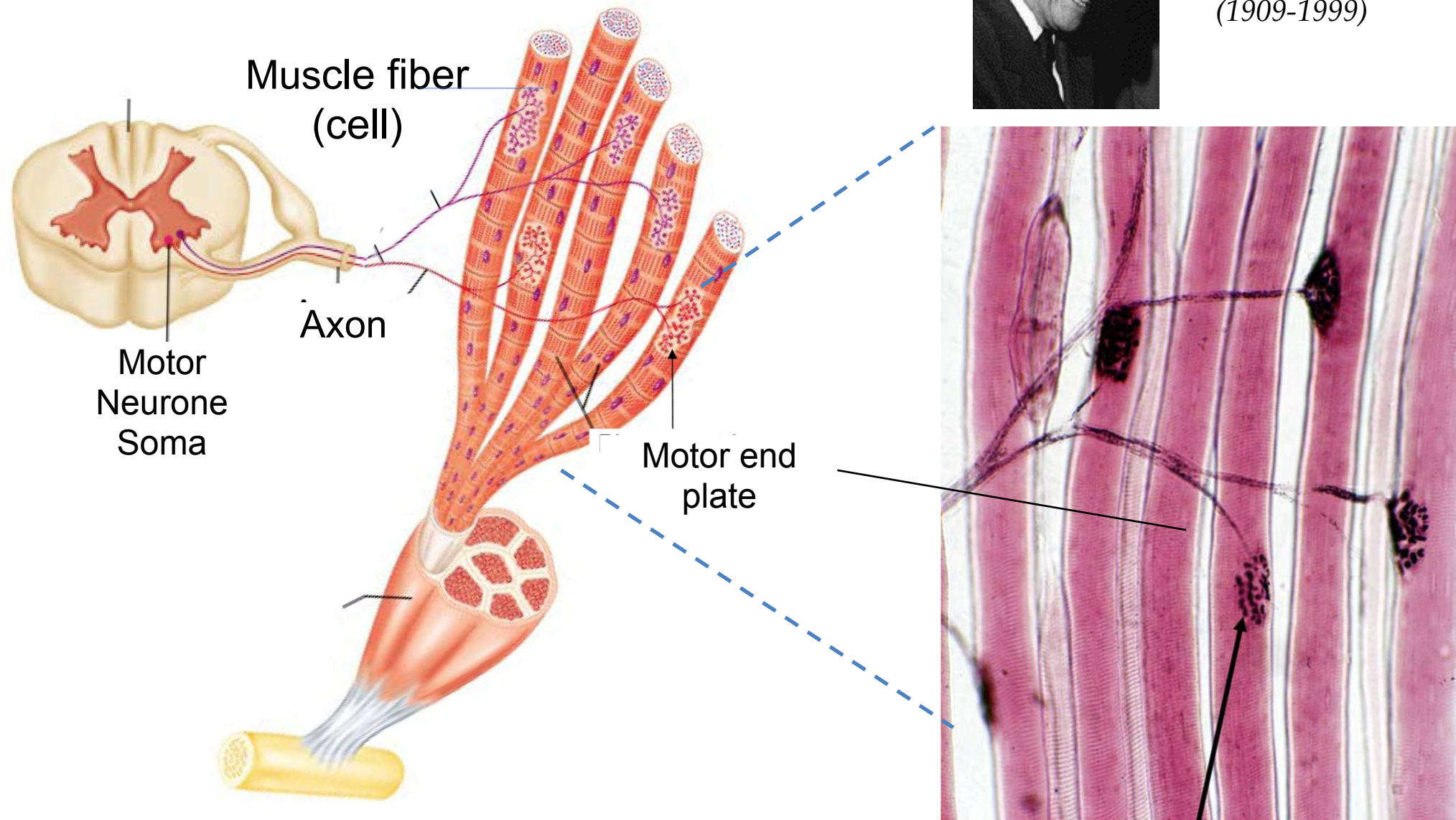
- Contraction function was recorded from an isolated frog ventricle perfused with blood or saline solutions.
- + => Cardiac contraction (measured by the volume changes in the solution).
- Full contractions were only seen when Ca^{2+} was present in the perfusate.

3 -Intracellular Calcium: Function

Excitation-Contraction Mechanism: Role of Acetylcholine & Ca



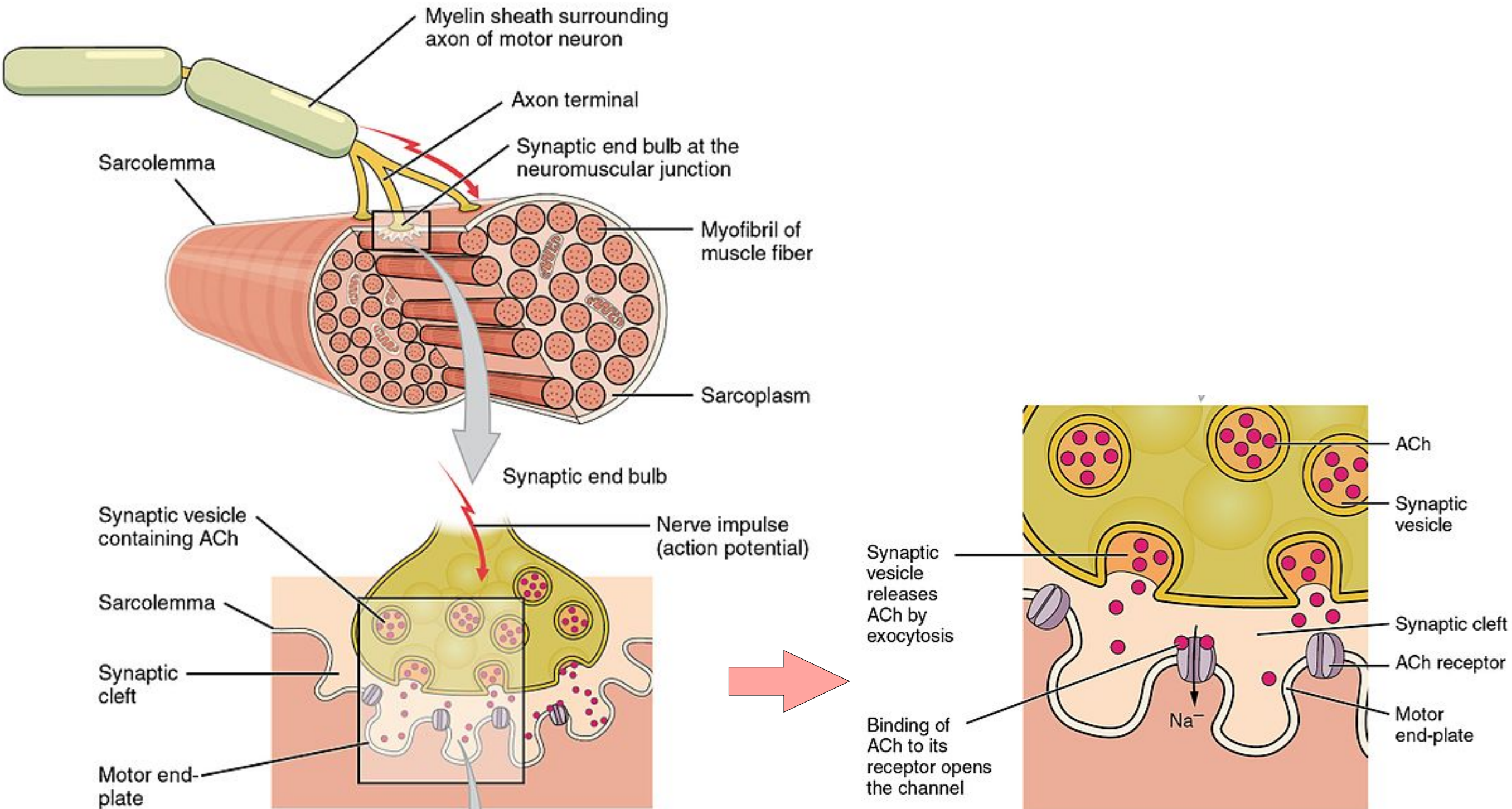
René Couteaux
(1909-1999)



Acetylcholine (Ach) is released in the motor end plate to trigger muscle contraction

3 -Intracellular Calcium: Function

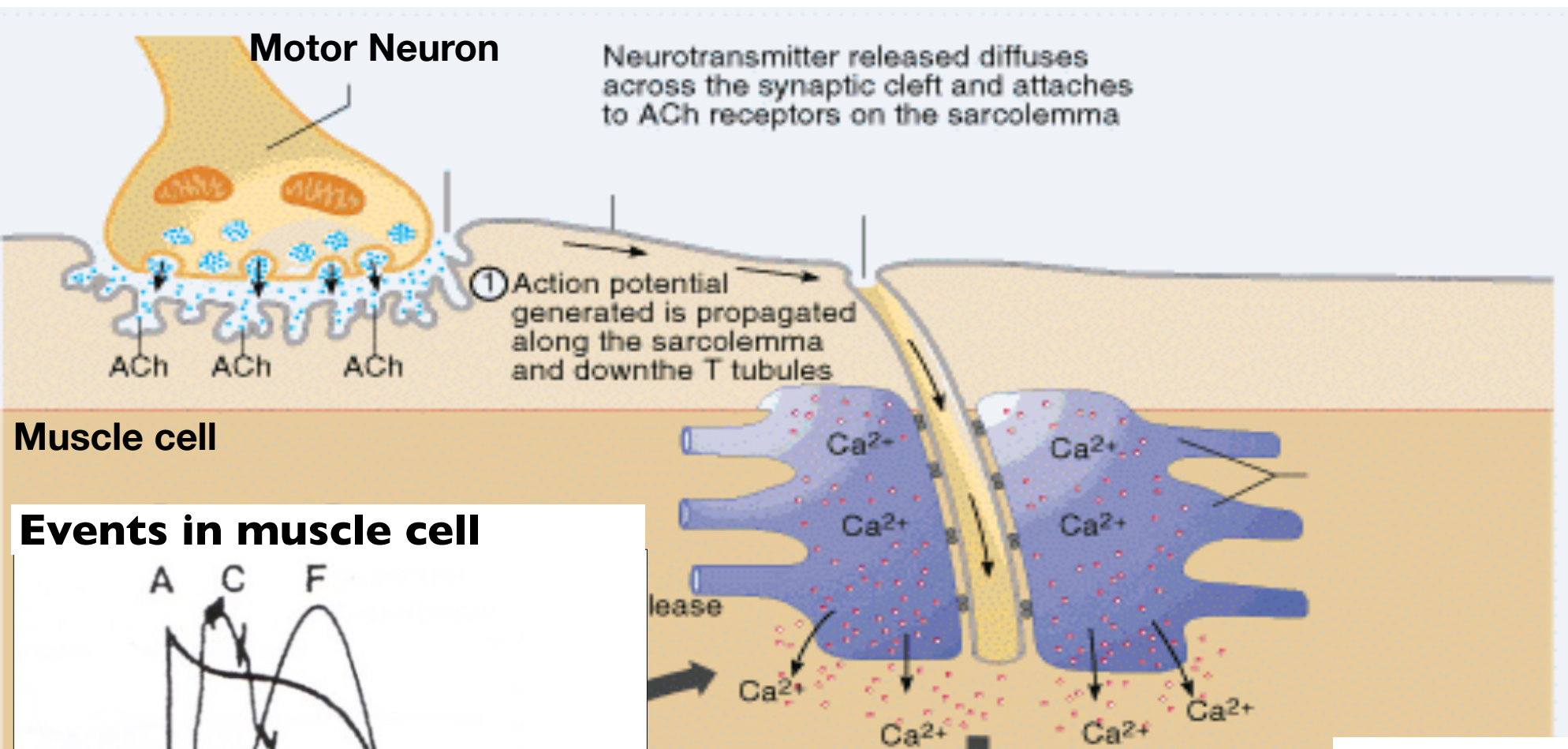
Excitation-Contraction Mechanism: Role of Acetylcholine & Ca



Note: Acetylcholine (ACh) receptor is a channel receptor => entry Na^+ ions => Dépolarisation => => Contraction

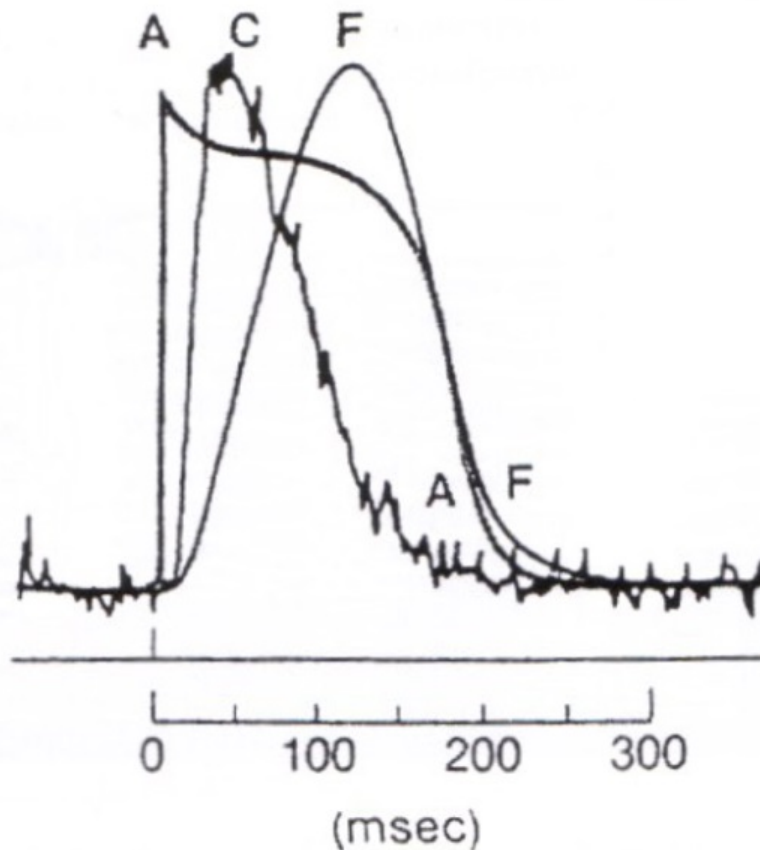
3 -Intracellular Calcium: Function

Excitation-Contraction Mechanism: Role of Acetylcholine & Ca

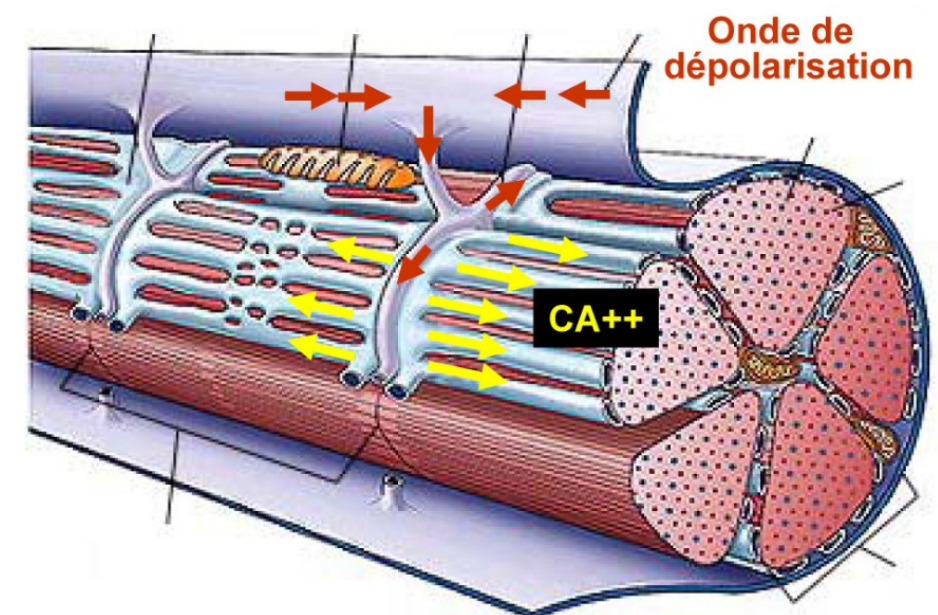


Muscle cell

Events in muscle cell

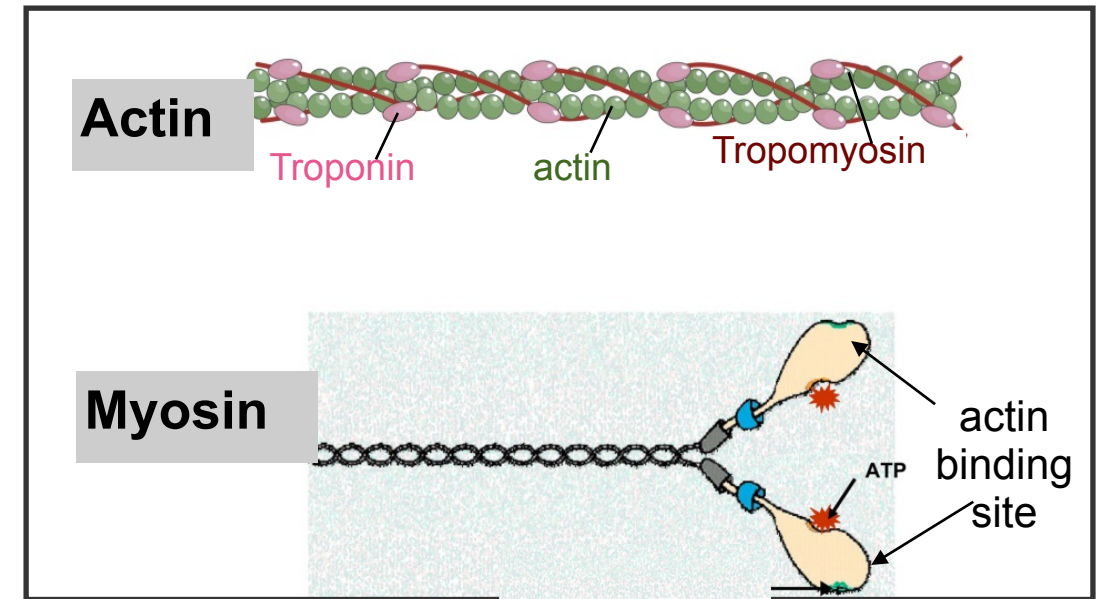
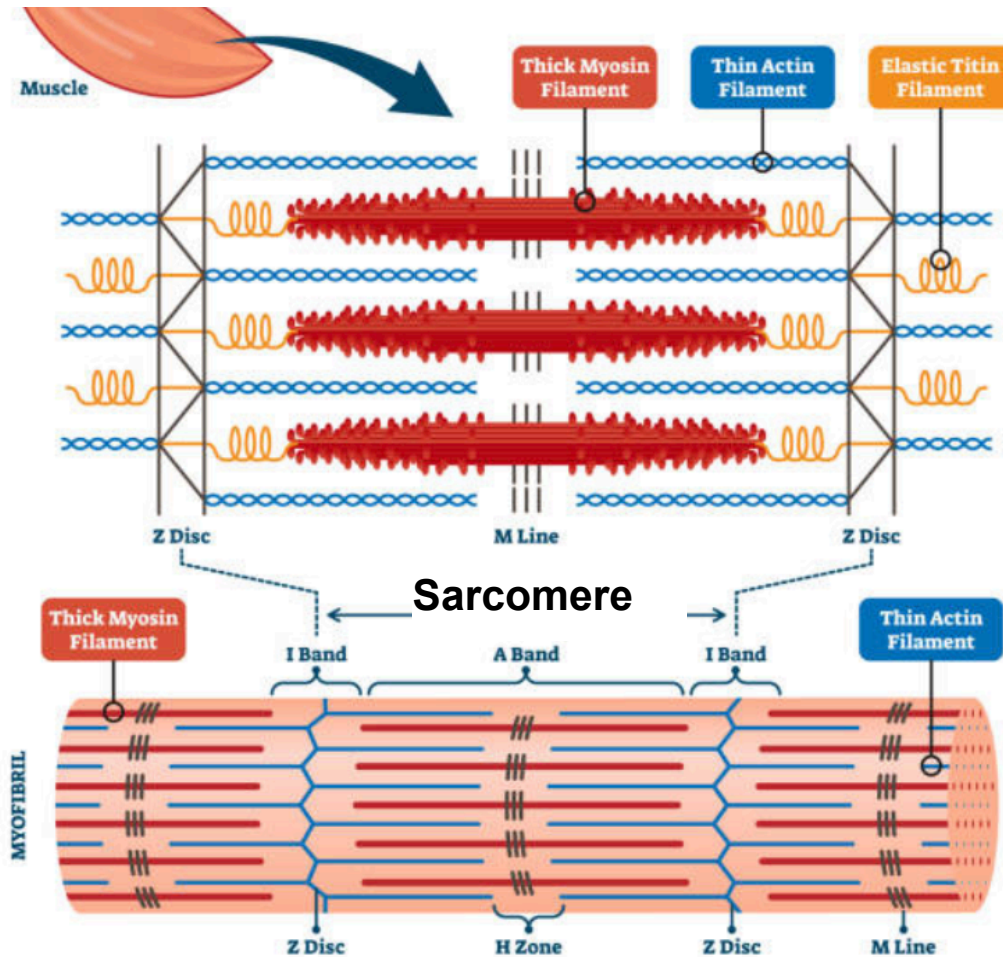


A : Action potential
C: Intracellular Calcium
F : Contraction strength



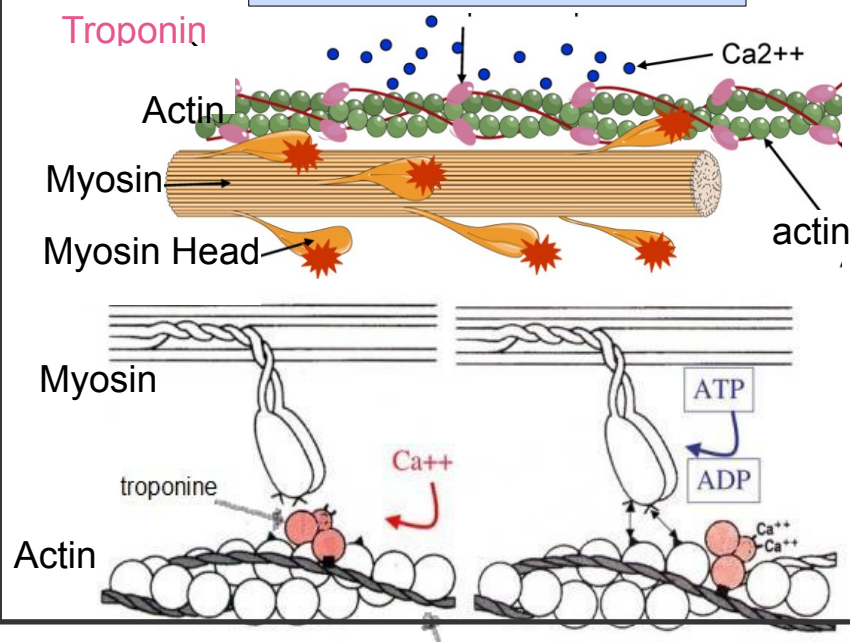
3 -Intracellular Calcium: Function

Excitation-Contraction Mechanism: Role of Acetylcholine & Ca^{2+}

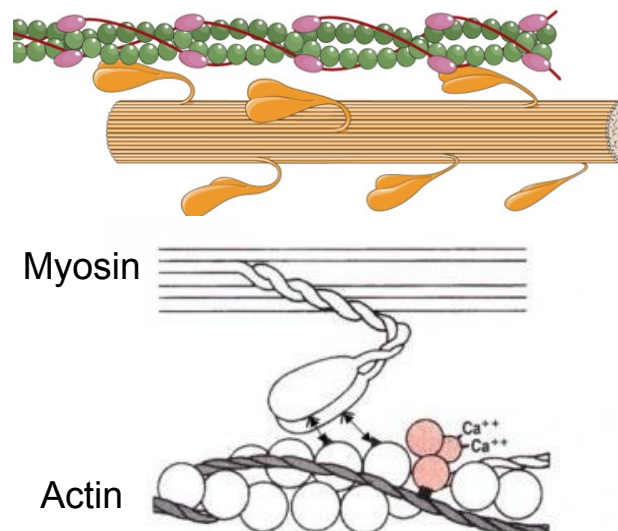


- 1- In basal states (low Ca^{2+}) tropomyosin blocks actin binding sites.
- 2- Acetylcholine increases cytosolic Ca^{2+} . Ca^{2+} binds to troponin C, leads to a conformational change that displaces the tropomyosin from the actin binding site.

Calcium Fixation



Contraction



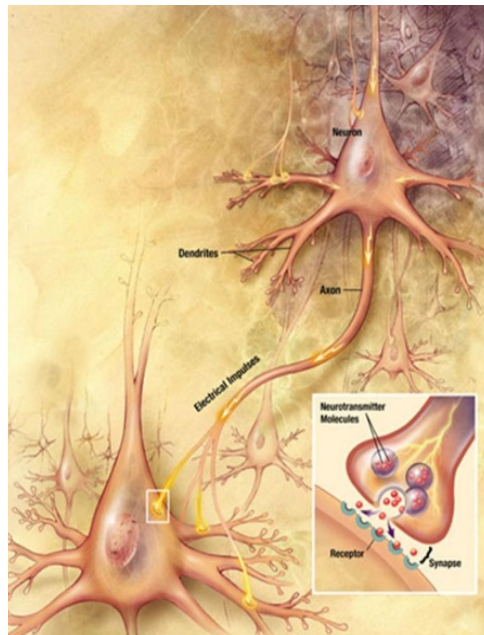
- 3-Actin and myosin binds to each other.

- 4- Myosin hydrolyses ATP, change conformation and moves the actin filament to the center of sarcomere => **Cell contraction**

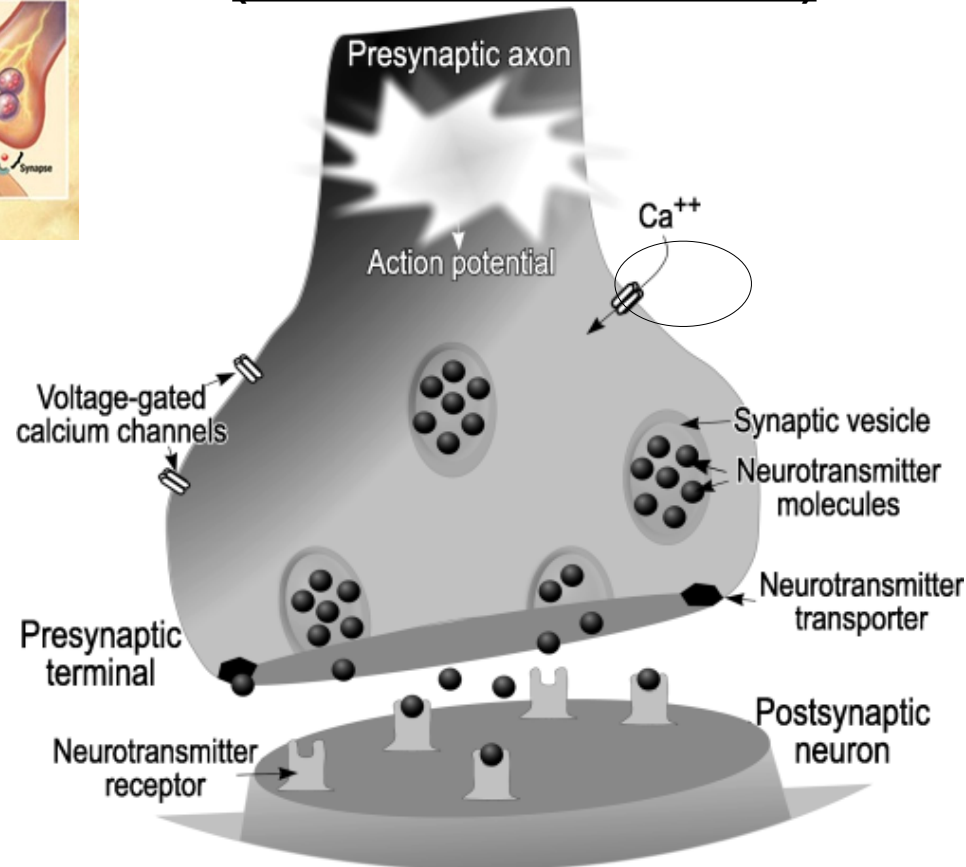
- 5- Decrease in intracellular Ca^{2+} disrupts actin myosin interaction - **Cell relaxation**

3 -Intracellular Calcium: Function

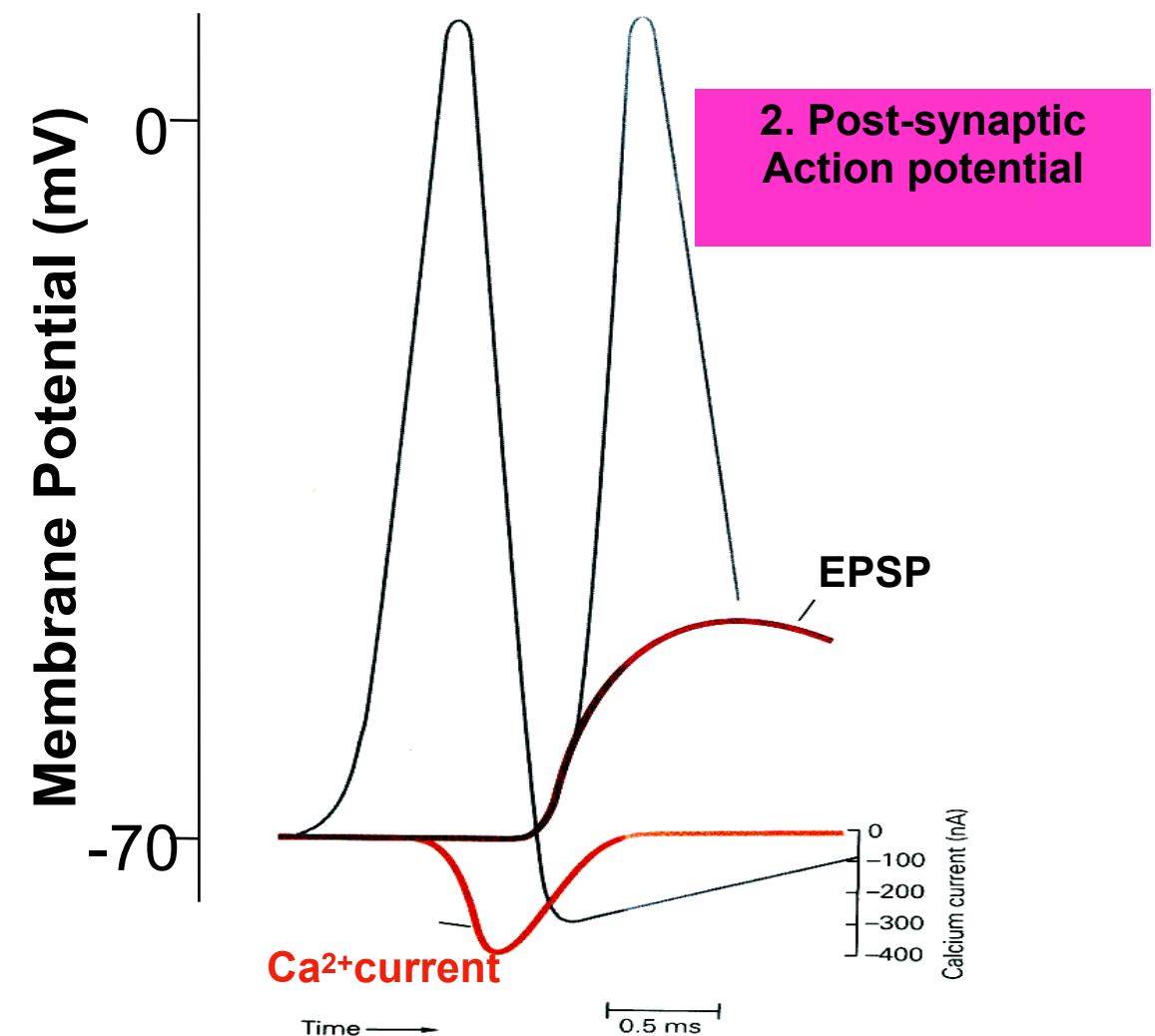
Exocytosis & Plasticity



Pre-synaptic level (Neurotransmitter release)



1. Pre-synaptic Action Potential



2. Post-synaptic Action potential

At presynaptic level => Ca^{2+} triggers vesicles exocytosis:

- Membrane depolarization and Action Potential
- Opening of Voltage-Gated Calcium channels (N or P/Q type)
- Increase in intracellular Ca^{2+} concentration
- Fusion of synaptic vesicle at the pre-synaptic membrane => **Exocytosis**
- Neurotransmitter Release (exemple: Glutamate...)
- EPSP at the postsynaptic membrane $\pm$ Action Potential => **Plasticity (LTP, LTD...)**

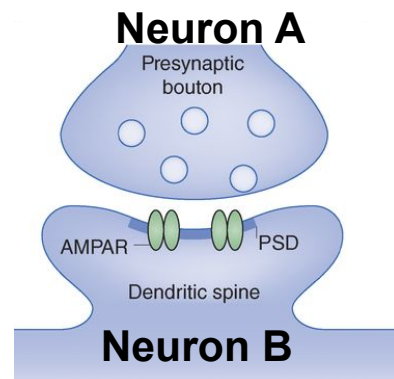
-EPSP= excitatory post-synaptic potential

3 -Intracellular Calcium: Function

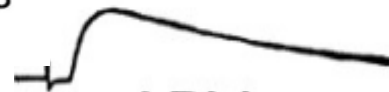
Exocytosis & Plasticity

Ca²⁺ effect at post-synaptic level

1

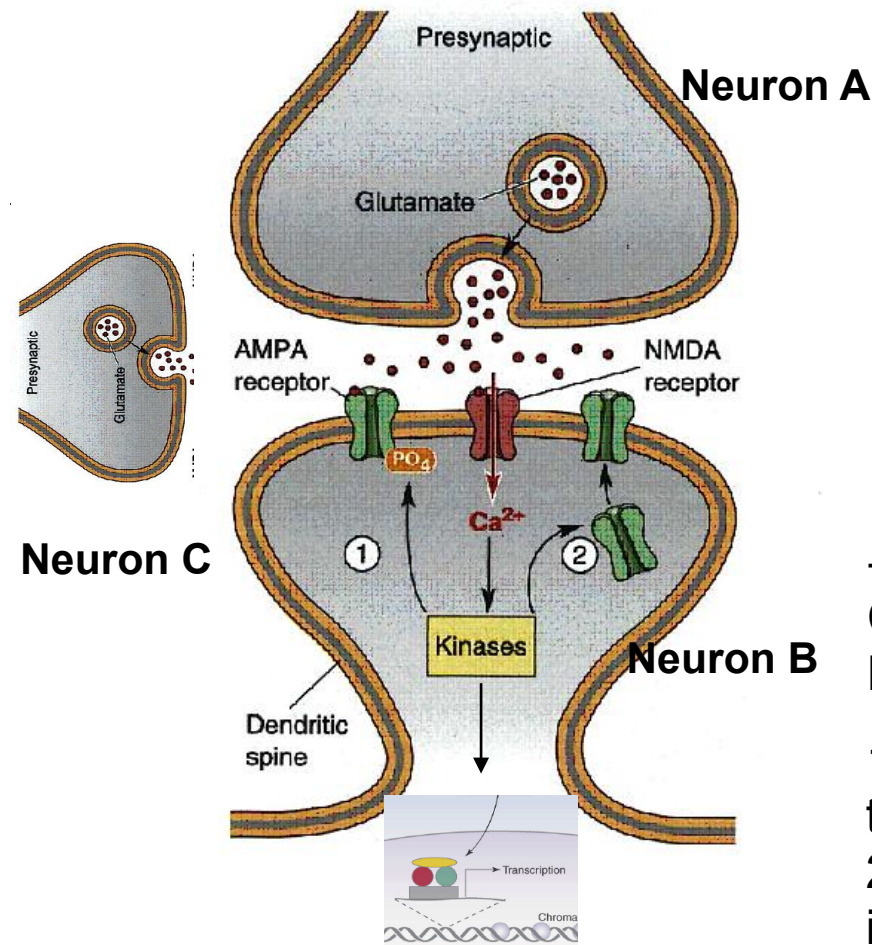


5 mV
20 ms



EPSP

Only postsynaptic AMPA Receptors responds to the release of Glutamate by an excitatory post-synaptic potential (EPSP)



2

Depolarization of the **postsynaptic membrane** occurs through additional stimulation of the B neuron by a C neuron, which acts simultaneously with the input from the A neuron and facilitates activation of NMDA receptors.

-NMDA receptors are activated => Ca²⁺ increases and activates Kinases that:

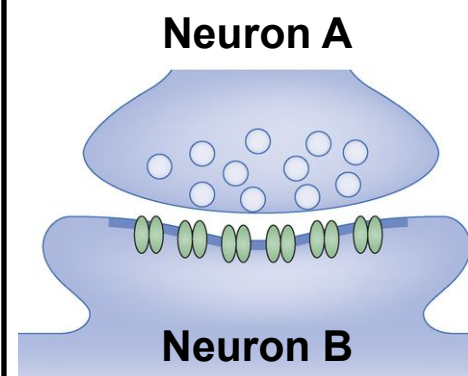
- 1- Phosphorylates AMPA receptors that translocates to the membrane.
- 2- Activates transcription factors to increase AMPA receptors synthesis.



Hebb's rule

3

The synapse is potentiated (LTP) and responds to the same release of Glutamate by a larger EPSP



larger EPSP

Revoir loi d'Ohm

$U = RI$

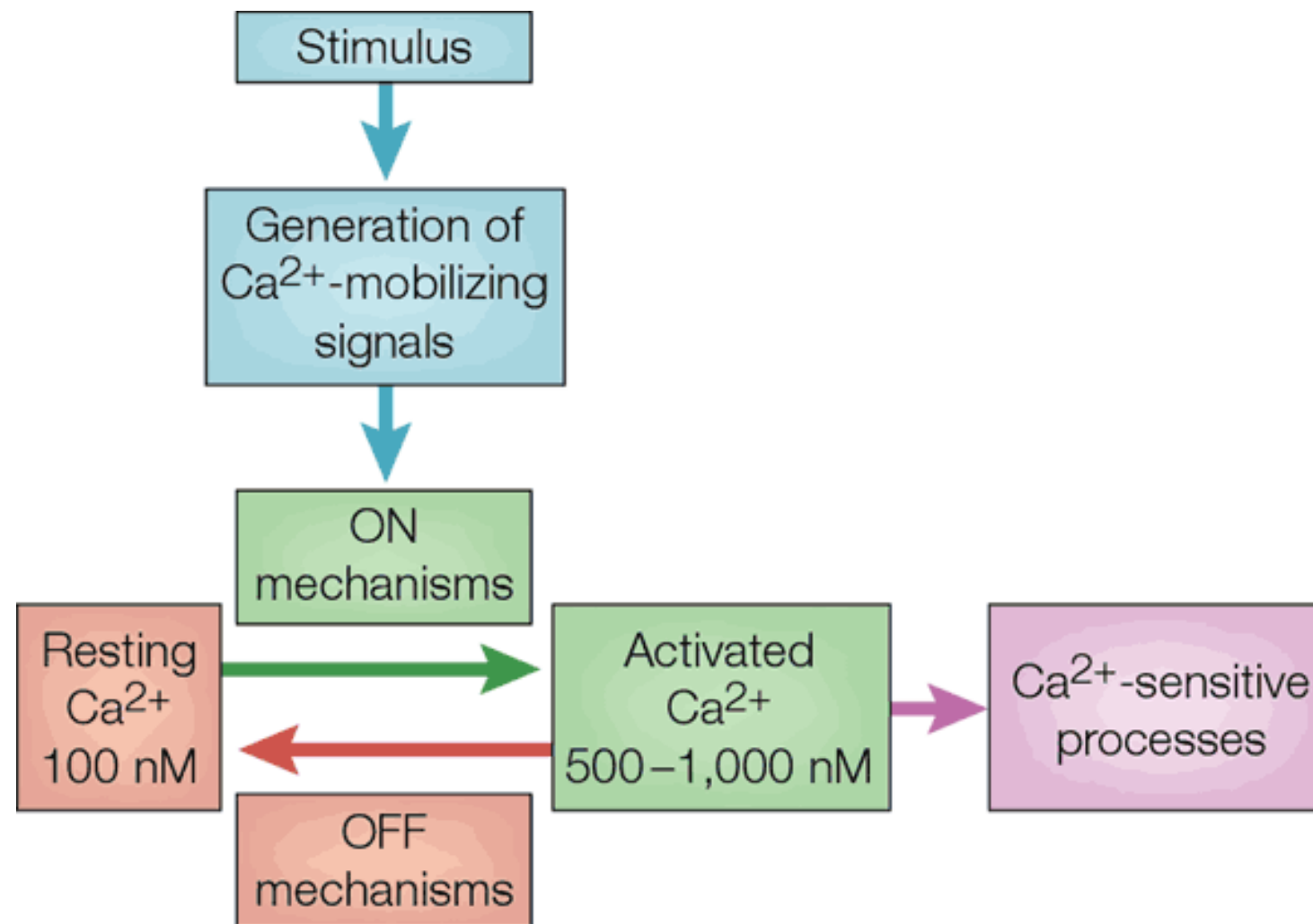
Une variation de courant entraîne une variation de potentiel

How is a single atomic element capable of accomplishing so many different functions at the cellular and integrated level?

4 -Intracellular Calcium: Regulatory mechanisms

REGULATORY MECHANISMS RESPONSIBLE FOR CALCIUM HOMEOSTASIS

ON and OFF MECHANISMS



Claude Bernard
(1813-1878)

Homeostasis = grec homoios (similar); stasis (stability).

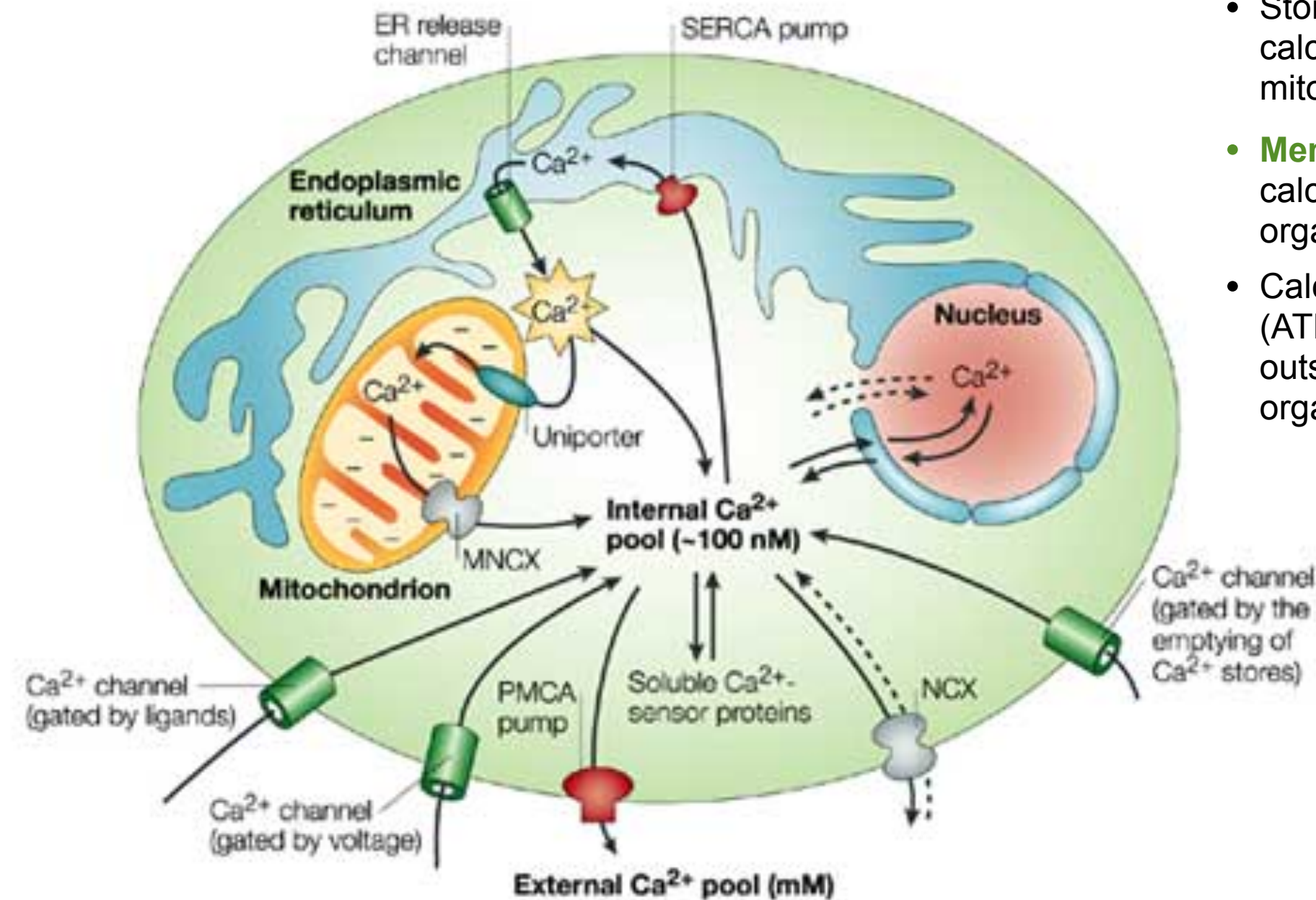
Described by Claude Bernard (1865).

Dynamic state of equilibrium: ex: Calcemia, glycemia, temperature, Sodium levels in the blood...

Each of these variables is controlled by one or more regulators (positive or negative regulatory loops), which allows the optimal function of the system.

4 -Intracellular Calcium: Regulatory mechanisms

REGULATORY MECHANISMS RESPONSIBLE FOR CALCIUM HOMEOSTASIS - TOOLKIT



- Storage compartments = intracellular calcium stores (ER cavity, nuclear envelope, mitochondria)
- **Membrane channels** = extracellular calcium entry or release from storage organelles
- Calcium transporters = **calcium pumps** (ATPases) Extrusion of cytosolic calcium outside the cell (PMCA) or storage in organelles (SERCA)
- **Na/Ca exchangers** = Extrusion of cytosolic calcium outside the cell (NCX) or used for rapid mobilization of calcium stores (MNCX)
- Protein sensors of the free cytosolic calcium

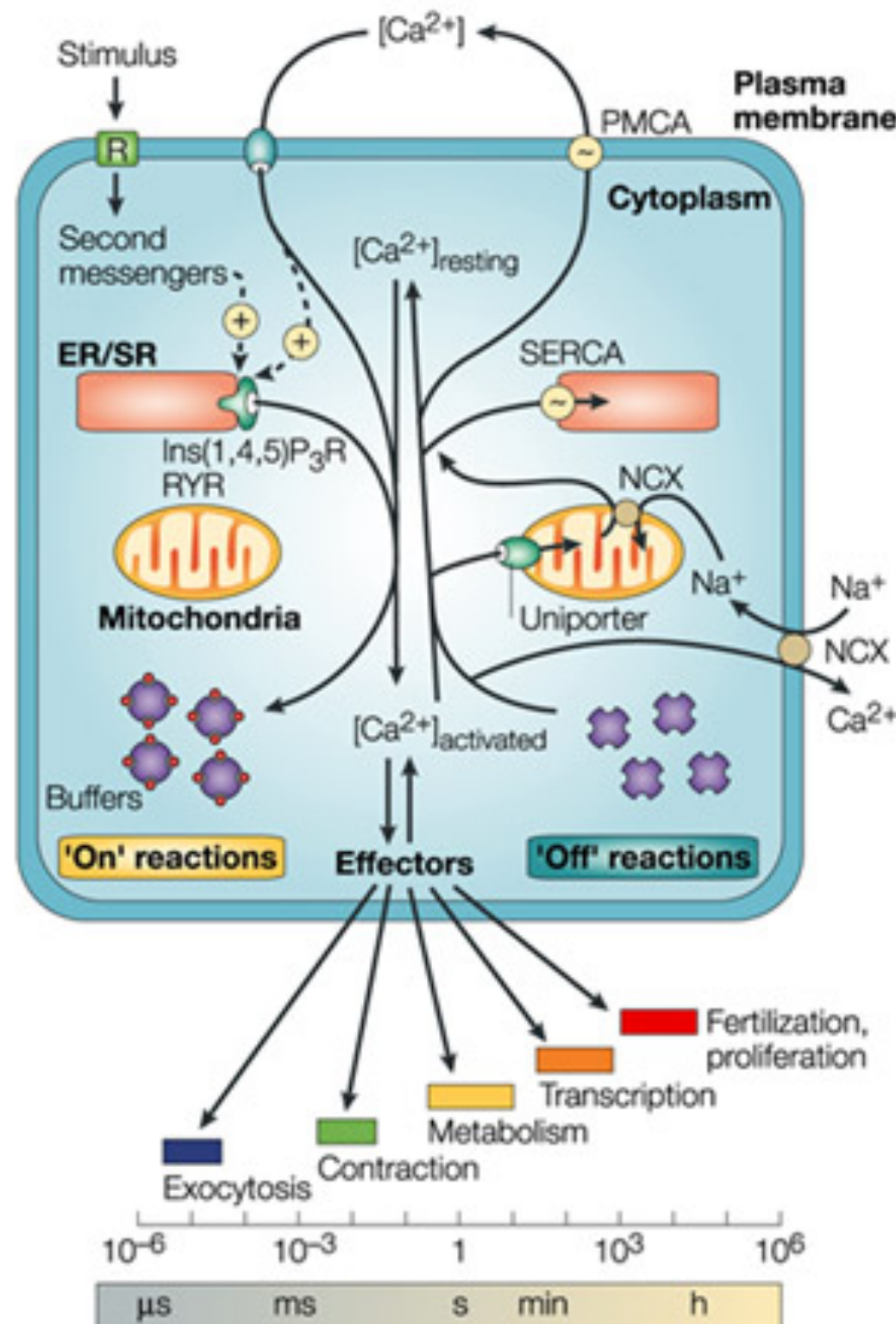
3 -Intracellular Calcium: Regulatory mechanisms

REGULATORS OF CALCIUM HOMEOSTASIS : 2 CATEGORIES

ON Reaction

Membrane channels:

- Entry of extracellular calcium or release from storage organelles.
- Ca^{2+} release from the endoplasmic/sarcoplasmic reticulum (ER/SR) can be triggered by 2nd messengers (IP3).
- Most of free Ca^{2+} binds to buffers/sensors
- A small fraction of free Ca^{2+} activates cellular processes

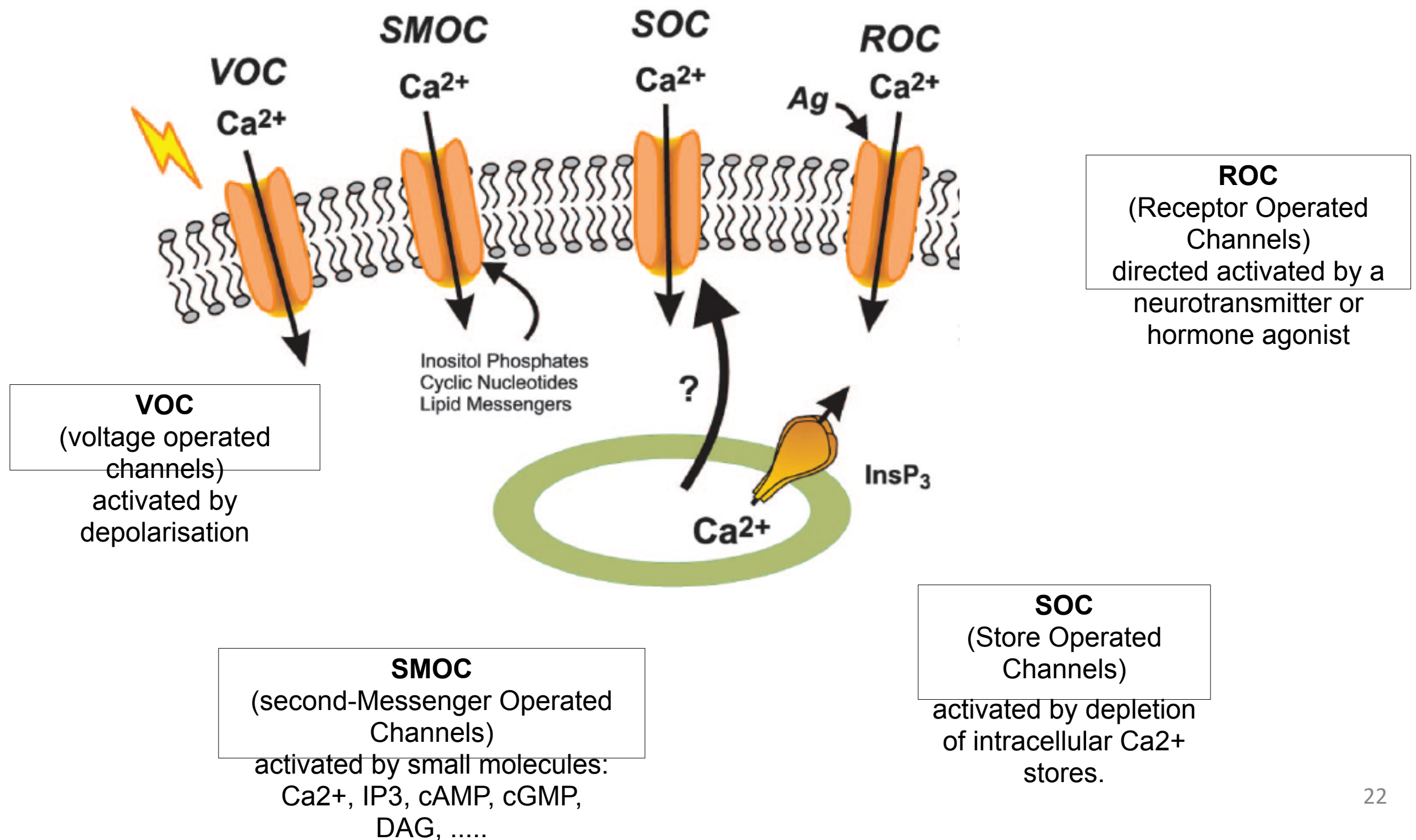


OFF Reaction

- Ca^{2+} is removed from the cytosol via calcium pumps (PMCA and SERCA) and Na/Ca exchangers (NCX)
- Mitochondria sequester Ca^{2+} rapidly through a uniporter.
- Ca^{2+} is then released more slowly back into the cytosol via a Na/Ca exchanger (NCX) for processing by SERCA and PMCA.

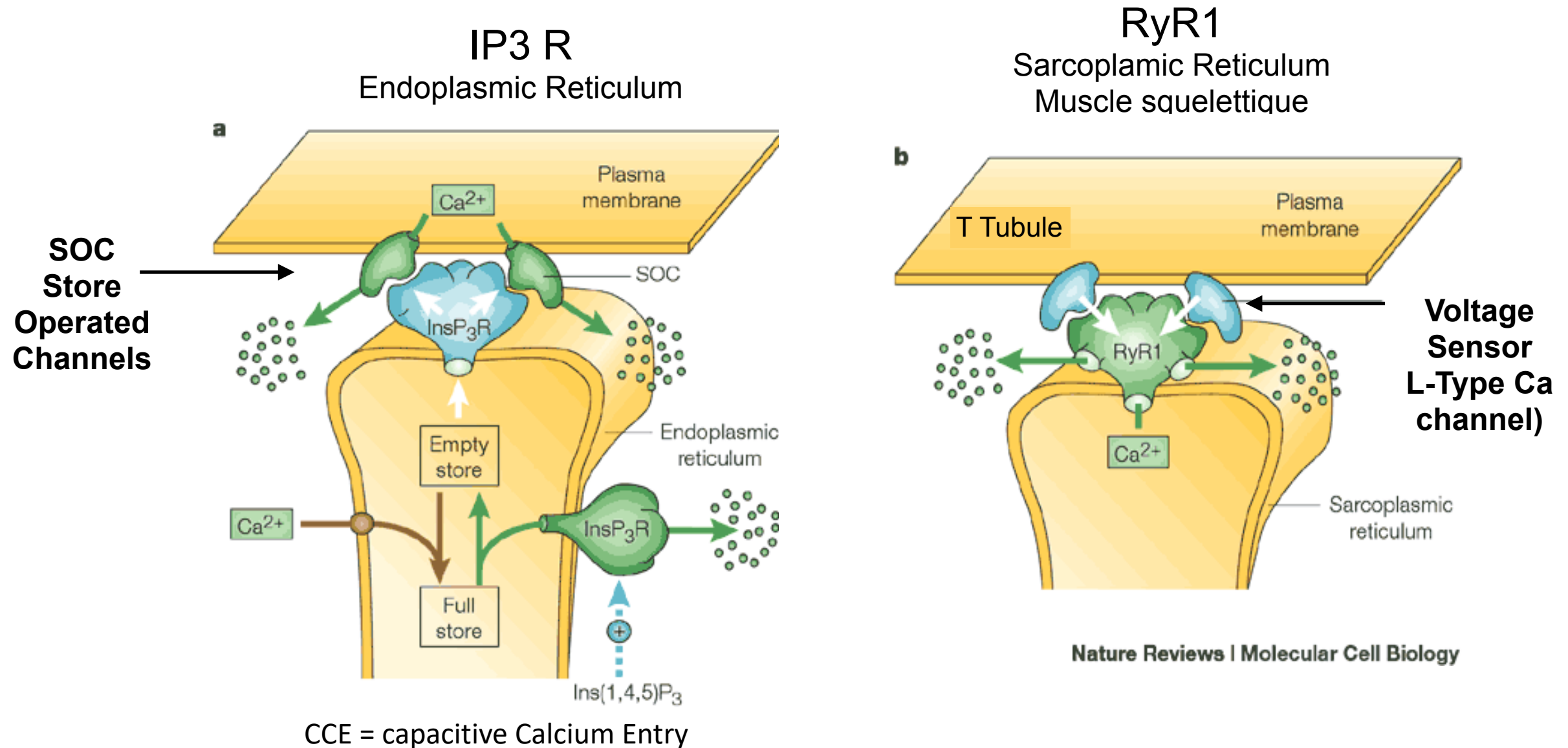
3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM CHANNELS AT THE PLASMA MEMBRANE



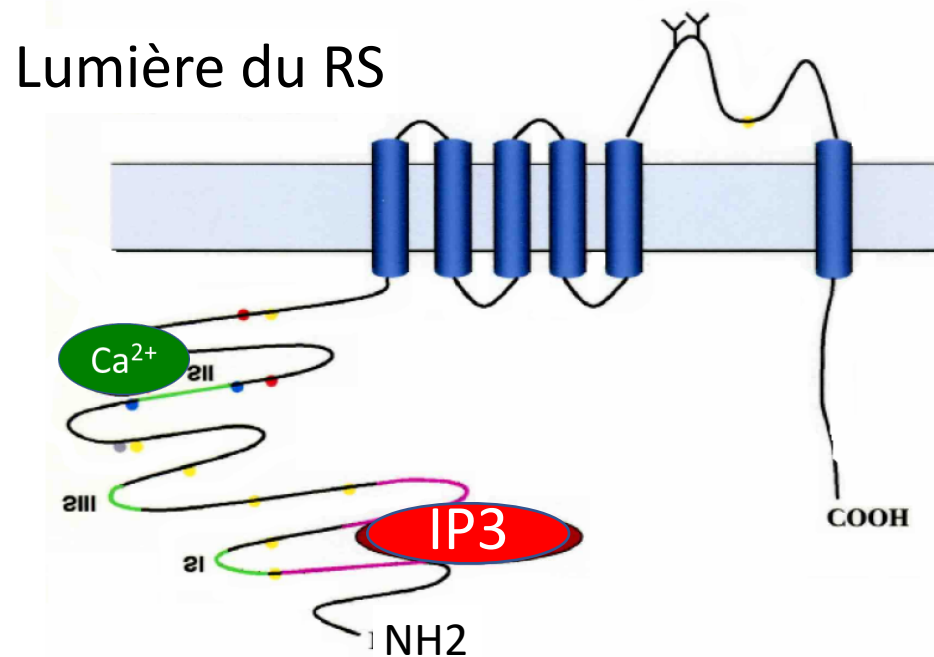
3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM SIGNALING by CONFORMATIONAL COUPLING USING MACROMOLECULAR COMPLEXES



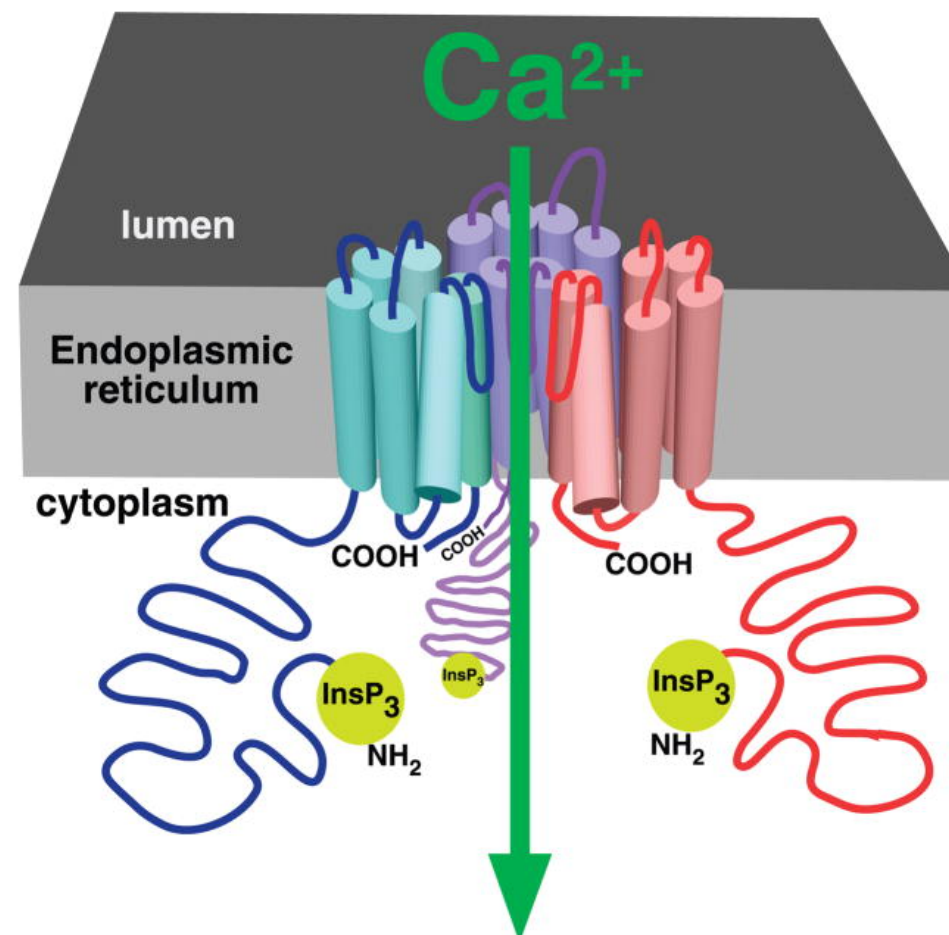
3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM CHANNELS IN THE SARCOPLASMIC RETICULUM Inositol Triphosphate Receptors (IP3R)



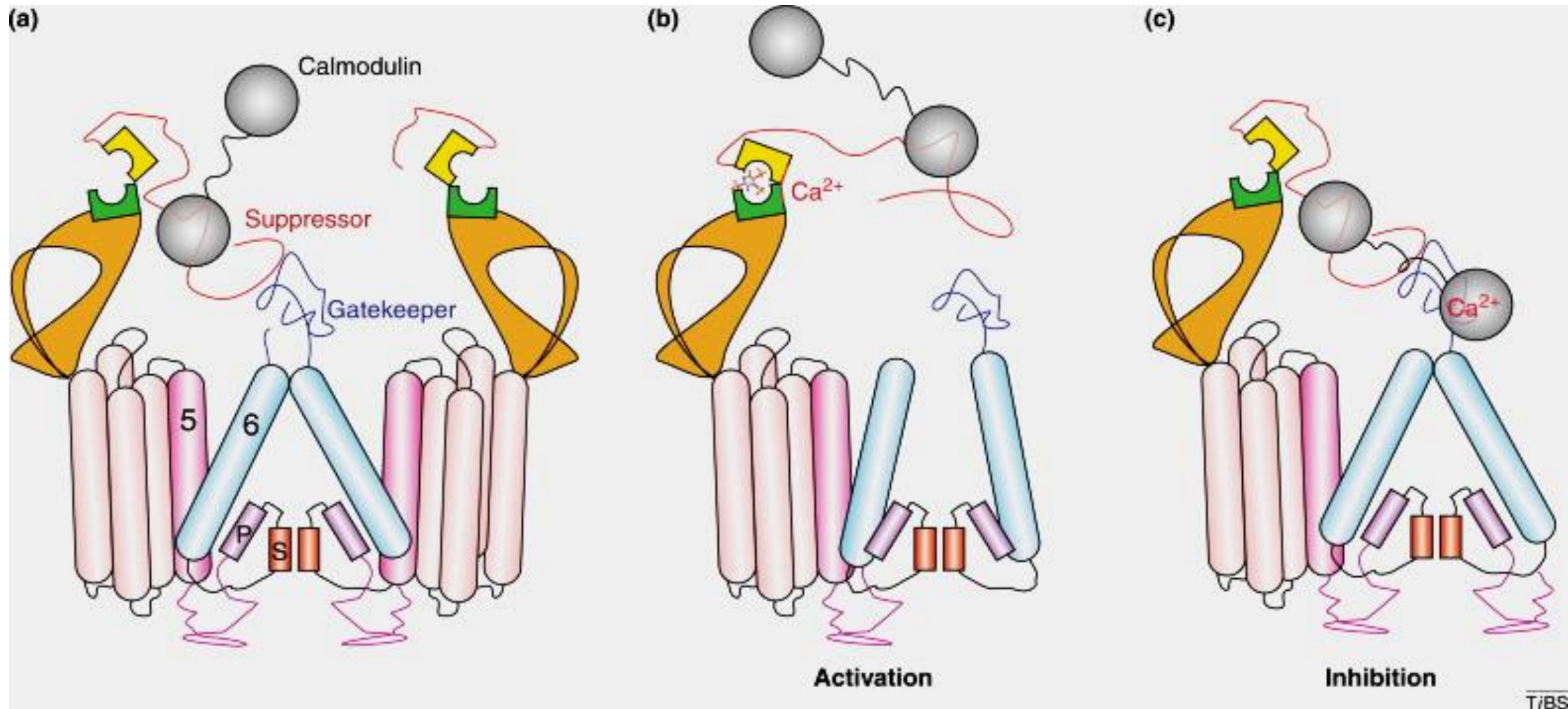
Inositol Triphosphate Receptors (IP3R):

- Ca²⁺ release from ER/SR
- homotetrameres
- 1 high affinity binding site for IP3 in each monomère
- activated by Low Ca²⁺ concentration
- Inhibited by > 300 nM Ca²⁺ concentration
- Conductance: 15-25 pS
- 3 isoforms IP3R1, IP3R2, IP3R3



3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM CHANNELS IN THE SARCOPLASMIC RETICULUM Inositol Triphosphate Receptors (IP3R) – 3 conformations



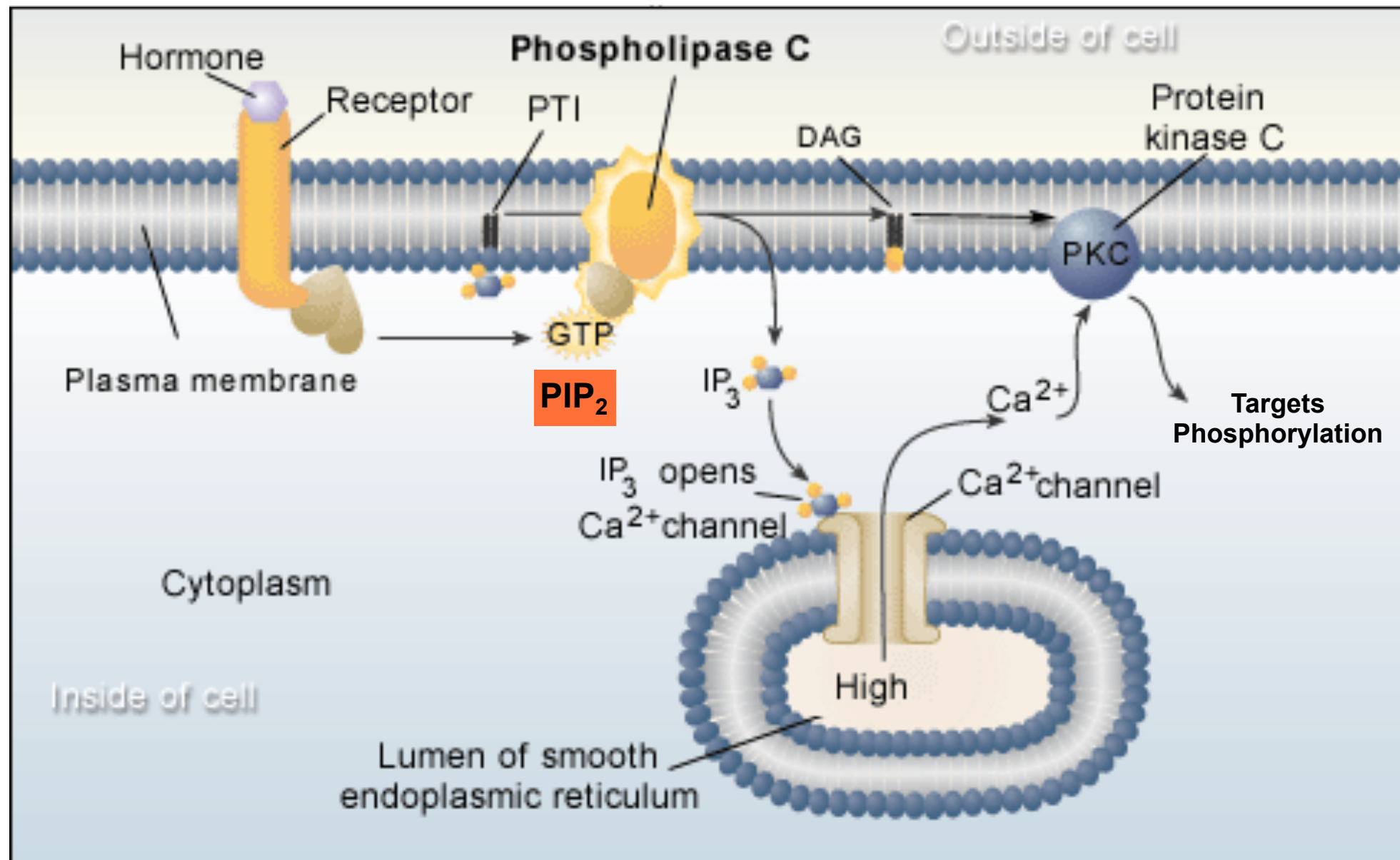
**Without IP3
and Without Ca^{2+} :
the channel is
closed**

**In the presence of
IP3 and Ca^{2+} the
channel opens**

**Without IP3,
 Ca^{2+} inhibits
the opening of
the channel**

3 -Intracellular Calcium: Regulatory mechanisms

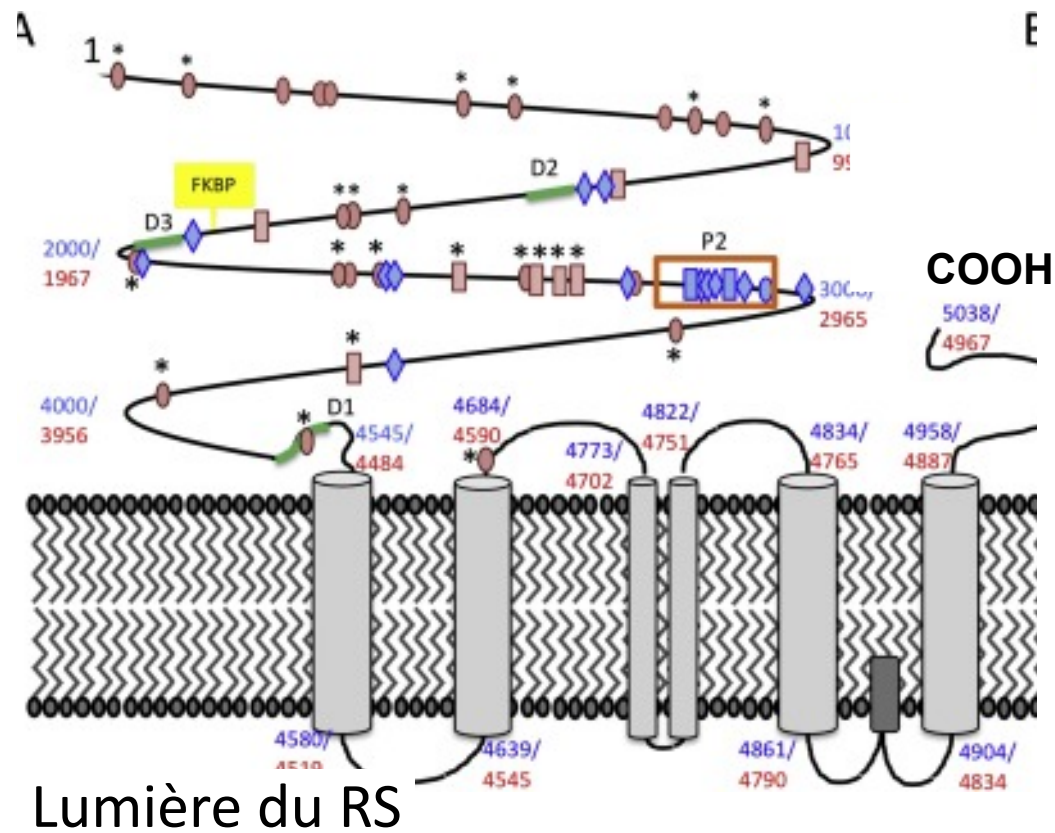
CALCIUM CHANNELS IN THE SARCOPLASMIC RETICULUM Inositol Triphosphate Receptors (IP3R)



- Phosphatidylinositol (PIP₂) is a phospholipid bound to the plasma membrane
- Phospholipase C degrades cell membrane PIP₂ releasing **IP₃** (inositol triphosphate) and diacylglycerol (DAG).
- Both are second messengers: IP₃ triggers the release of Ca²⁺ and DAG activates protein kinase C (PKC), a kinase modulated by Ca²⁺.

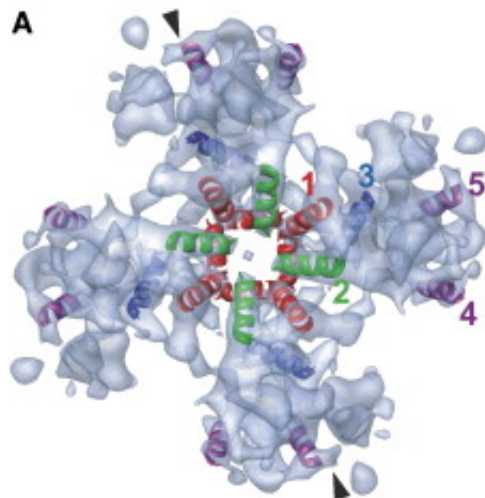
3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM CHANNELS IN THE SARCOPLASMIC RETICULUM Ryanodine Receptor (RyR)



Denniss, 2018

RyR1 (top view)

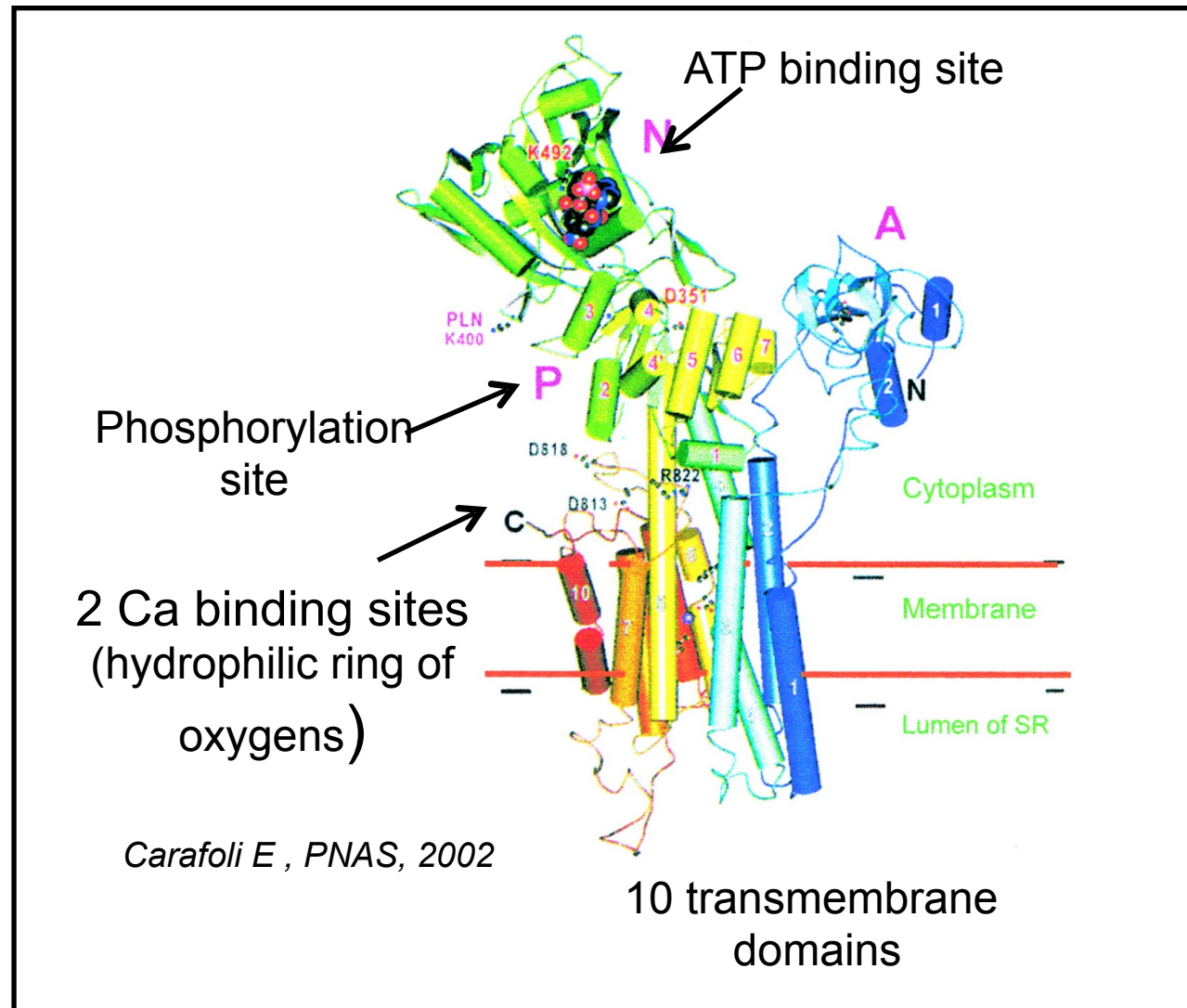


Van Petegem, 2012

- Function of RyR => Ca^{2+} release from ER/SR
- Homotetramer with mushroom-like shape activated by ryanodine ($<10 \mu\text{M}$), by high (μM) Ca^{2+} levels and ATP...
- Inhibited by ryanodine at high concentrations ($>100 \mu\text{M}$),
- 3 isoforms
 - RyR1 (muscle)
 - RyR2 (myocardium)
 - RyR3 (brain)
- each subunit is closely associated to regulatory proteins
- mutations (muscle disorders, arrhythmia)

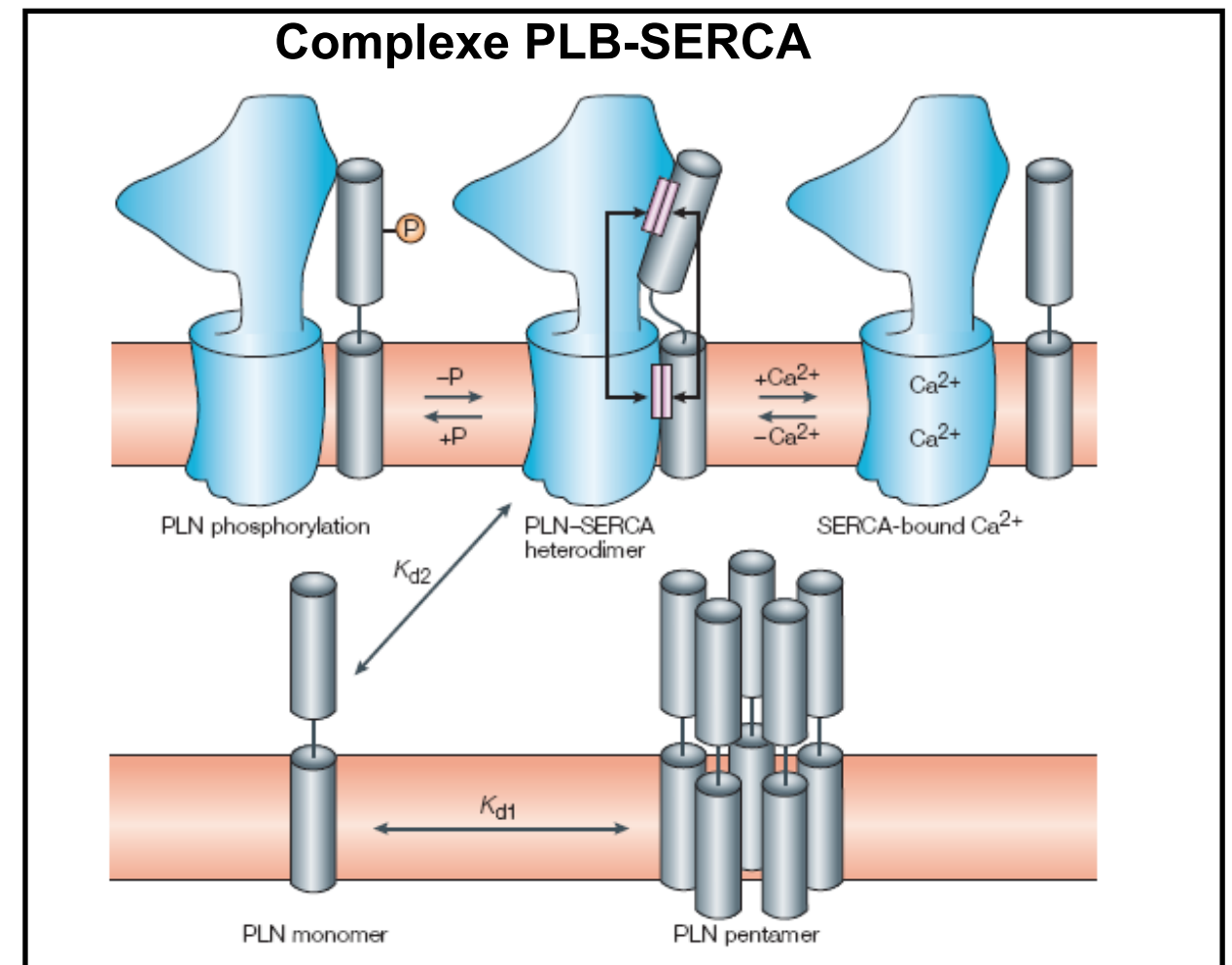
3 -Intracellular Calcium: Regulatory mechanisms

DECREASE IN CYTOSOLIC Ca^{2+} : SERCA PUMP



SERCA (Sarco/Endoplasmic Reticulum Ca^{2+} ATPase) PUMP:

- pumps Ca^{2+} from the cytosol into ER/SR lumen
- 3 isoforms
 - SERCA1 (muscle, heart)
 - SERCA2a (heart)
 - SERCA2b (ubiquitous)
 - SERCA3 (intestin, lymphatic tissues, platelets...)
- low affinity, high capacity => **High Ca required**
- role in muscle relaxation



Phospholamban (PLB or PLN) in cardiac muscle cells:

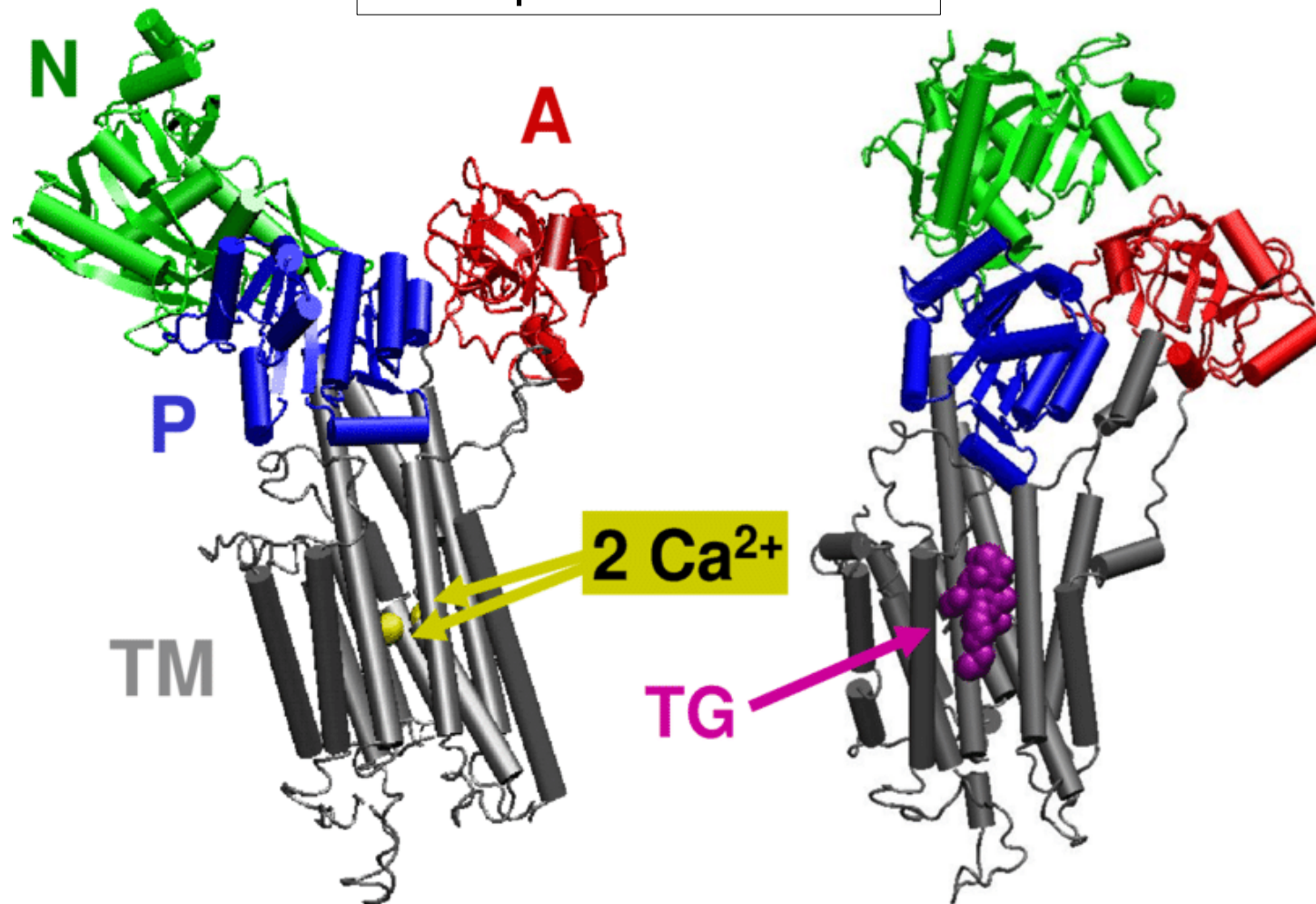
- Small transmembrane peptide of the SR;
- Regulator of SERCA activity: PLB inhibits SERCA in basal conditions (unphosphorylated)
- PLB loses its inhibitory effect on SERCA when phosphorylated by PKA
- Faster Ca^{2+} uptake into the SR, increase in the relaxation rate (lusitropic effect)
- Mutation > human cardiomyopathy

3 -Intracellular Calcium: Regulatory mechanisms

DECREASE IN CYTOSOLIC Ca^{2+} : SERCA PUMP

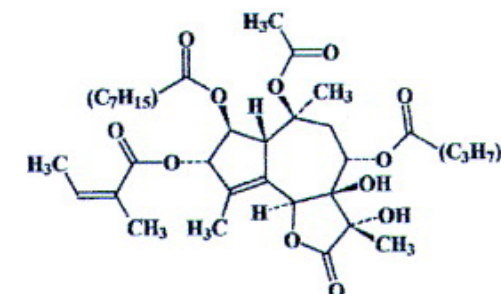
Crystal structure of SERCA pump

In the presence of Ca^{2+}



With Thapsigargin (TG)
SERCA non-competitive inhibitor

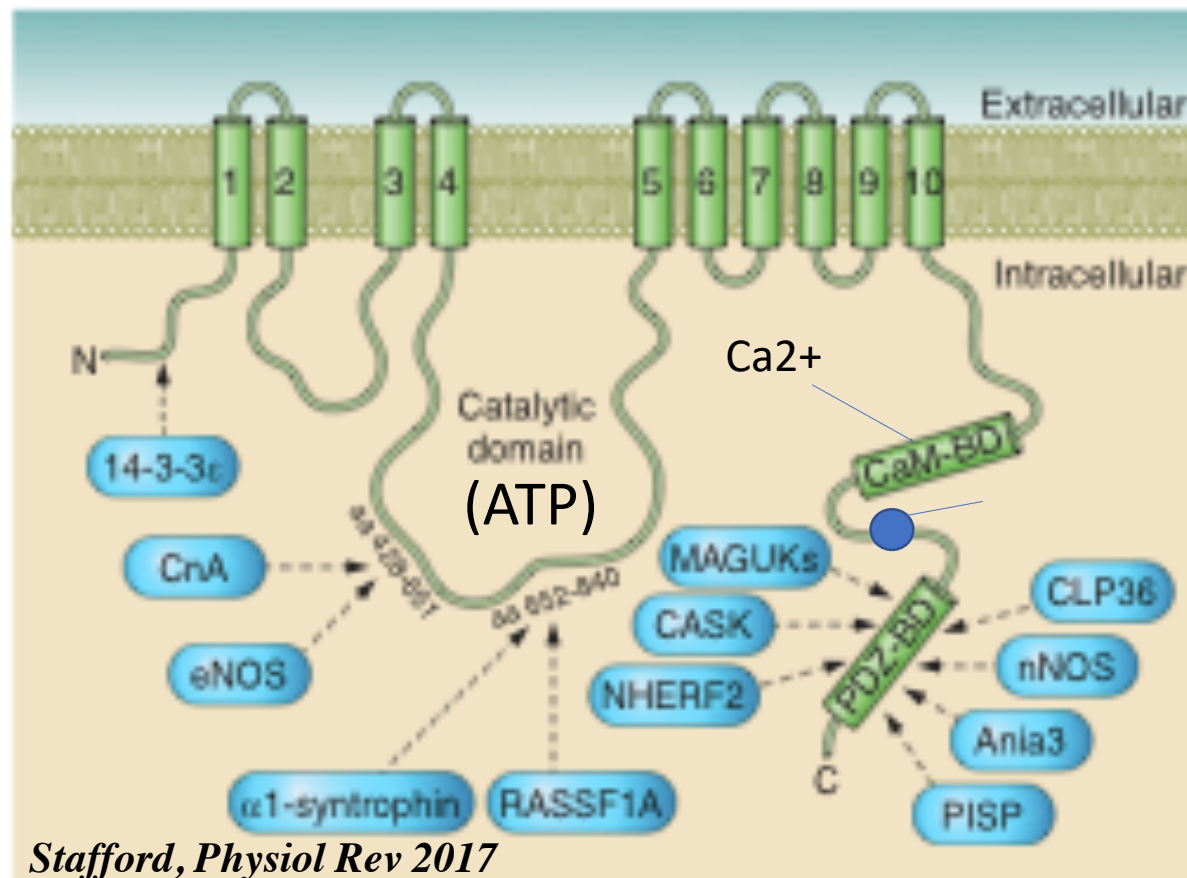
Thapsia garganica



> TG action : slowly deplets the Ca^{2+} stores // leak Ca channels

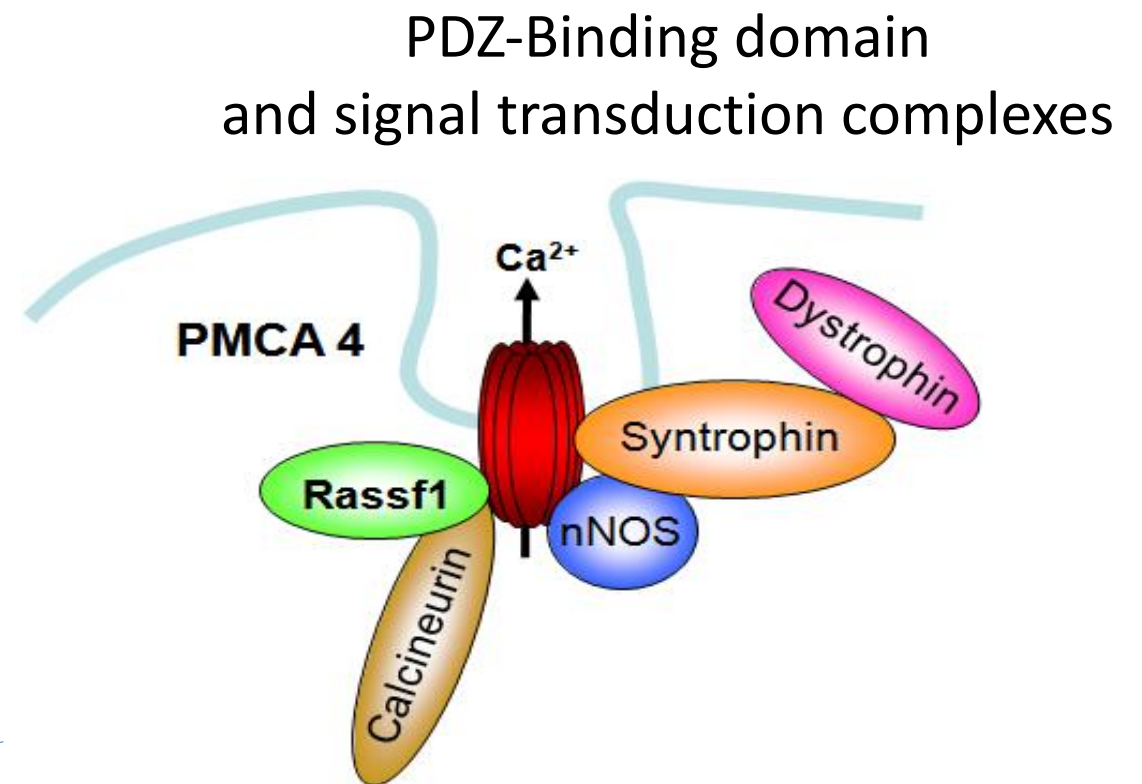
3 -Intracellular Calcium: Regulatory mechanisms

DECREASE IN CYTOSOLIC Ca^{2+} : PMCA PUMP



Stafford, *Physiol Rev* 2017

● Phosphorylation site (PKC, PKA)

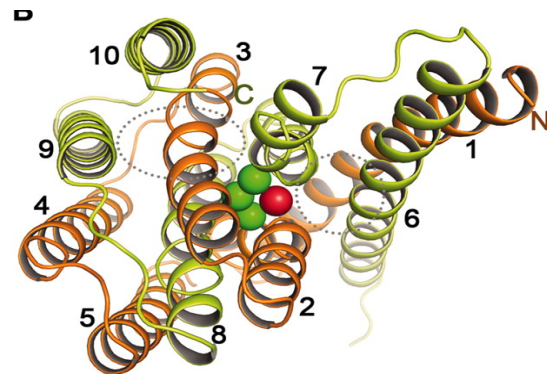
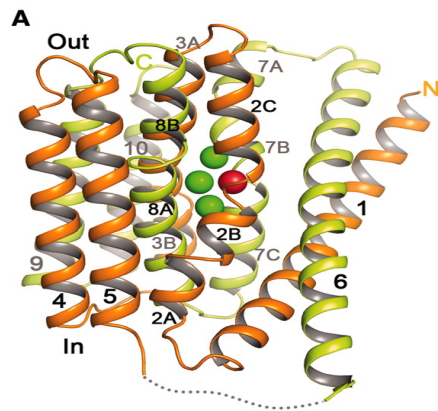


PMCA (plasma membrane calcium ATPase):

- PMCA extrudes calcium from the cytosol
- activated by the complex calcium calmoduline (Ca-CaM)
- 4 isoforms: PMCA 1-4
 - PMCA1 et 4 : cardiomyocytes
 - PMCA2 et 3 : Central Nervous System
- high Ca affinity => Activated by low Ca levels
- low transport rate (single system involved in non-excitable cells)
-

3 -Intracellular Calcium: Regulatory mechanisms

DECREASE IN CYTOSOLIC Ca^{2+} : $\text{Na}^+/\text{Ca}^{2+}$ exchanger (NCX)

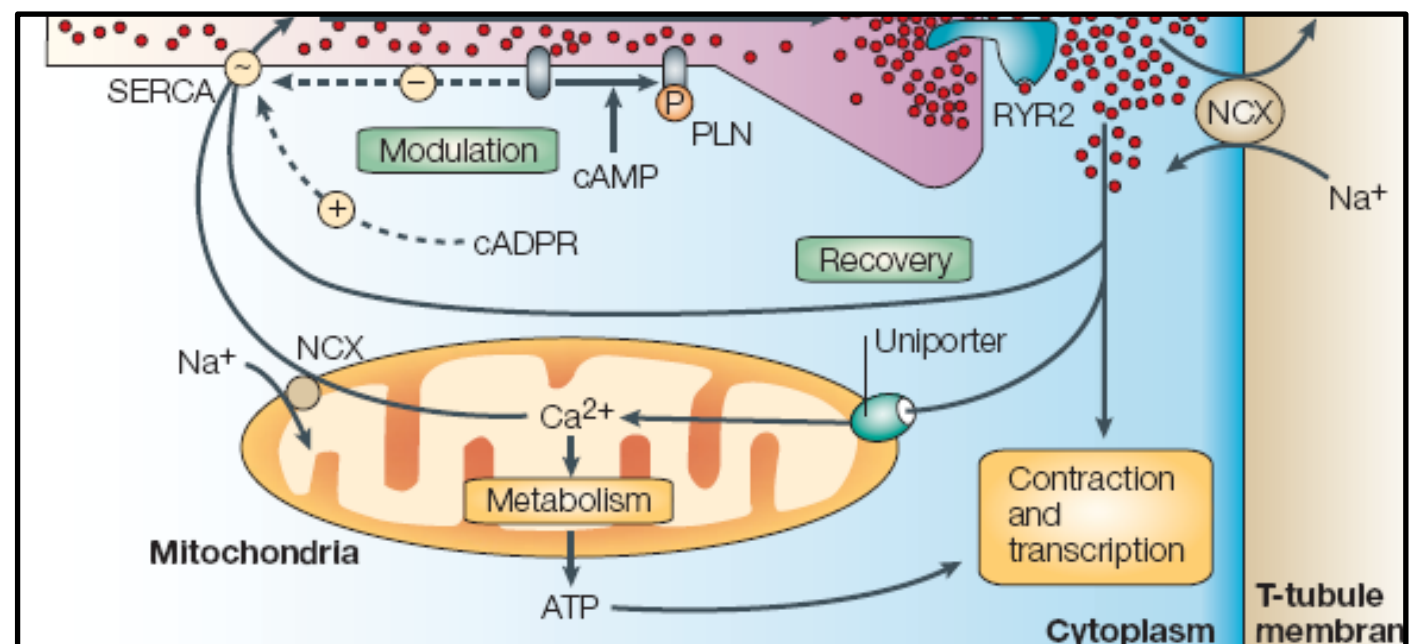
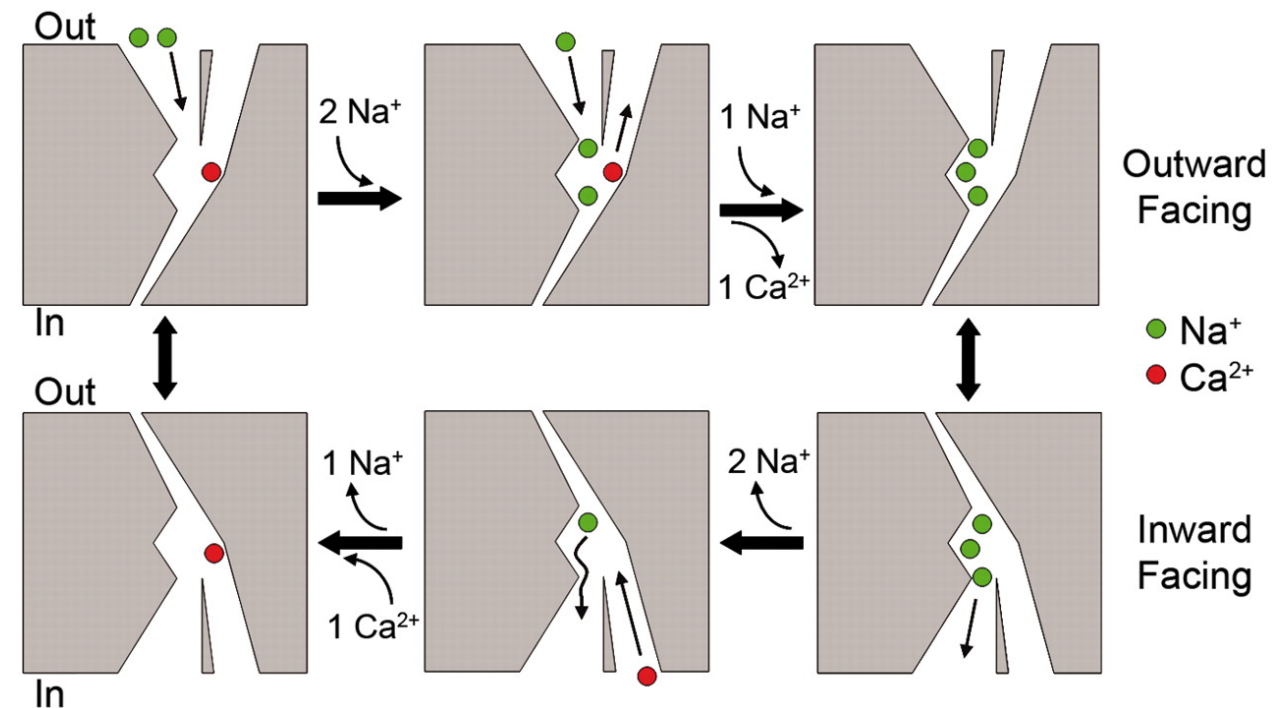


Crystal structure of NCX

- 10 transmembrane helices.
- 4 binding sites for ions (1 Ca^{2+} et 3 Na^+)
- Bidirectional during a few msec during AP
- NCX 1 : ubiquitous and cardiac cells
- NCX2 and NCX3 : brain and skeletal muscle
- Low Ca affinity => high capacity (5000 $\text{Ca}^{2+}/\text{sec}$)

Expression of NCX:

- Plasma membrane
- Mitochondria



3 -Intracellular Calcium: Regulatory mechanisms

Summary on the toolkit story

Cellular stimulation (ligand, voltage, intra 2nd mess, intra Ca stores)



Activation of Ca channels on cellular membrane



Direct increase in cytosolic calcium



Contribution of intracellular stores (homotetrameres IP3R and RyR)
Activated via CICR and/or macromolecular interactions



Local decrease in cytosolic calcium via PMCA, SERCA and NCX
(single 10 transmembrane domains molecules)



CaBP read the free Ca to activate cellular processes
(ex: activation of ion channels > membrane potential changes
Activation of gene transcription)

ON reactions

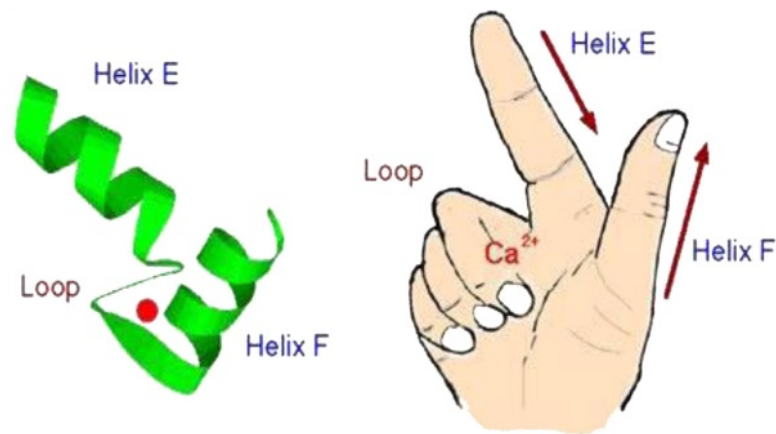
OFF reactions

IP3R and PMCA
Resting level
High Ca affinity
Activated for Low Ca
Low capacity

RyR and SERCA, NCX
Activated cell³²
Low Ca affinity
High Ca required
High capacity

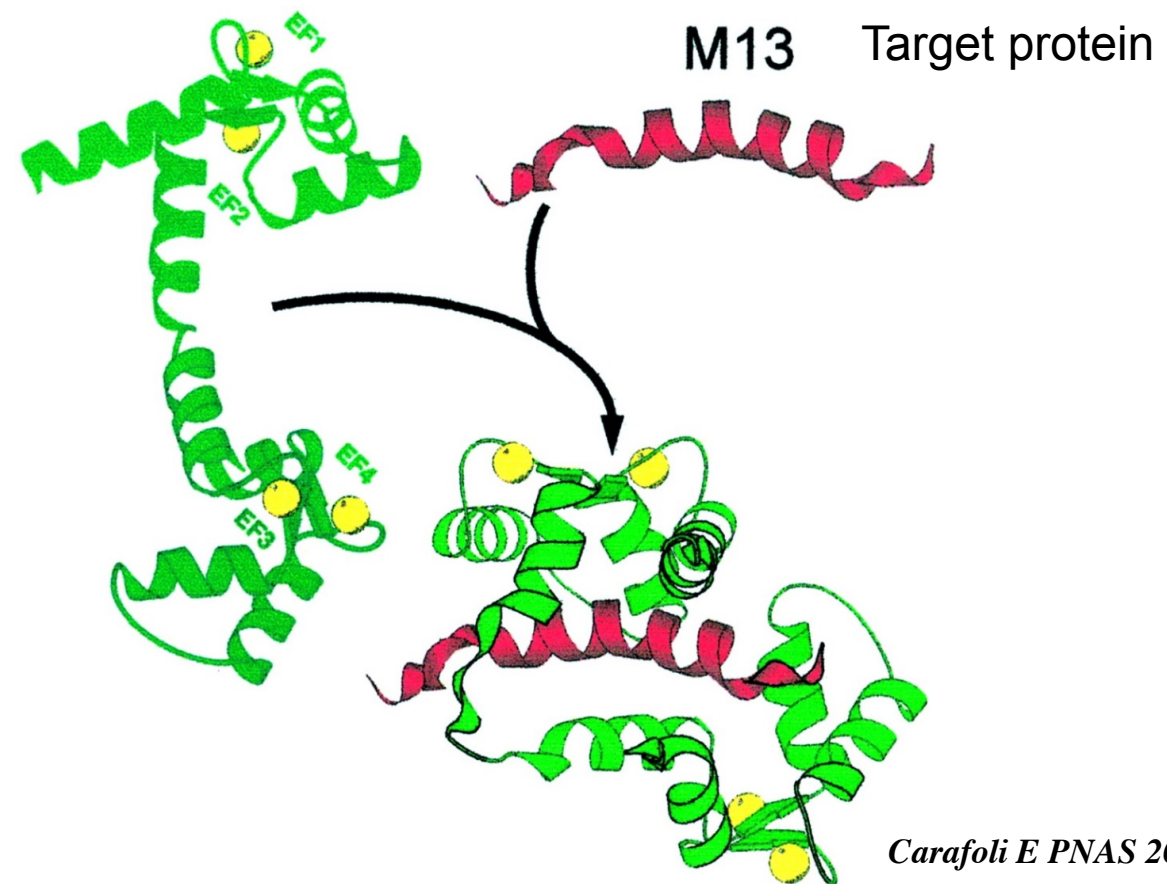
3 -Intracellular Calcium: Regulatory mechanisms

**INTRACELLULAR CALCIUM SENSORS ARE EF-HAND PROTEINS
= Calcium Binding Proteins (CaBP)**



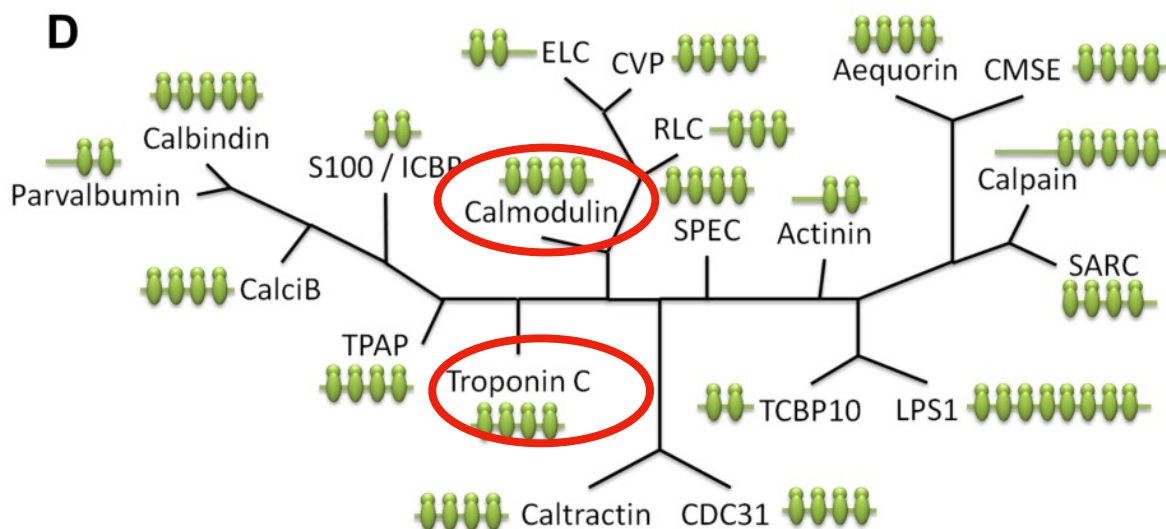
EF-hand domain :
Helix-loop-helix motifs
29 aminoacids
12 in Ca binding loop

Many different Ca²⁺ binding protein with variable number of EF hand domains (up to 8)



Carafoli E PNAS 2002;99:1115-1122

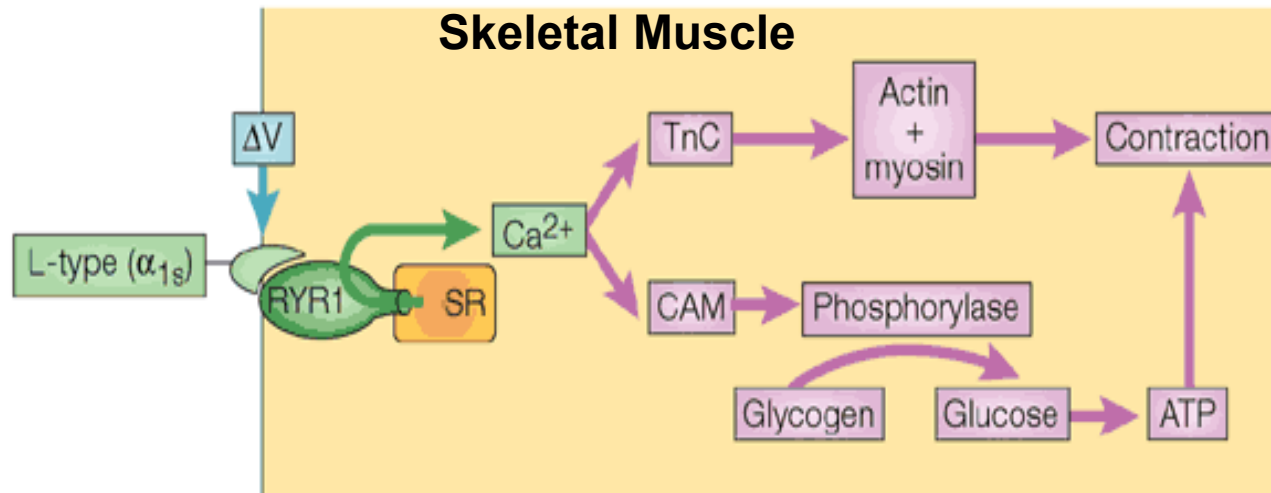
- By binding 4 Ca²⁺, CaM changes its conformation
- CaM binds target protein, for exemple of M13 (myosine light chain kinase), and regulates its activity.
- Many CaM-dependant enzyme activation is regulated by Ca²⁺



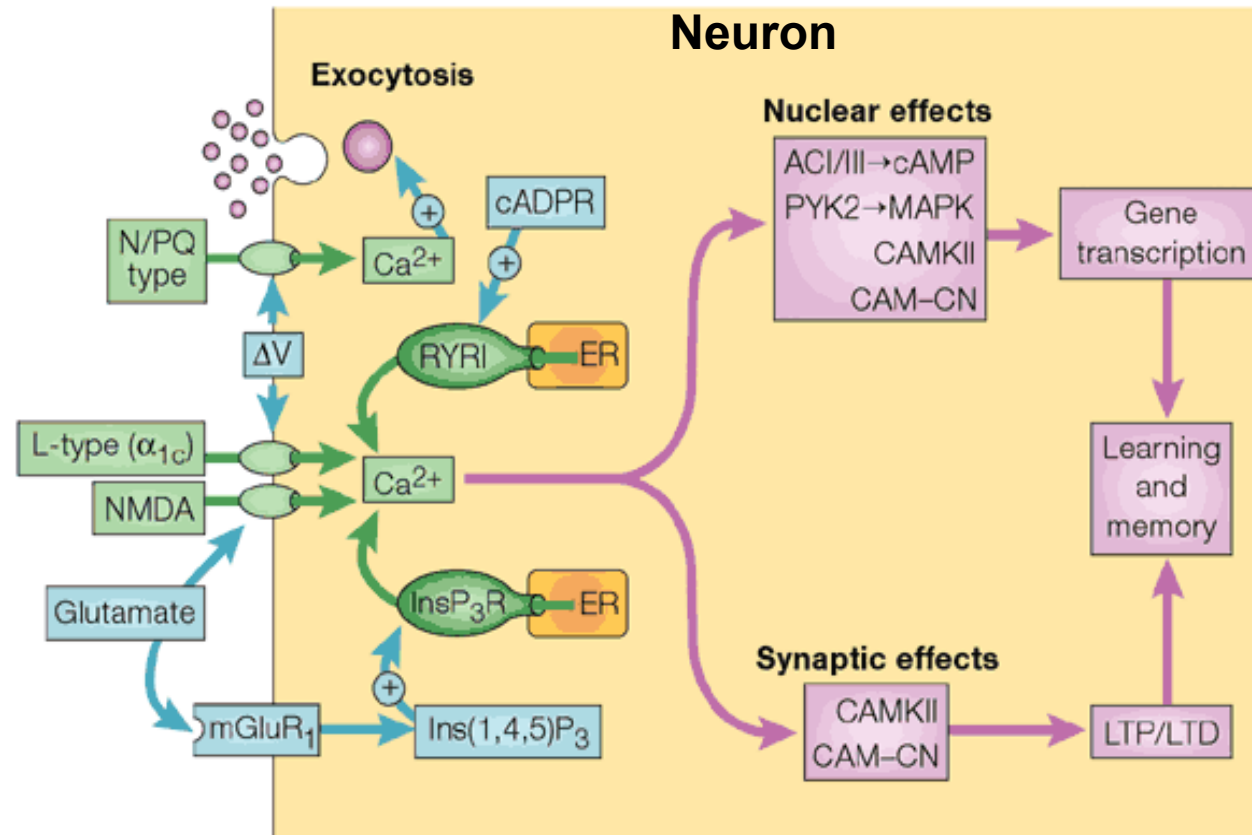
3 -Intracellular Calcium: Regulatory mechanisms

CALCIUM SIGNALING TOOLKIT TO REGULATE DIFFERENT CELLULAR PROCESSES

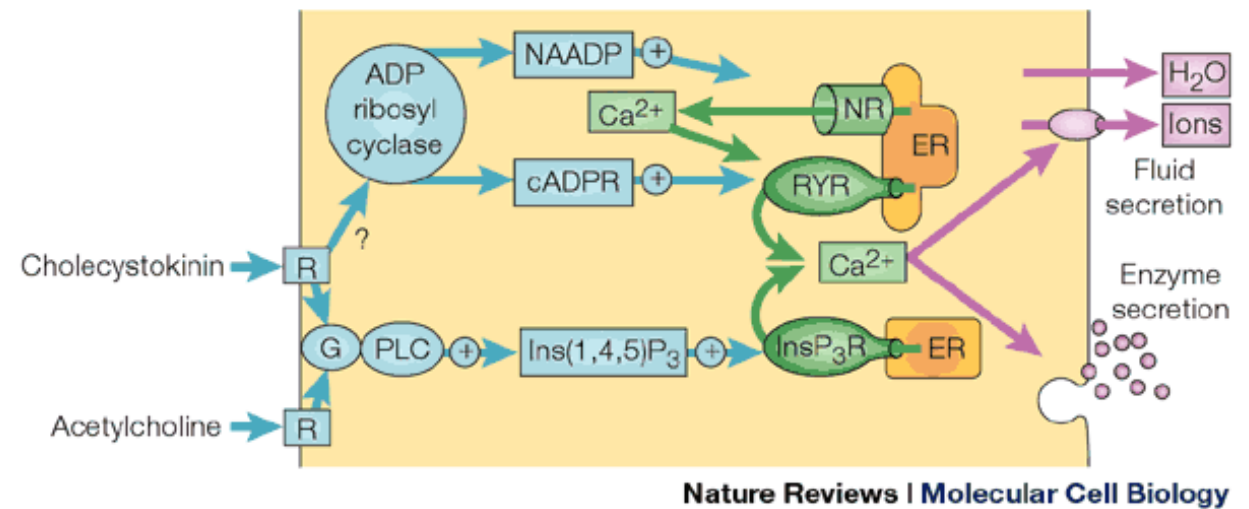
Skeletal Muscle



Neuron



Exocrine Pancreas cell



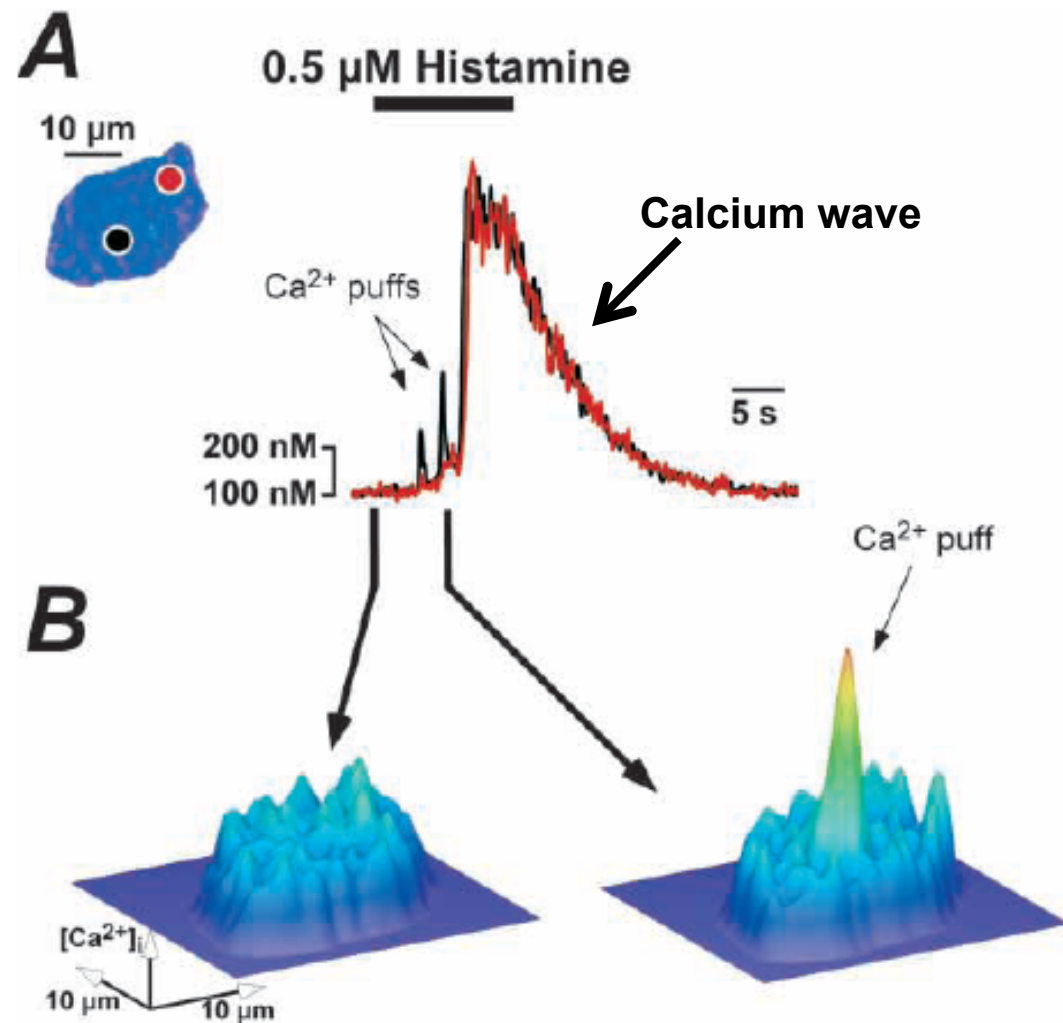
Berridge et al., 2000

4. Spatiotemporal modulation of Ca^{2+} signal in different cell types

5 - Spatio-temporal profile of Ca²⁺ signals

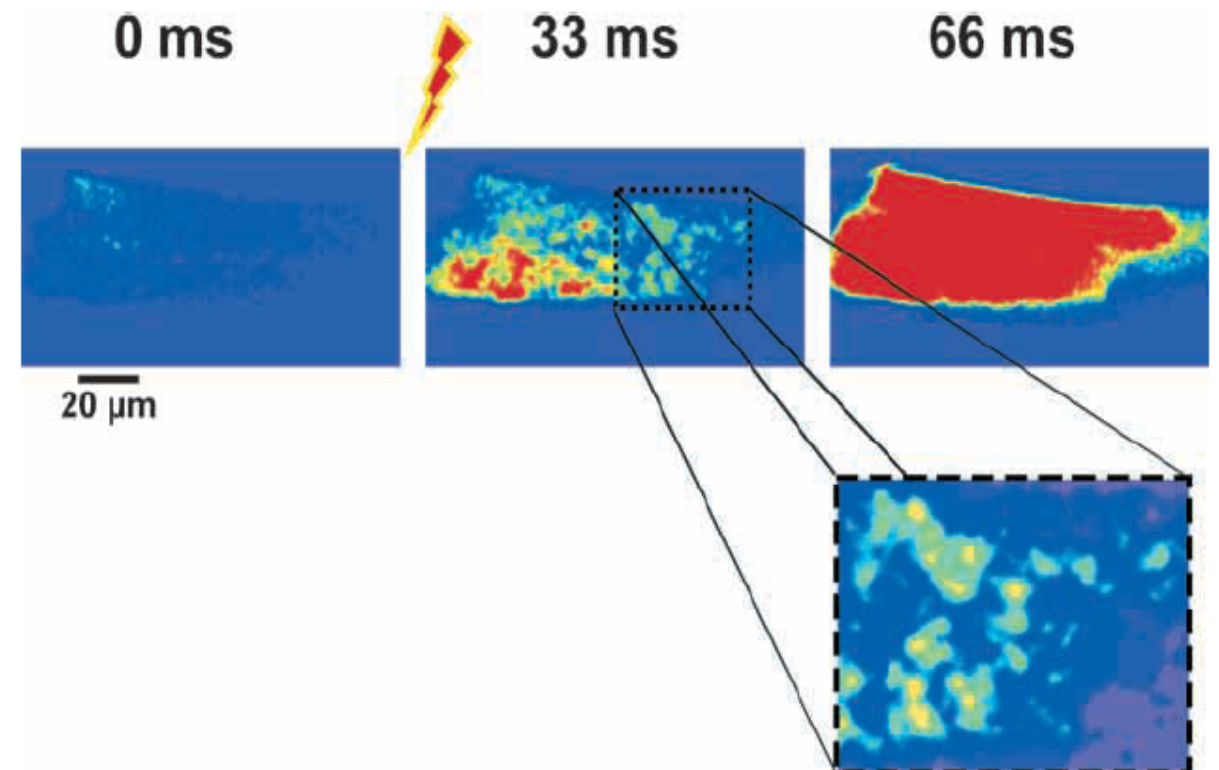
PUFFS vs SPARKS

Non excitable cell (HeLa cell)



- Fluo3-AM sensor (Exc 488 nm/ Emi 505 nm)
- Ca²⁺ increase in the microdomains = **Ca²⁺ puffs**
- IP₃R sites (50-600 nM, 1 sec, 6µm)
- Spatio-temporal coordination of Ca²⁺ puff results in a Calcium wave (example: oocytes; pancreatic acinar cells...)

Excitable cell (cardiac cell)



- Ca²⁺ increase in the microdomains = **Ca²⁺ sparks**
- RyR sites (faster, 1-3µm diameter)

**Details of a similar
experience (excitable cell)
in the next slide**

5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling during Excitation-Contraction Coupling in excitable cell (cardiac cells)

Calcium imaging with biosensors

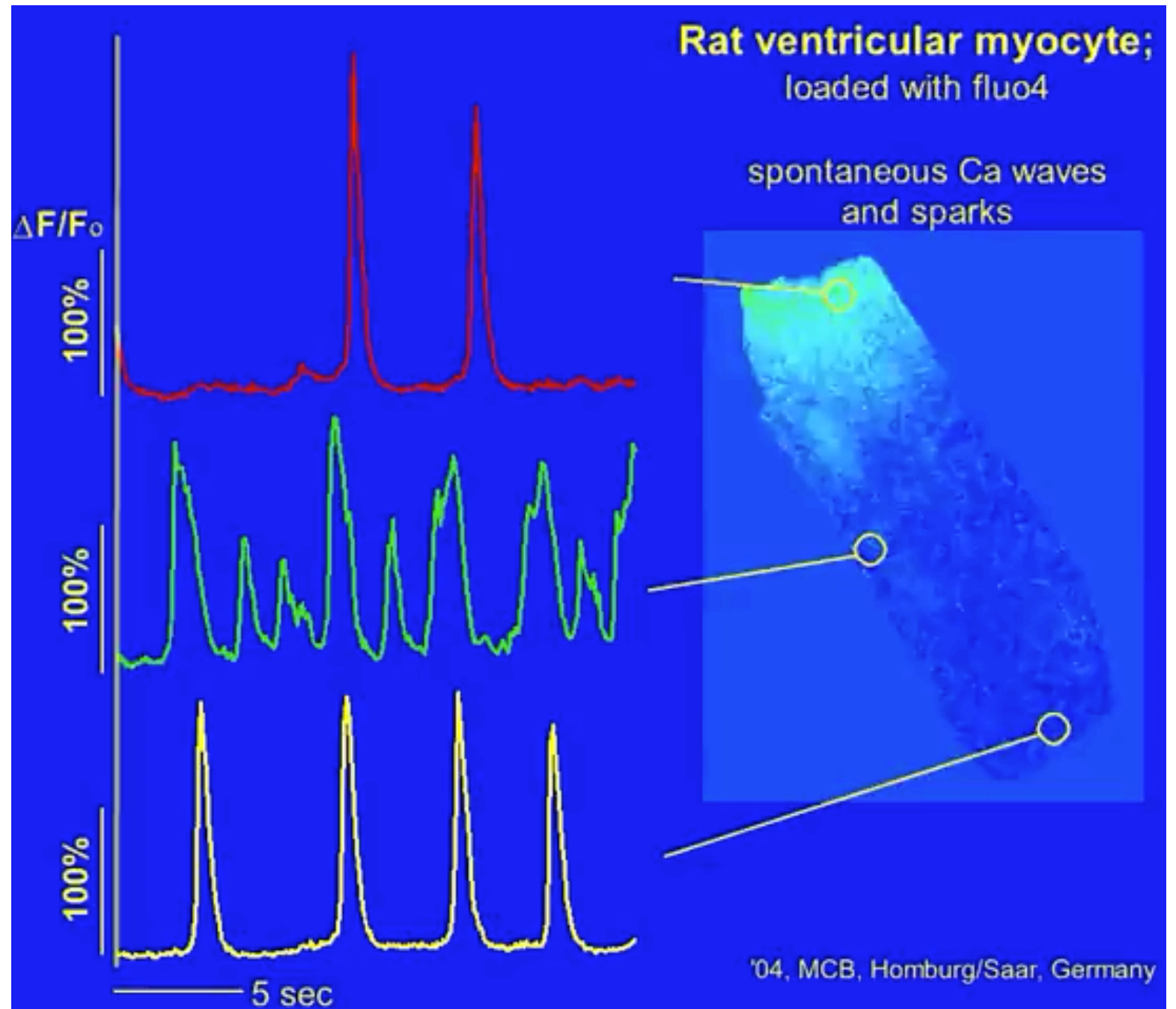
Variation of fluorescence intensity overtime in diferent ROI

Still image from a videorecording

Imaging setup



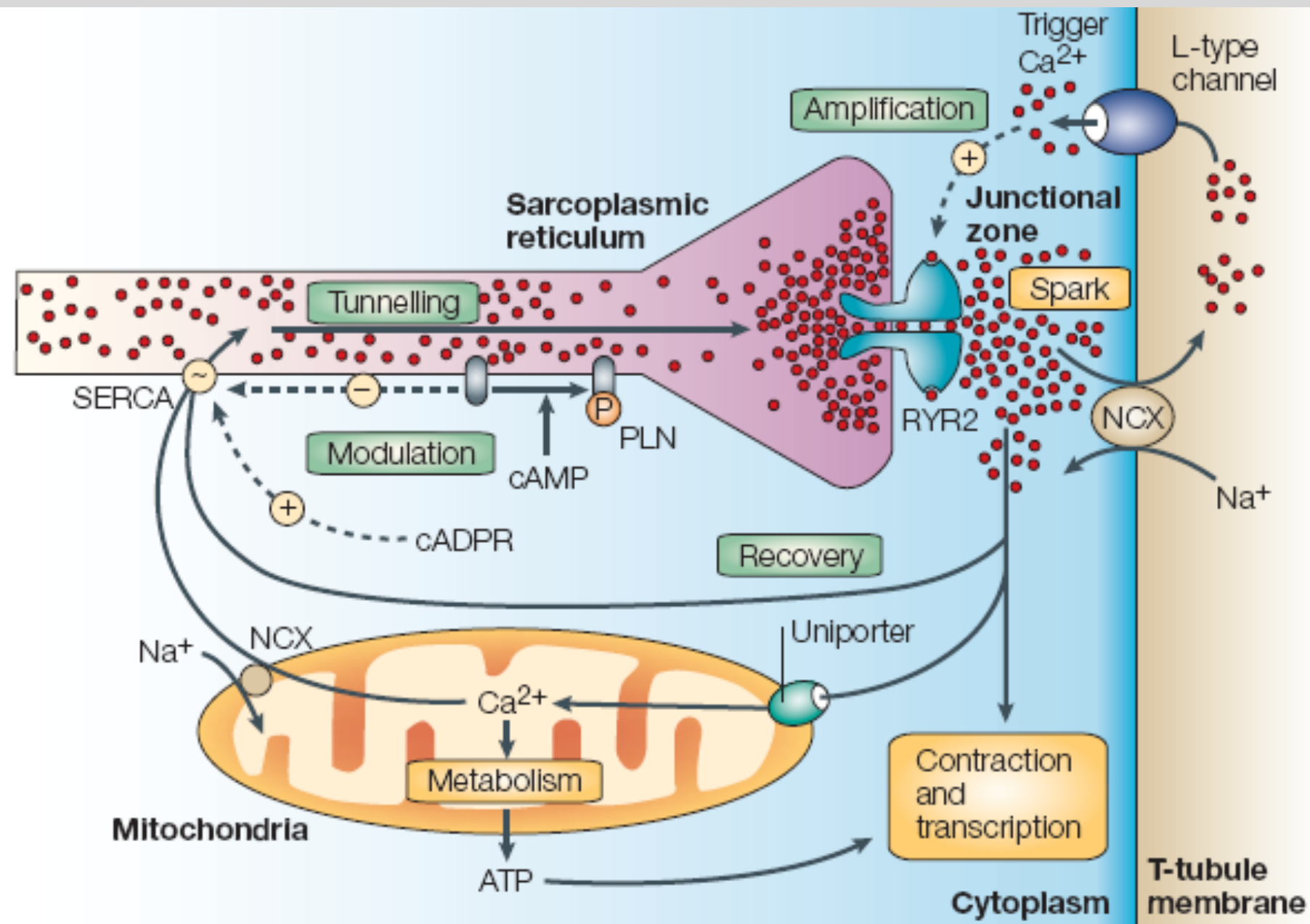
Epifluorecent microscope



VIDEO 1 on moodle

6 - Spatio-temporal profile of Ca^{2+} signals

Calcium signaling during Excitation-Contraction Coupling in cardiac muscle cells



- A proportion of cytosolic Ca^{2+} travels through the mitochondria, during which time it stimulates the metabolism to provide the ATP that is necessary to maintain contraction and transcription.
- The Ca^{2+} , that is returned to the SR, 'tunnels' back to the junctional zone to be used again for subsequent heart beats.
- Adrenaline effect: Activation of (+AMPc/PKA signaling) $\Rightarrow$ phosphorylation of LTTC, RyR, PLN, Troponine C $\Rightarrow$ Rapid/larger Ca^{2+} increase & Rapid decrease $\Rightarrow$ **Increase strength and frequency of cardiac contraction**

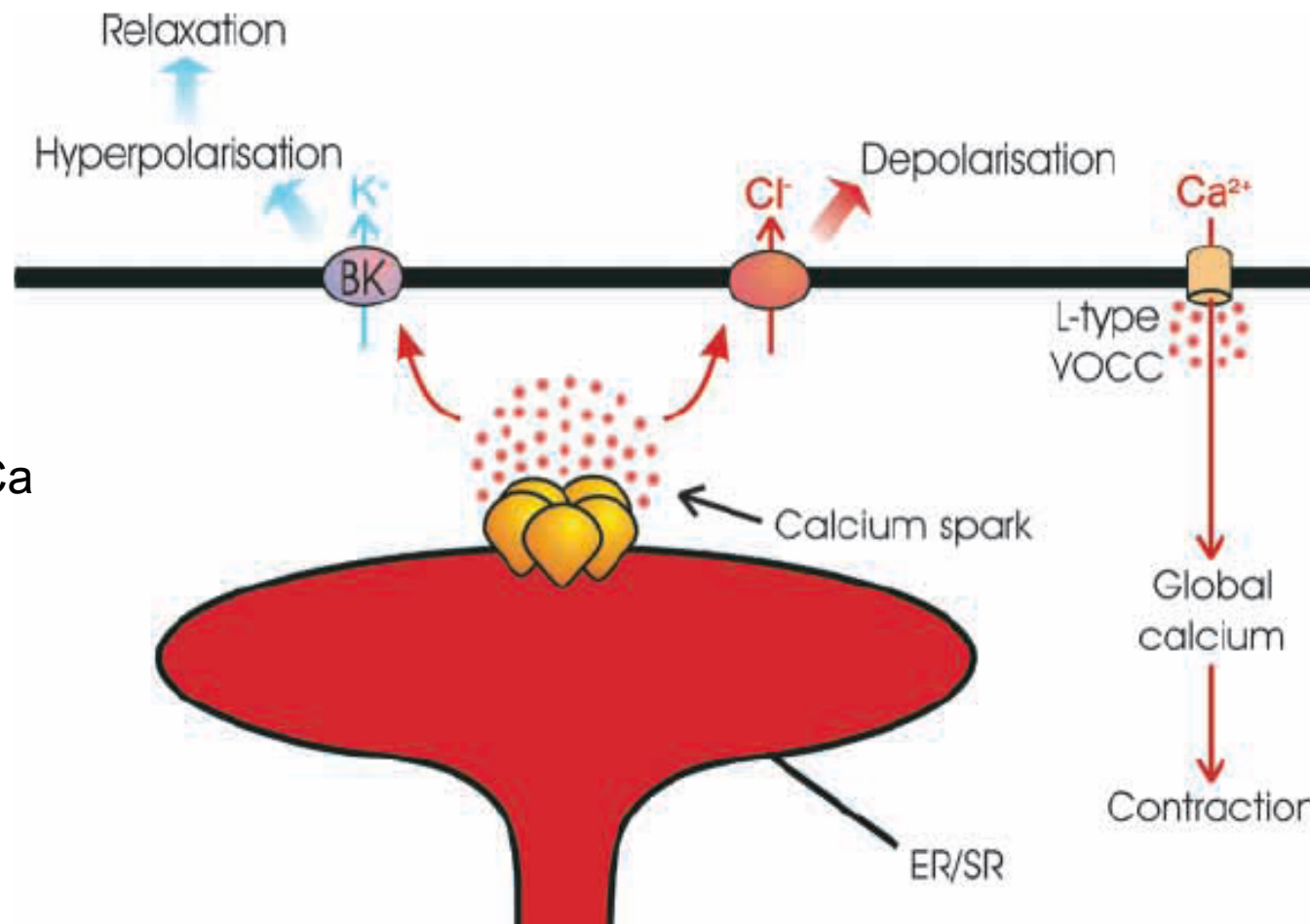
5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in vascular smooth muscle cells

Sparks induits par les RyR

BK = Big potassium
Large conductance
calcium-activated
Potassium channels

Outward current K⁺
Low sensitivity to cytosolic Ca
[Ca²⁺]_i > 1 μM



Inward current Cl⁻
Higher sensitivity to
cytosolic Ca
(Not as widely expressed
Role to stabilize membrane
potential)

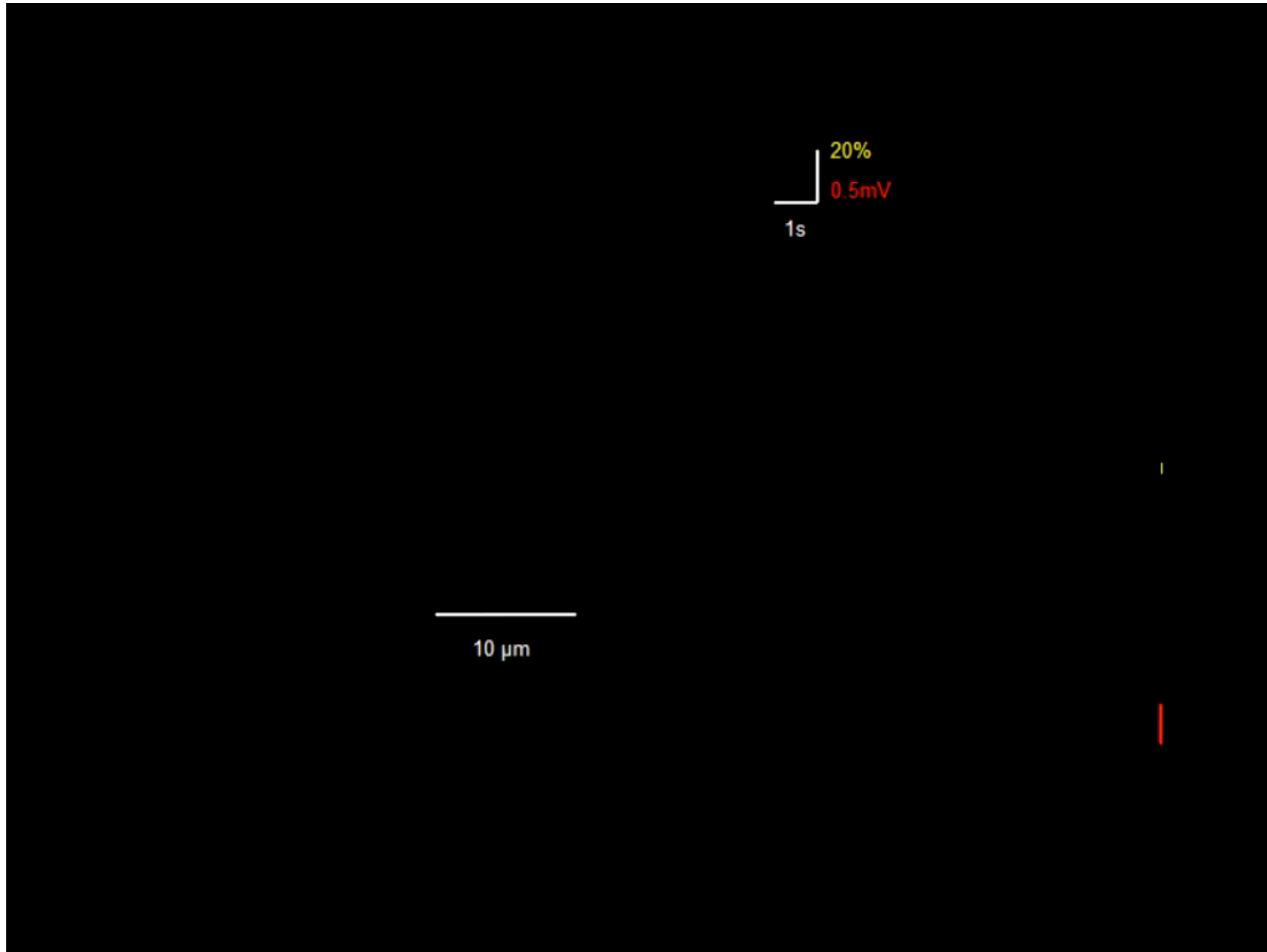
Spontaneous Ca²⁺ spark originating from RyRs can activate BK channels (hyperpolarisation) and Cl⁻ channels (depolarization) => **vascular tonus equilibrium**

Note: Here Ca²⁺ entry via L-type VOCC does not recruit such Ca²⁺ spark sites but leads to global Ca²⁺ increases that trigger **contraction of the vessel**.

5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in neurons

The amplitude of calcium signal depends on the number of action potentials.



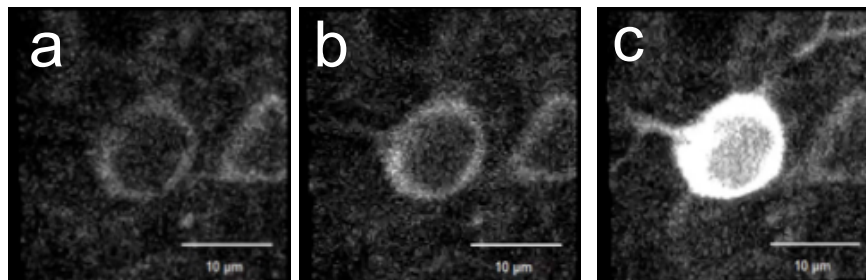
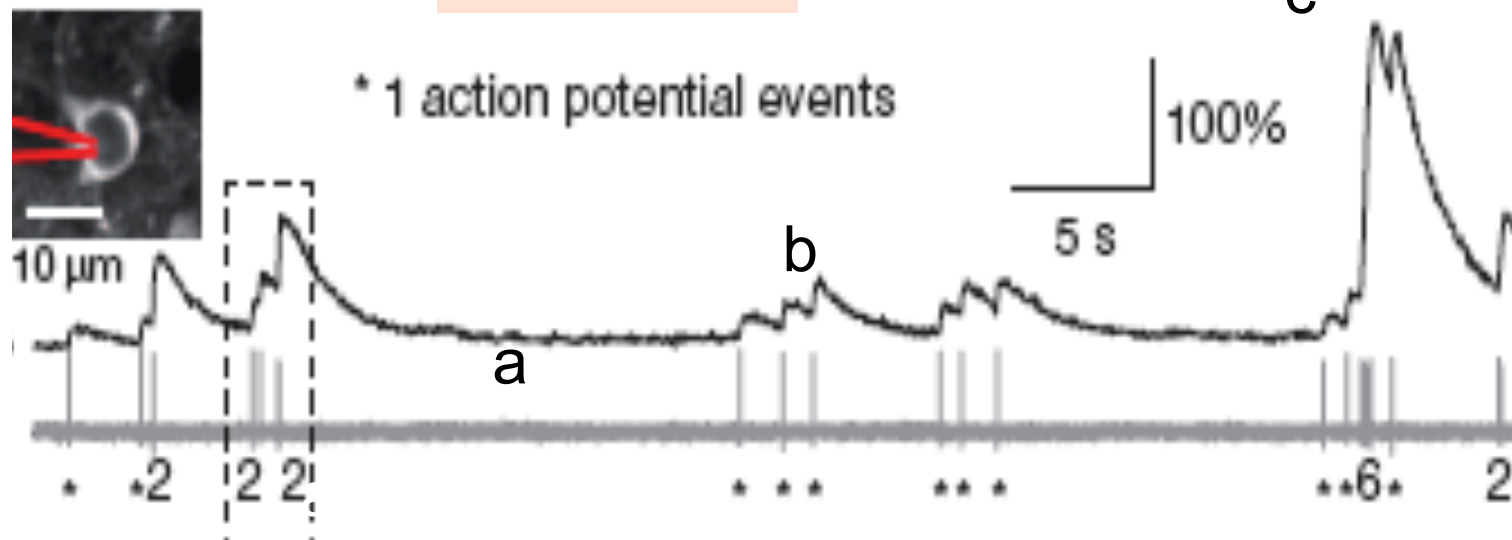
Details of this
experience in next
slide

5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in neurons

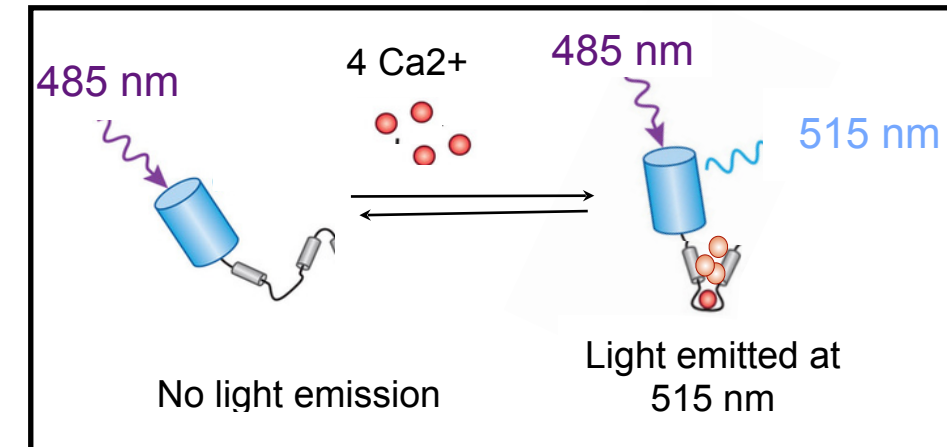
- Stereotaxic Injection of the calcium sensor GCaMP in the visual cortex (vector: virus AAV, brain slices)
- 3 weeks later: simultaneous recording of electrical activity and calcium signal in brain slices.

GCaMP sensor

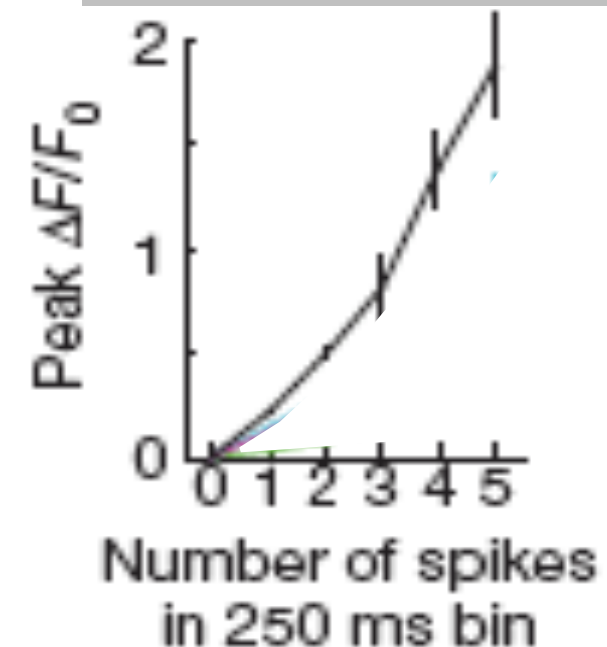


VIDEO 2, moodle

GCaMP Ca²⁺ sensor



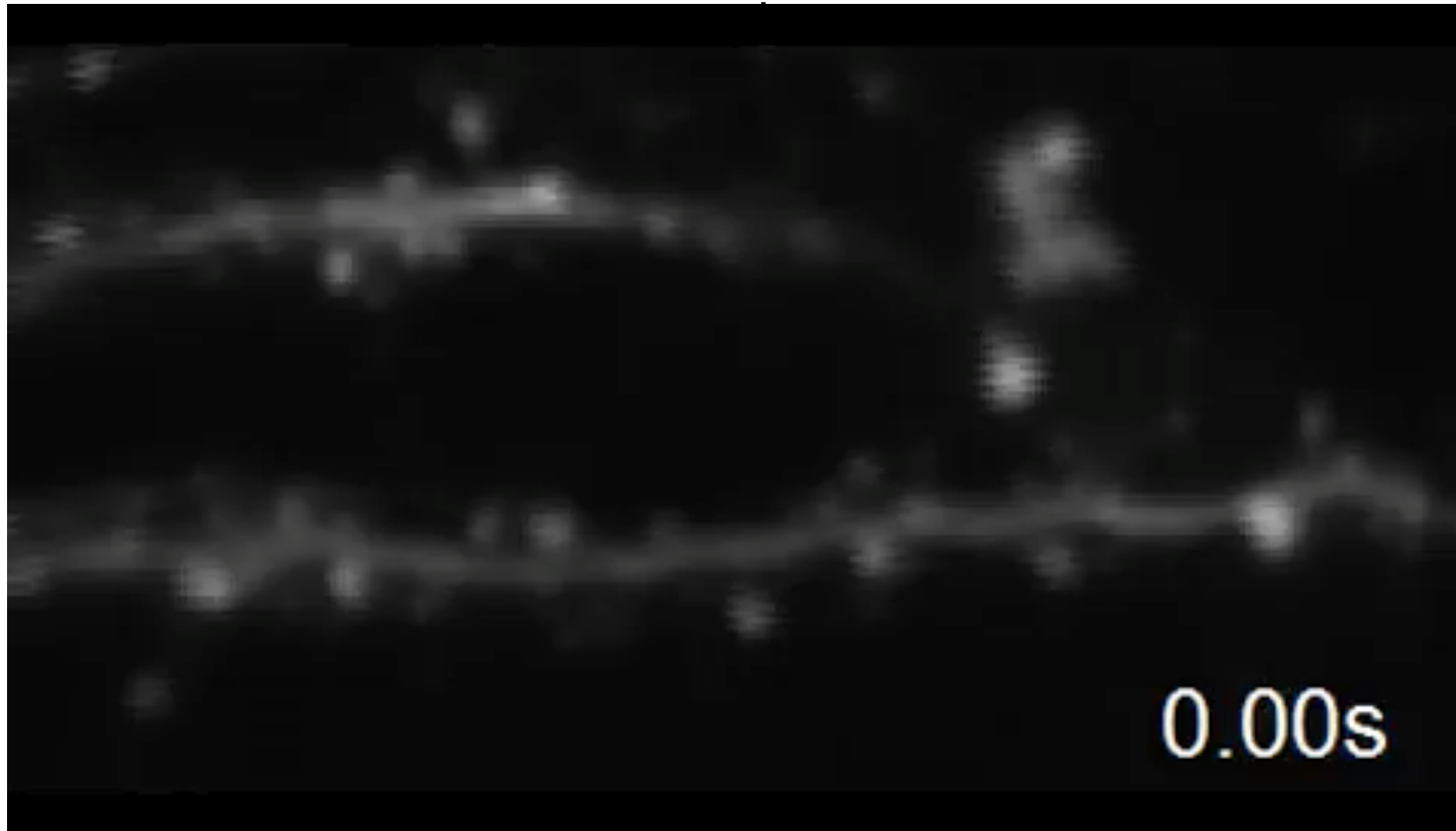
The amplitude of calcium signal is directly proportional to the number of action potentials.



5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in neurons

Spatial features of *in vivo* Calcium signaling in



Details in next slide

VIDEO 3, moodle

- Expression of the calcium sensor GCaMP in the visual cortex (vector: virus AAV) 

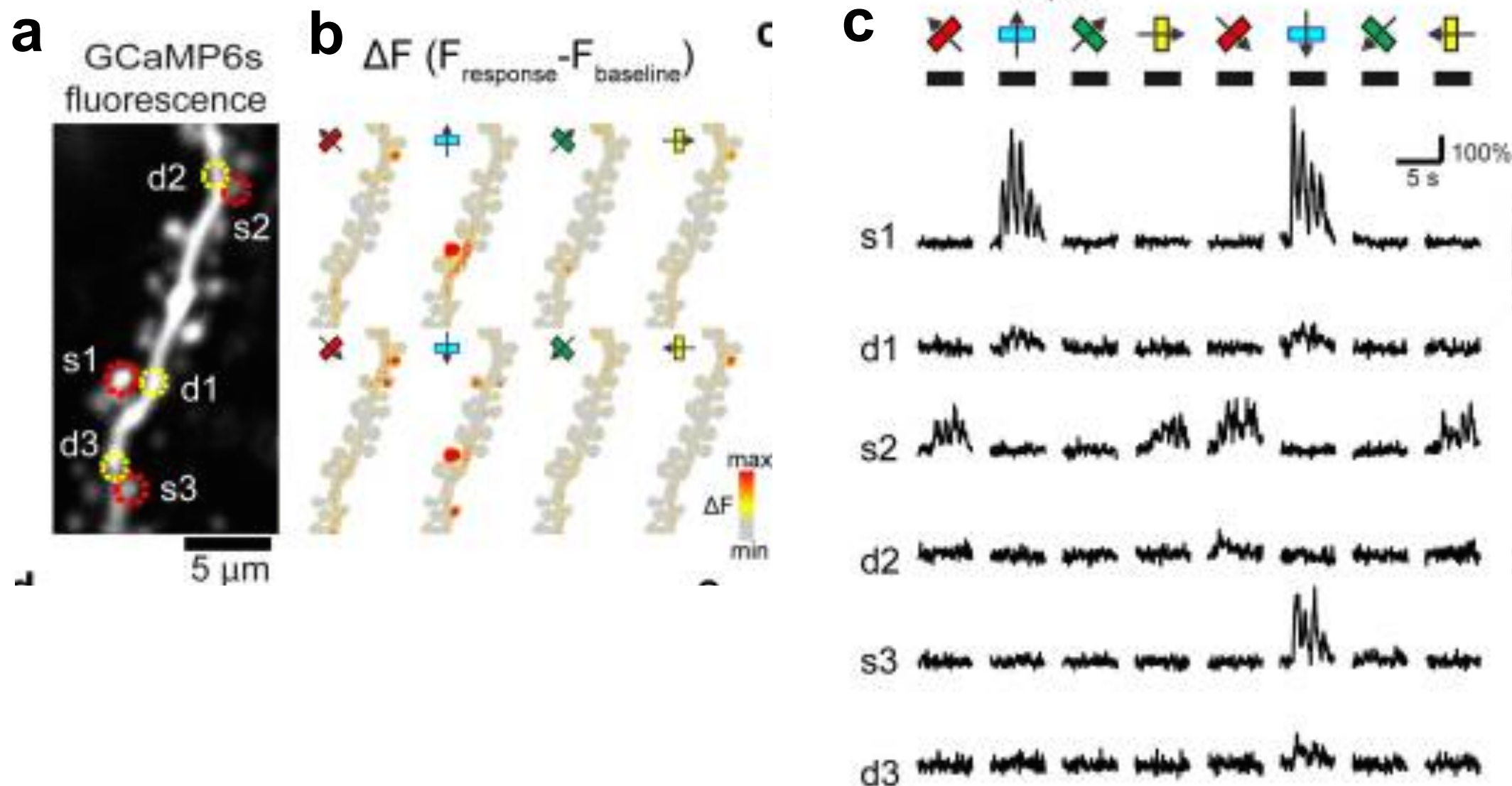
- 3 weeks later: recording of the calcium signal in vivo in response to different orientation of visual stimuli (moving grating):

Each dendritic spine «responds» to specific visual stimulus

5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in neurons

Spatial features of Calcium signaling in spines



- Expression of the calcium sensor GCaMP in the visual cortex (vector: virus AAV)
- 3 weeks later: recording of the calcium signal in vivo in response to different orientation of visual stimuli (moving grating):

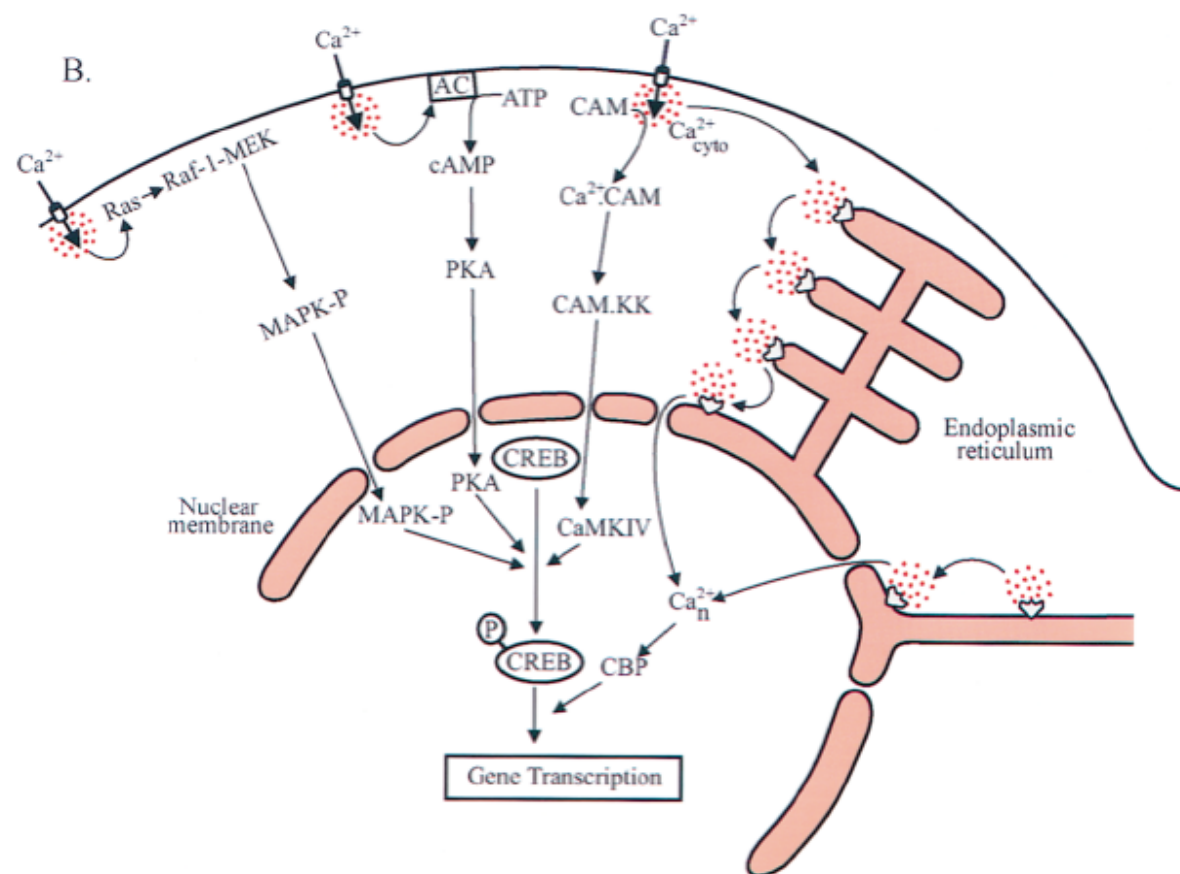
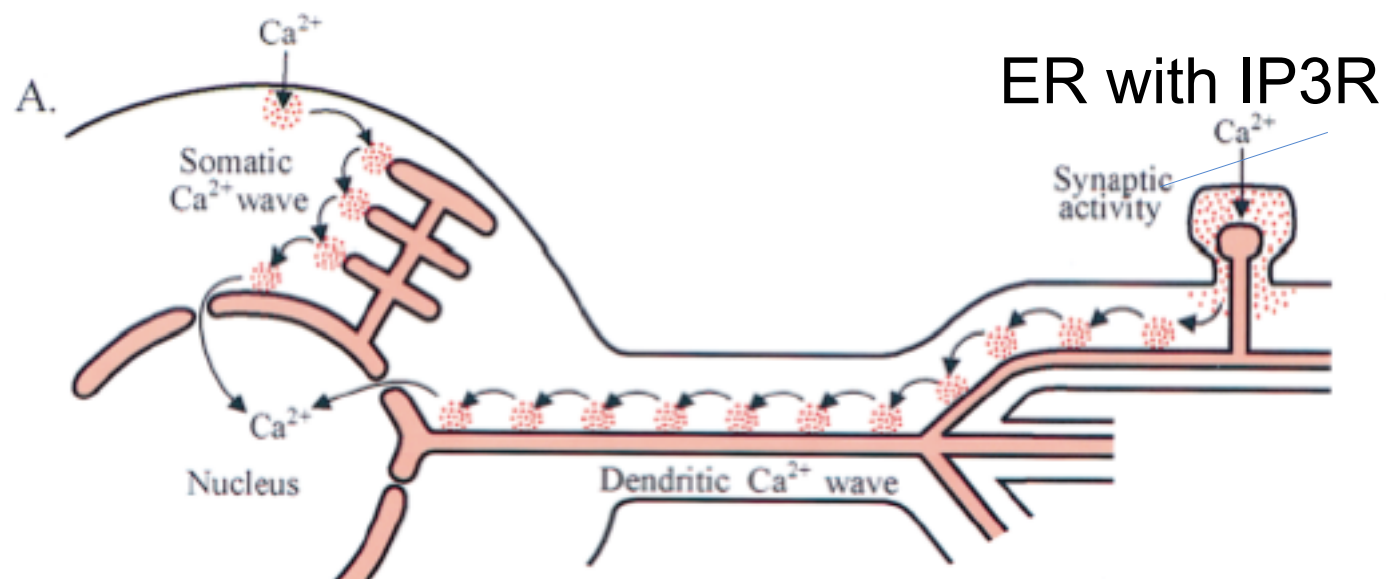
Each dendritic spine «responds» to specific visual stimulus

VIDEO 3, moodle

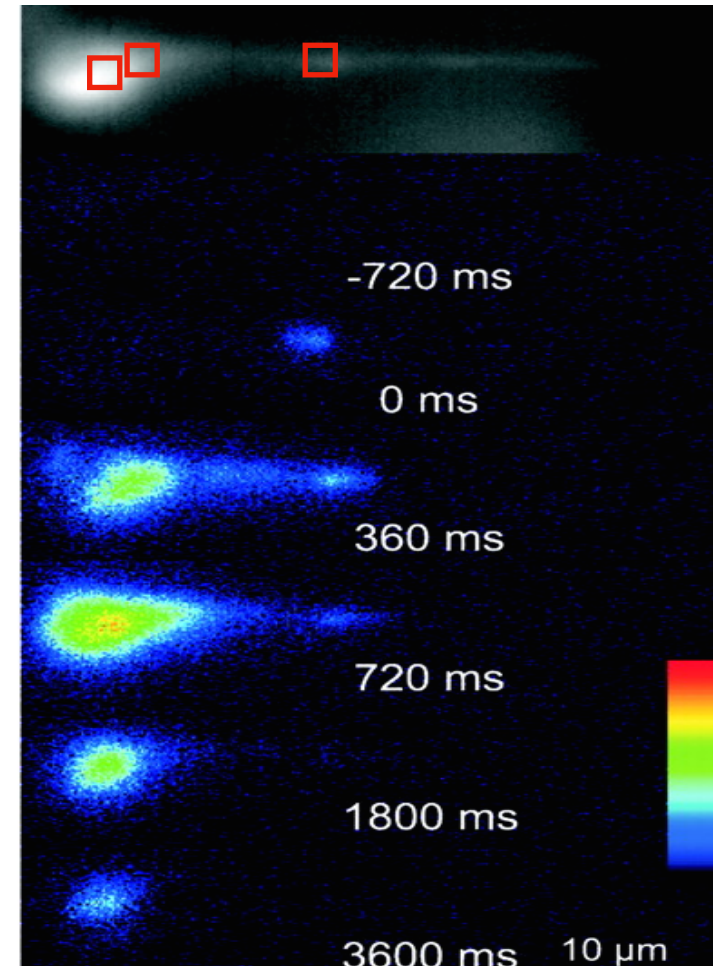
5 - Spatio-temporal profile of Ca²⁺ signals

Calcium signaling in neurons

Calcium signaling in neurons : propagation from spine to the nucleus

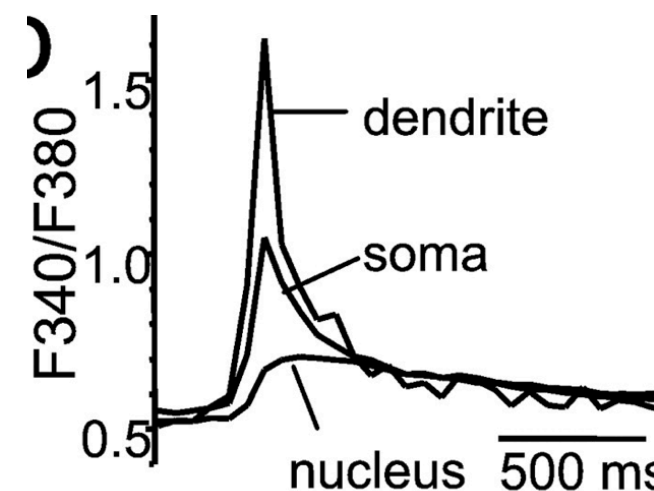


Hippocampal neuron loaded with Fluo 3
Local release of Acetylcholine (0msec)



1 - Acetylcholine triggers Ca²⁺ signal in spines

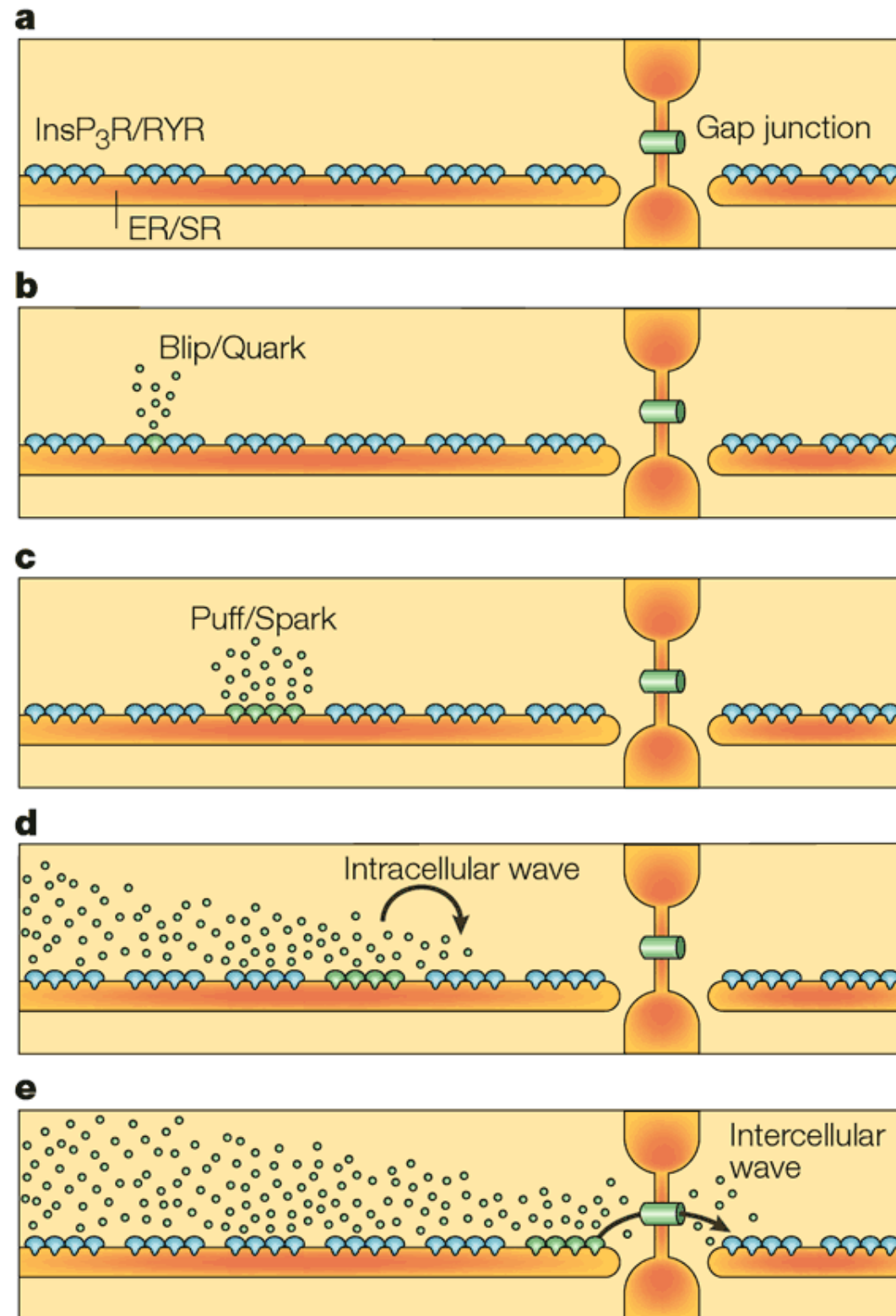
2- Ca²⁺ spreads from dendrites to soma/nucleus where it regulates gene transcription



The intensity of the Ca²⁺ signal decreases

5 - Spatio-temporal profile of Ca^{2+} signals

INTRACELLULAR and INTERCELLULAR PROPAGATION OF Ca^{2+} SIGNAL



a. IP_3R and RyR are distributed over the surface of the endoplasmic and/or sarcoplasmic reticulum (ER/SR)

b. In response to weak stimuli, individual channels open to give either blips (IP_3R) or quarks (RyR s).

c. At higher levels of stimulation, groups of IP_3R or RyR open together to produce puffs or sparks, respectively.

d. When cells are fully excitable, Puffs or Sparks can excite neighbouring receptors through a process of **Ca^{2+} -induced Ca^{2+} release** to set up an intracellular wave

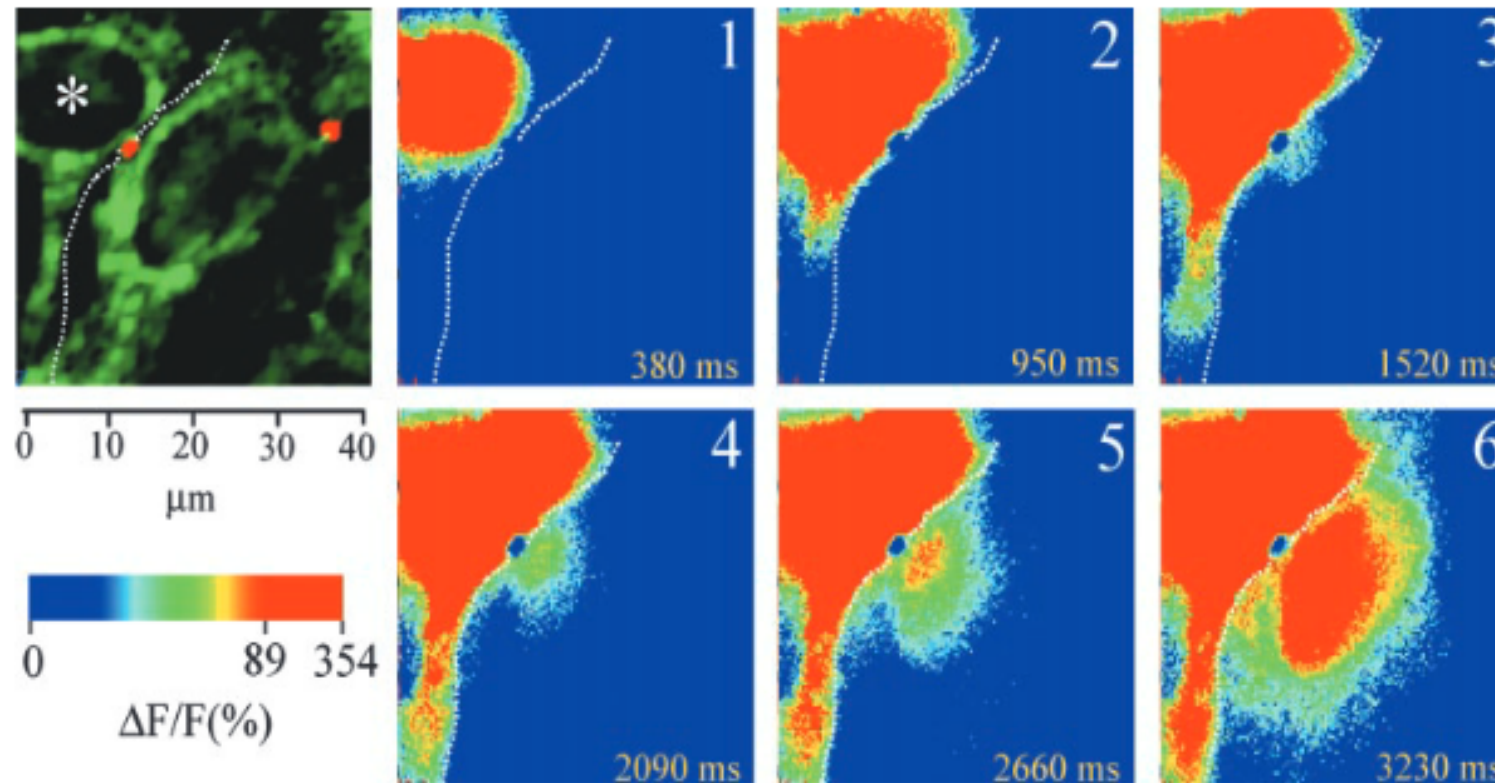
e. When gap junctions connect cells, waves can travel from one cell to the next to set up an intercellular wave

5 - Spatio-temporal profile of Ca²⁺ signals

INTERCELLULAR Ca²⁺ WAVE PROPAGATION BETWEEN CELLS : GAP JUNCTIONS

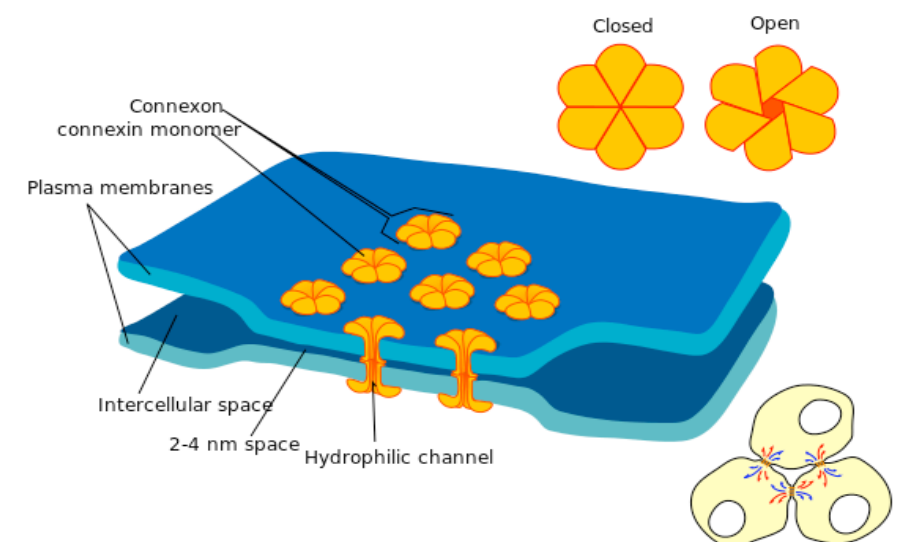
Histology-reconstruction

(HeLa cells transfected with Cx43-GFP
+ live-cell stain ER tracer)



An intercellular calcium wave
propagating via a gap junction
between two HeLa cells
(after mechanical stimulation)

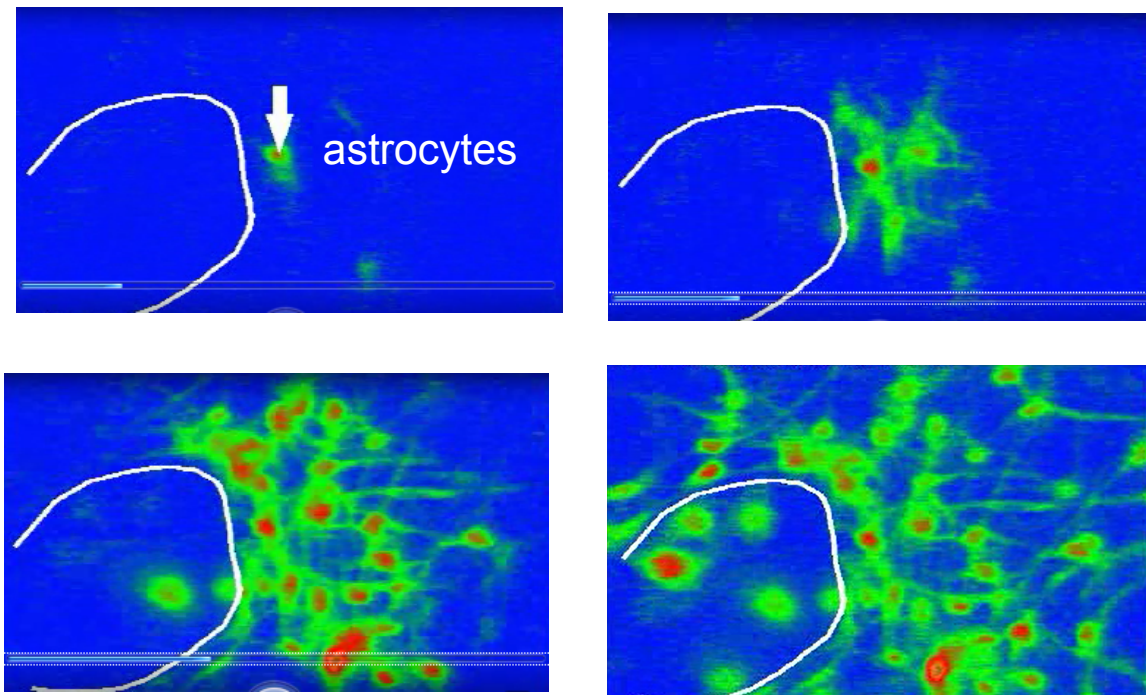
- Cells were loaded with fura-2 AM Ca²⁺ sensor
- HeLa cells without GAP junctions: absence of Ca²⁺ wave propagation.



5 - Spatio-temporal profile of Ca²⁺ signals

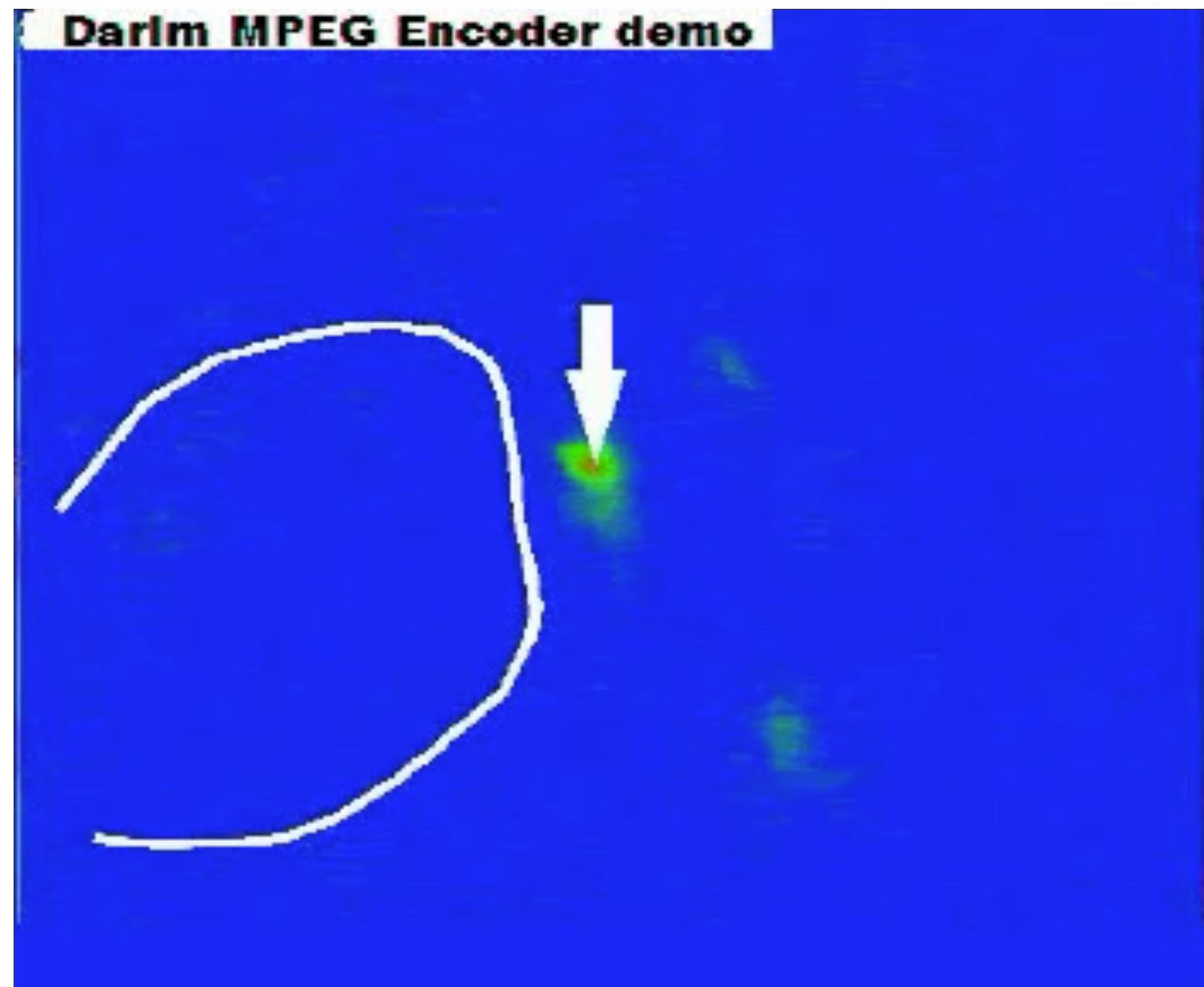
CALCIUM WAVE FROM ASTROCYTES TO ENDOTHELIAL CELLS IN CULTURE

Co-culture:
Astrocytes and Endothelial cells



AM loading

- Loading of the cells with fluo3-AM
- Mechanical stimulation of Astrocytes
- Fast propagation of the calcium signal in glial cells (100µm/sec)
- Propagation of the Ca²⁺ signal to adjacent endothelial cells (inside the white circle).
- Role in Vasoconstriction/Relaxation**



VIDEO 4, moodle

Conclusion

Because calcium level is finely regulated in the cells and serves as a **2nd messenger**, monitoring cellular calcium transients and waves (**calcium imaging**) with Ca^{2+} waves is a good way to follow and learn about involvement of a given cell or cell-type in a population activity (network function) as well as cellular processes triggered inside the cells, in **specific subcellular compartments**.